Linear Algebra

Lecture Notes for MATH2922

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Introduction

Linear algebra is often perceived as difficult and unmotivated by many students, even though it is a central tool in virtually all areas of mathematics, from algebra to differential equation, computing, statistics and more. One reason the subject might be perceived difficult is that it deals with rather more abstract "objects" than say calculus. Before we continue we first answer some frequently asked questions.

What is linear algebra? Linear algebra is the study of abstract *vector spaces* and *linear operators*.

What is a vector space? Roughly speaking, a vector space is a set of objects that behave like "vectors".

Why introduce abstract vector spaces? There are many problems where objects with "vector like" properties occur including geometry, algebra and number theory, solution of systems of linear equations, partial differential equations, Fourier series and harmonic analysis, Statistics and probability and more. This wide occurrence of the same basic features justify the introduction of an abstract structure.

What are abstract structures? The idea of abstract structures is to extract the essential features of problems from seemingly unrelated contexts. Then investigate these structures with or without reference to specific applications. If it is a "good" structure, these investigations feed back to and unify approaches to these seemingly unrelated problems.

A sensible consequence of the above approach is the need for an *axiomatic approach*. We define a minimal set of properties for a structure. Then we use them to derive more properties, and finally apply the results in different contexts. One particularly useful and important structure is that of a *vector space*. In the next section we discuss minimal properties of vectors from which we can deduce all other properties we know already in specific cases. It took a long time for the notion of vector space to be widely accepted. The first with a full theory of vector spaces was the German mathematician Hermann Günter Grassmann (1809–1877) who spent large parts of his working life on developing the theory, and proved many theorems we also encounter in this course. The only problem was that his work was almost completely ignored. In the foreword to

his main work "Die Ausdehnungslehre: Vollständig und in strenger Form bearbeitet" published in 1862 he expressed his frustration[1]:

> *I remain completely confident that the labour I have expended on the science presented here and which has demanded a significant part of my life as well as the most strenuous application of my powers, will not be lost. It is true that I am aware that the form which I have given the science is imperfect and must be imperfect. But I know and feel obliged to state (though I run the risk of seeming arrogant) that even if this work should again remain unused for another seventeen years or even longer, without entering into the actual development of science, still that time will come when it will be brought forth from the dust of oblivion and when ideas now dormant will bring forth fruit. I know that if I also fail to gather around me (as I have until now desired in vain) a circle of scholars, whom I could fructify with these ideas, and whom I could stimulate to develop and enrich them further, yet there will come a time when these ideas, perhaps in a new form, will arise anew and will enter into a living communication with contemporary developments. For truth is eternal and divine and no phase of it ... can pass without a trace; it remains in existence even if the cloth in which weak mortals dress it disintegrates into dust.*

He was right in his judgement about the importance of linear algebra, but it took until 1947 with the publication of Bourbaki's volume on algebra which got abstract linear algebra fully established in mainstream mathematics. Many theorems Grassmann proved were rediscovered and named after other people who did their work many decades later! Today, we could not imagine mathematics without abstract linear algebra. "Nicolas Bourbaki" is a pseudonym under which a group of select French mathematicians sought to unify and publish all mathematics from an axiomatic point of view. They had tremendous influence on how we approach and view mathematics to this day.

The shift to an axiomatic approach

High school mathematics and most of first year mathematics emphasises a mostly informal approach with emphasis on certain objects. For instance, in geometry these objects might be triangles, conics and curves. Calculus works with functions and the related graphs, tangents, areas and so on.

The linear algebra presented in these notes is based on a significant shift in point of view. We move away from specific objects and are not so much concerned what they are, but *how they interact*. The essence of such an axiomatic approach is as follows.

> Emphasis is on a set of rules governing the *interaction between the objects*, not what the objects actually are.

[1] Translation from www-history.mcs.st-and.ac.uk/Extras/Grassmann_1862.html

The advantage for applications is obvious: Whenever we find a set of objects that interact in the way required, all conclusions of the theory are valid, so "object" becomes a place holder for many different objects. This approach makes mathematics more universally applicable and that is in particular true for linear algebra. The objects in linear algebra might be functions, matrices, solutions to ordinary and partial differential equations, wave functions in quantum mechanics, objects in conjunction of image processing, web indexing or the coding of CD's and DVD's. The ideas and concepts of linear algebra generalise in many different directions such as in algebra, functional analysis, computing, statistics and more.

The shift from an empirical to an axiomatic approach took part between about 1880 and 1930. It was not established without conflict since "proofs" given by the greatest and most influential mathematicians working with a more "intuitive" and 'empirical approach saw their work in doubt and under scrutiny with the new methodology and rigour based on very precise definitions and sets of rules. However the new very rigorous approach turned out to be extremely successful and lead to an explosion of knowledge in and reliability of mathematics. In particular it solved the foundational crisis in many areas of mathematics.

Similarly as for the mathematicians of the early 20th century, for many students it is a personal struggle to change the point of view from an empirical to a more abstract and rigorous approach, but certainly worth the effort.

In this course we follow the extremely successful axiomatic approach of linear algebra. As always when following an axiomatic approach it is of great importance to keep examples in mind to make sure not to get lost in "abstract nonsense". Examples and applications are a great source of motivation and guidance on where an abstract approach should go, and how proofs are found and motivated.

Chapter I

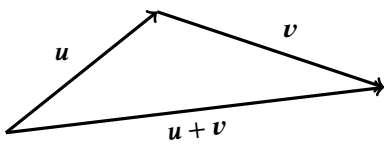
Fields and Vector Spaces

1 Vectors and Scalars

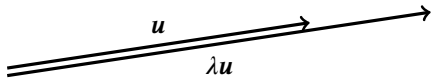
In this section we discuss essential properties of vectors. As explained in the introduction it turns out that these properties are shared by many different objects, not just vectors in the usual sense. This will lead to the definition of an abstract vector space.

On a vector space two operations are defined. They involve two different kinds of objects, namely *vectors* and *scalars*:

- *Addition* of vectors: $(\boldsymbol{u}, \boldsymbol{v}) \mapsto \boldsymbol{u} + \boldsymbol{v}$

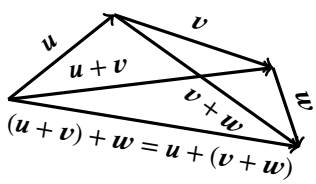


- *Multiplication* of a vector *by a scalar*: $(\lambda u, \boldsymbol{u}) \mapsto \lambda \boldsymbol{v}$

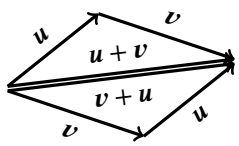


"Scalars" behave like numbers, whereas "vectors" have direction and magnitude. They satisfy the following set of rules. In all statements $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are vectors and λ, μ are scalars.

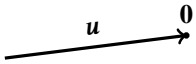
(V1) $(\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \boldsymbol{u} + (\boldsymbol{v} + \boldsymbol{w})$ (Associative law)



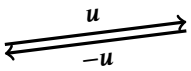
(V2) $\boldsymbol{u} + \boldsymbol{v} = \boldsymbol{v} + \boldsymbol{u}$ (Commutative law)



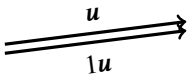
(V3) $\boldsymbol{u} + \mathbf{0} = \boldsymbol{u}$ (Existence of a zero vector)



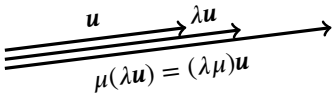
(V4) $\boldsymbol{u} + (-\boldsymbol{u}) = \mathbf{0}$ (Existence of opposite vector)



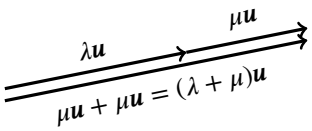
(V5) $1\boldsymbol{u} = \boldsymbol{u}$ (Multiplication by scalar 1)



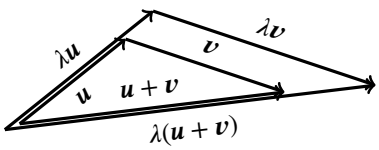
(V6) $\lambda(\mu\boldsymbol{u}) = (\lambda\mu)\boldsymbol{u}$ for all scalars λ, μ



(V7) $(\mu + \lambda)\boldsymbol{u} = \mu\boldsymbol{u} + \lambda\boldsymbol{u}$ (Distributive law)



(V8) $\lambda(\boldsymbol{u} + \boldsymbol{v}) = \lambda\boldsymbol{u} + \lambda\boldsymbol{v}$ (Distributive law)



As we just saw, in the theory of vectors we deal with two kinds of objects, namely "vectors" and "scalars". Scalars can be elements of $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$, depending on the context. If we decide to take an axiomatic approach to vectors it seems reasonable to take an axiomatic approach to scalars as well. Hence, somewhat imprecisely, we make the following definitions:

- Any set of objects with two operations called addition and multiplication that satisfy the usual rules of arithmetic is called a *field*.
- Any set of objects that follow the same rules as vectors is called a *vector space*

As you see, the emphasis here is on the "rules", not on the "objects". The objects become place holders in the same way a variable x in an equation is a place holder for a number. We can do calculations and conclusions without knowing what the oject of interest are. The advantage is that such conclusions can be applied to many different objects exploiting common structures arising in a variety of contexts. We make this more precise below.

2 Fields and abstract vector spaces

A field is a set of "objects" with two operations that follow the usual rules of arithmetic. These rules are given below, and all others follow from them.

Definition 2.1 (field)**.** A field $\mathbb{F}$ is a set with two operations, namely

$$(x, y) \mapsto x + y \quad \text{(addition)} \qquad (x, y) \mapsto x \cdot y \quad \text{(multiplication)}$$

It has two special elements, 0 and 1 such that for all $x, y, z \in \mathbb{F}$

(F1) $x + y = y + x$ (commutative law of addition)

(F2) $(x + y) + z = x + (y + z)$ (associative law of addition)

(F3) $x + 0 = x$ (neutral element for addition)

(F4) for every $x \in \mathbb{F}$ there exists $y \in \mathbb{F}$ with $x + y = 0$ (existence of additive inverse)

(F5) $x \cdot y = y \cdot x$ (commutative law of multiplication)

(F6) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ (associative law of multiplication)

(F7) $1 \cdot x = x$ (neutral element for multiplication)

(F8) for every $x \in \mathbb{F} \setminus \{0\}$ there exists $y \in \mathbb{F}$ with $x \cdot y = 1$ (existence of multiplicative inverse)

(F9) $x \cdot (y + z) = x \cdot y + x \cdot z$ (distributive law)

Before giving some examples let us have a look at the axioms of a field. Axioms (F1)–(F4) relate to addition only. They are the axioms of a commutative group with neutral element 0. We call it the *additive group*. Axioms (F5)–(F8) are the axioms of a group with neutral element 1 if we remove $x = 0$. We call $\mathbb{F} \setminus \{0\}$ the *multiplicative group*. Finally, Axiom (F9) connects properties of the additive and multiplicative group and provides some compatibility between the two.

We give some examples of fields. They will be the only examples we use in this course. There are many more which appear in various applications.

Examples 2.2. (a) The rational numbers $\mathbb{Q} = \{\frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0\}$.

(b) The real numbers $\mathbb{R}$.

(c) The complex numbers $\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\}$.

(d) There are also *finite fields* such as $\mathbb{Z}_2 = \{0, 1\}$ with the operations defined in the tables below:

Addition:

$+$	0	1
0	0	1
1	1	0

Multiplication:

$\cdot$	0	1
0	0	0
1	0	1

A more general class of which $\mathbb{Z}_2$ is a special case is $\mathbb{Z}_p$.

Example 2.3 (The fields $\mathbb{Z}_p$). Let $p \geq 2$ be a positive integer and set

$$\mathbb{Z}_p := \{0, 1, 2 \ldots, p - 1\}$$

Define addition and multiplication in the following way:

- perform addition or multiplication as usual;
- then take the remainder after dividing by p.

For example, if $p = 5$ and $a, b \in \mathbb{Z}_5$ then $2 + 3 = 0$ in $\mathbb{Z}_5$ since the remainder of $(2+3)/5$ is zero, and $2 \cdot 3 = 1$ in $\mathbb{Z}_5$ since the remainder of $(2 \cdot 3)/5$ is one. Addition and multiplication are best given in a table. For $\mathbb{Z}_3$ we get

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$\cdot$	0	1	2
0	0	0	0
1	0	1	2
2	0	2	1

For $\mathbb{Z}_4$ we get

$+$	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

$\cdot$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

$\mathbb{Z}_4$ is not a field since 2 has no multiplicative inverse! ($2k$ never has remainder one when divided by 4.) But $\mathbb{Z}_5$ is a field as one can see from the tables:

$+$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\cdot$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

One can show that $\mathbb{Z}_p$ is a field if and only if p is prime (see tutorials). The fields $\mathbb{Z}_p$ are used extensively in number theory and also in coding theory.

Once we have defined a field, we can define what we mean by a vector space.

Definition 2.4 (vector space)**.** A *vector space* V over a field $\mathbb{F}$ is a set with two operations:

$$\begin{aligned} &V \times V \to V, \quad (\boldsymbol{u}, \boldsymbol{v}) \mapsto \boldsymbol{u} + \boldsymbol{v} && \text{addition} \\ &\mathbb{F} \times V \to V, \quad (\lambda, \boldsymbol{u}) \mapsto \lambda \boldsymbol{u} && \text{multiplication by scalars} \end{aligned}$$

The operations satisfy the axioms (V1)–(V8) above for all $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in V$ and all $\lambda, \mu \in \mathbb{F}$.
The elements of V are called *vectors* and elements of $\mathbb{F}$ are called *scalars*.

☞ *Remark* 2.5. Elements of V are called *vectors* and elements of $\mathbb{F}$ are called *scalars* regardless of whether or not they are actually scalars or vectors in the usual sense. The language of linear algebra is inspired by geometry which helps to keep a picture or model example in mind even when dealing with more abstract objects.

Remark 2.6. Some vector spaces have other operations like *dot product* or *cross product* or even multiplication. At the moment we only deal with the bare bone structure defined above, but later we will consider vector spaces later with such additional structures, in particular with the dot product.

When working with vectors or scalars we use many properties not stated in the axioms above. In the next section we show that they are a consequence of the axioms.

3 Consequences of the field and vector space axioms

The axioms state a minimal number of properties of vectors and scalars. All other properties not explicitly stated, even if looking familiar and obvious, need to be derived from the axioms. For instance we take it for granted that $0\boldsymbol{w} = \mathbf{0}$ for all $w \in V$. However, the axioms do not state that so we need to prove this from the axioms. Similarly we need to prove that $-\boldsymbol{u} = (-1)\boldsymbol{u}$. Recall that every field is a vector space over itself, so everything we prove about vector spaces applies to fields as well.

Proposition 3.1. *Let V be a vector space over the field $\mathbb{F}$. Then*

(i) *The zero vector is unique;*

(ii) *The* 1 *in $\mathbb{F}$ is unique;*

(iii) $0\boldsymbol{u} = \mathbf{0}$ *for all* $\boldsymbol{u} \in V$ *and* $\lambda\mathbf{0} = \mathbf{0}$ *for all* $\lambda \in \mathbb{F}$*;*

(iv) *For every* $\boldsymbol{u} \in V$ *there is a unique vector* $\boldsymbol{v}$ *with* $\boldsymbol{u} + \boldsymbol{v} = \mathbf{0}$*. We set* $-\boldsymbol{u} := \boldsymbol{v}$*;*

(v) $(-\lambda)\boldsymbol{u} = \lambda(-\boldsymbol{u}) = -(\lambda\boldsymbol{u})$ *for all* $\lambda \in \mathbb{F}$ *and* $\boldsymbol{u} \in V$*;*

(vi) $(-\lambda)(-\boldsymbol{u}) = \lambda\boldsymbol{u}$ *for all* $\lambda \in \mathbb{F}$ *and* $\boldsymbol{u} \in V$*;*

(vii) *If* $\lambda\boldsymbol{u} = \mathbf{0}$*, then* $\lambda = 0$ *or* $\boldsymbol{u} = \mathbf{0}$*.*

Proof. (i) Let $\mathbf{0}, \tilde{\mathbf{0}} \in V$ be zero vectors. Then

$$\mathbf{0} \overset{(V3)}{=} \mathbf{0} + \tilde{\mathbf{0}} \overset{(V2)}{=} \tilde{\mathbf{0}} + \mathbf{0} \overset{(V3)}{=} \tilde{\mathbf{0}}$$

(ii) We proceed in exactly the same way as in (i) replacing addition by multiplication and zero by one. Let $1, \tilde{1} \in \mathbb{F}$ be scalars with the property (F7). Then

$$1 \overset{(F7)}{=} 1 \cdot \tilde{1} \overset{(F5)}{=} \tilde{1} \cdot 1 \overset{(F7)}{=} \tilde{1}$$

(iii) If $\boldsymbol{u} \in V$ and $0 \in \mathbb{F}$, then

$$0\boldsymbol{u} \overset{(F3)}{=} (0+0)\boldsymbol{u} \overset{(V7)}{=} 0\boldsymbol{u} + 0\boldsymbol{u}$$

By (V4) there exists $\boldsymbol{v} \in V$ with $\mathbf{0} = 0\boldsymbol{u} + \boldsymbol{v}$. Hence

$$\mathbf{0} = 0\boldsymbol{u} + \boldsymbol{v} = (0\boldsymbol{u} + 0\boldsymbol{u}) + \boldsymbol{v} \overset{(V1)}{=} 0\boldsymbol{u} + (0\boldsymbol{u} + \boldsymbol{v}) = 0\boldsymbol{u} + \mathbf{0} \overset{(V3)}{=} 0\boldsymbol{u}$$

A similar argument using (F4) and (V8) instead of (V7) and (F3), respectively, proves that $\lambda\mathbf{0} = \mathbf{0}$ for all $\lambda \in \mathbb{F}$.

(iv) By (V4) there exists $\boldsymbol{w} \in V$ with $\boldsymbol{u} + \boldsymbol{w} = \mathbf{0}$, so we only need to show the uniqueness. Hence let $\boldsymbol{w} \in V$ be such that $\mathbf{0} = \boldsymbol{u} + \boldsymbol{w}$. Then

$$\boldsymbol{v} = \boldsymbol{v} + \mathbf{0} = \boldsymbol{v} + (\boldsymbol{u} + \boldsymbol{w}) \overset{(V1)}{=} (\boldsymbol{v} + \boldsymbol{u}) + \boldsymbol{w} \overset{(V2)}{=} (\boldsymbol{u} + \boldsymbol{v}) + \boldsymbol{w} = \mathbf{0} + \boldsymbol{w} \overset{(V2)}{=} \boldsymbol{w} + \mathbf{0} \overset{(V3)}{=} \boldsymbol{w}$$

so $\boldsymbol{x} = \boldsymbol{v} = \boldsymbol{w}$ as claimed.

(v) By (iii) and (iv) we have $\mathbf{0} = (\lambda - \lambda)\boldsymbol{u} \overset{(V7)}{=} \lambda\boldsymbol{u} + (-\lambda)\boldsymbol{u}$ and hence again by (iii) we get $-(\lambda\boldsymbol{u}) = (-\lambda)\boldsymbol{u}$. Similarly by (iii) and (iv) $\mathbf{0} = \lambda(\boldsymbol{u} - \boldsymbol{u}) \overset{(V8)}{=} \lambda\boldsymbol{u} + \lambda(-\boldsymbol{u})$ and by (iii) we get $-(\lambda\boldsymbol{u}) = \lambda(-\boldsymbol{u})$.

(vi) From from (iv) we have $\mathbf{0} = \boldsymbol{u} + (-\boldsymbol{u}) \overset{(V2)}{=} (-\boldsymbol{u}) + \boldsymbol{u}$ and so $-(-\boldsymbol{u}) = \boldsymbol{u}$. Hence using (vi) we get $(-\lambda)(-\boldsymbol{u}) = \lambda(-(-\boldsymbol{u})) = \lambda\boldsymbol{u}$ as claimed.

(vii) We give a proof by contrapositive and assume that $\lambda \neq 0$ and $\boldsymbol{u} \neq 0$. Then using (F8) there exists μ with $\lambda\mu = 1$. Hence

$$\mu(\lambda\boldsymbol{u}) \overset{(V6)}{=} (\mu\lambda)\boldsymbol{u} = 1\boldsymbol{u} \overset{(V5)}{=} \boldsymbol{u} \neq \mathbf{0}.$$

Hence $\lambda\boldsymbol{u} \neq 0$ as otherwise (iii) implies that $\lambda\boldsymbol{u} = 0$. □

As stated in the proposition above every $\boldsymbol{u} \in V$ has a unique additive inverse $-\boldsymbol{u}$. This allows us to define *subtraction*, which is not part of the axioms explicitly. We define

$$\boldsymbol{v} - \boldsymbol{u} := \boldsymbol{v} + (-\boldsymbol{u}).$$

In a similar way we define subtraction in a field. In a field we can also define *division* using multiplication with the multiplicative inverse.

4 Examples of vector spaces

The most familiar example of vector spaces is $\mathbb{R}^2$ and $\mathbb{R}^3$ studied in first year. We can generalise these vector spaces as follows. Let $\mathbb{F}$ be a field and $n \geq 1$ an integer. We set

$$\mathbb{F}^n := \{\boldsymbol{x} = (x_1, x_2, \dots, x_n) : \ x_k \in \mathbb{F},\ k = 1, \dots, n\}$$

For $\boldsymbol{x} = (x_1, x_2, \dots, x_n)$, $\boldsymbol{y} = (y_1, \dots, y_n) \in \mathbb{F}^n$ and $\lambda \in \mathbb{F}$ we define addition and multiplication by scalars as follows:

$$\begin{aligned} \boldsymbol{x} + \boldsymbol{y} &:= (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \\ \lambda\boldsymbol{x} &:= (\lambda x_1, \lambda x_2, \dots, \lambda x_n) \end{aligned}$$

The space $\mathbb{F}^n$ is a vector space over $\mathbb{F}$. Most of the time we consider vectors in $\mathbb{F}^n$ to be *column vectors*, that is, vectors of the shape

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

In particular, for $n = 1$ we can view $\mathbb{F} = \mathbb{F}^1$ as a vector space over $\mathbb{F}$. To prove that $\mathbb{F}^n$ is a vector space we would need to check that all eight vector space axioms are satisfied. We do that for a more general example, namely the set of functions from some set into a vector space. The above is covered as well since we can viewed $\mathbb{F}^n$ as the set of functions from $\{1, \dots, n\}$ into $\mathbb{F}$.

Proposition 4.1. *Let X be a set and V a vector space over the field $\mathbb{F}$. Consider the set of functions*

$$F(X, V) := \{u \mid u\colon X \to V \text{ is a function}\}.$$

For $u, v \in F(X, V)$ and $\lambda \in \mathbb{F}$ we define addition and multiplication by scalars as follows:

$$\begin{aligned} (u + v)(x) &:= u(x) + v(x) && \text{for all } x \in X \\ (\lambda u)(x) &:= \lambda u(x) && \text{for all } x \in X. \end{aligned}$$

Then the space $F(X, V)$ is a vector space over $\mathbb{F}$. We explicitly allow $V = \mathbb{F}$.

Proof. Note that a function is given if we know its value at every point in its domain. Since addition and multiplications by scalars is defined pointwise, it is sufficient to check (V1)–(V8) for every $x \in X$. We let $u, v, w \in F(X, V)$ and $\lambda, \mu \in \mathbb{F}$

(V1) By (V1) in the vector space V

$$\begin{aligned} \big((u + v) + w\big)(x) &= (u(x) + v(x)) + w(x) \\ &= u(x) + (v(x) + w(x)) = \big(u + (v + w)\big)(x) \end{aligned}$$

for all $x \in X$. If $V = \mathbb{F}$ we use (F2) instead of (V1).

(V2) By (V1) in the vector space V

$$(u + v)(x) = u(x) + v(x) = v(x) + u(x) = (v + u)(x)$$

for all $x \in X$. If $V = \mathbb{F}$ we use (F1) instead of (V2).

(V3) Define the *zero function* by $\mathbf{0}(x) := 0$ for all $x \in X$, where $0 \in V$ is from (V3). If $V = \mathbb{F}$ use $0 \in \mathbb{F}$ from (F3).

(V4) From (V4) in V we can define $(-u)(x) := -u(x)$ for all $x \in X$. If $V = \mathbb{F}$ we use (F4) instead.

(V5) From (V5) in V

$$(1u)(x) = 1\big(u(x)\big) = u(x)$$

for all $x \in X$. If $V = \mathbb{F}$ we use (F7) instead of (V5).

(V6) From (V6) in V

$$\big(\mu(\lambda u)\big)(x) = \mu(\lambda u)(x) = \mu\big(\lambda u(x)\big) = (\mu\lambda)u(x) = \big((\mu\lambda)u\big)(x)$$

for all $x \in X$. If $V = \mathbb{F}$ we use (F6) instead of (V6).

(V7) From (V7) in V

$$\big((\mu + \lambda)u\big)(x) = (\mu + \lambda)u(x) = \mu u(x) + \lambda u(x) = (\mu u)(x) + (\lambda u)(x)$$

for all $x \in X$. If $V = \mathbb{F}$ we use (F9) instead of (V7).

(V8) From (V8) in V

$$\big(\lambda(u + v)\big)(x) = \lambda\big(u(x) + v(x)\big) = \lambda u(x) + \lambda v(x) = \big(\lambda u + \lambda v\big)(x)$$

for all $x \in X$. If $V = \mathbb{F}$ we use (F9) instead of (V8).

Having checked all axioms the proof of the proposition is complete. □

We now discuss more examples of vector spaces.

Example 4.2. The space $\mathrm{Mat}(m \times n, \mathbb{F})$ of $m \times n$ matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

with $a_{jk} \in \mathbb{F}$. We often write

$$A = [a_{jk}]_{1 \le j \le m,\, 1 \le k \le n} \qquad \text{or simply} \qquad A = [a_{jk}].$$

Then $\mathrm{Mat}(m \times n, \mathbb{F})$ is a vector space over $\mathbb{F}$ if we define

$$A + B = [a_{jk}] + [b_{jk}] := [a_{jk} + b_{jk}]$$
$$\lambda A = \lambda[a_{jk}] := [\lambda a_{jk}]$$

To prove that $\mathrm{Mat}(m \times n, \mathbb{F})$ is a vector space just note that $\mathrm{Mat}(m \times n, \mathbb{F}) = F(X, \mathbb{F})$ with $X = \{1, \dots, m\} \times \{1, \dots, n\}$.

As learnt in first year linear algebra matrices of appropriate sizes can be multiplied. We do not want to do that here, but defer it to later when we look at linear operators.

Another example of a vector space is the space of polynomials with coefficients in a given field.

Example 4.3. Let $\mathbb{F}$ be an arbitrary field and X a variable. We call

$$a_0 + a_1 X + a_2 X^2 + \cdots + a_n X^n$$

a (*formal*) *polynomial* with coefficients in $a_0, \dots, a_n \in \mathbb{F}$. If $a_n \neq 0$ we call n the *degree* of the polynomial. The set of polynomials over $\mathbb{F}$

$$\mathbb{F}[X] := \{a_0 + a_1 X + a_2 X^2 + \cdots + a_n X^n : \ a_i \in \mathbb{F}, \ i = 0, \dots, \ n \in \mathbb{N}\}$$

is a vector space if we define addition and multiplication by scalars in the obvious way. The zero polynomial is the one with all coefficients equal to zero. Each polynomial corresponds to a sequence $(a_k)_{k \in \mathbb{N}}$ of coefficients, where at most finitely many are non-zero.

Next we look at sums or products of vector spaces.

Example 4.4 (product of vector spaces). Let $V_1, \dots, V_n$ be vector spaces over a field $\mathbb{F}$. Similar to $\mathbb{F}^n$ we can form a new vector space

$$V_1 \times V_2 \cdots \times V_n := \{(v_1, \dots, v_n) : \ v_k \in V_i, k = 1, \dots, n\}.$$

This is a vector space if we define the operations component-wise, that is,

$$(v_1, \dots, v_n) + (w_1, \dots, w_n) := (v_1 + w_1, \dots, v_n + w_n)$$

and

$$\alpha(v_1, \dots, v_n) := (\alpha v_1, \dots, \alpha v_n).$$

Many vector spaces are inside a bigger vector space and called *subspaces*. Let V be a vector space over $\mathbb{F}$. We consider a non-empty set $W \subset V$ satisfying

(i) $\boldsymbol{u} + \boldsymbol{v} \in W$ for all $\boldsymbol{u}, \boldsymbol{v} \in W$;

(ii) $\lambda \boldsymbol{u} \in W$ for all $\boldsymbol{u} \in W$ and $\lambda \in \mathbb{F}$;

Then naturally, all axioms of a vector space are fulfilled except possibly (V3) and (V4). Now (V3) follows since Proposition 3.1 and (ii) imply $\mathbf{0} = 0\boldsymbol{w} \in W$. Similarly (V4) follows since Proposition 3.1 and (i) imply that $-\boldsymbol{w} = (-1)\boldsymbol{w} \in W$.

Proposition 4.5 (Subspace of a Vector Space)**.** *A non-empty subset $W \subseteq V$ is a* subspace *of the vector space V if*

(i) $\boldsymbol{u} + \boldsymbol{v} \in W$ *for all* $\boldsymbol{u}, \boldsymbol{v} \in W$*;*

(ii) $\lambda \boldsymbol{u} \in W$ *for all* $\boldsymbol{u} \in W$ *and* $\lambda \in \mathbb{F}$*;*

The easiest way to show W is non-empty is to check that $0 \in W$.

Examples 4.6. (a) Lines and planes in $\mathbb{R}^3$ passing through the origin.

(b) Sets of the form

$$\{x = (x_1, \dots, x_n) \in \mathbb{F}^n : \lambda_1 x_1 + \lambda_2 x_2 + \cdots + \lambda_n x_n = 0\}$$

are subspaces of $\mathbb{F}^n$. This is an "$(n-1)$-dimensional" subspace of $\mathbb{F}^n$, often called a *hyperplane* in $\mathbb{F}^n$. The space is non-empty since it contains the zero vector.

(c) Intersections of lines and planes in $\mathbb{F}^n$ form a subspace. For instance the intersection of hyperplanes which can be described by two or more simultaneous equations

$$\begin{aligned} 0 &= \lambda_1 x_1 + \lambda_2 x_2 + \cdots + \lambda_n x_n \\ 0 &= \mu_1 x_1 + \mu_2 x_2 + \cdots + \mu_n x_n \end{aligned}$$

Clearly the zero vector is a solution, so the set is non-empty. This is illustrated in Figure 4.1.

(d) Polynomials of degree n or less form a subspace of $\mathbb{F}[X]$.

(e) The space $C((a,b),\mathbb{R})$ of continuous functions from (a,b) to $\mathbb{R}$ is a subspace of $F((a,b),\mathbb{R})$.

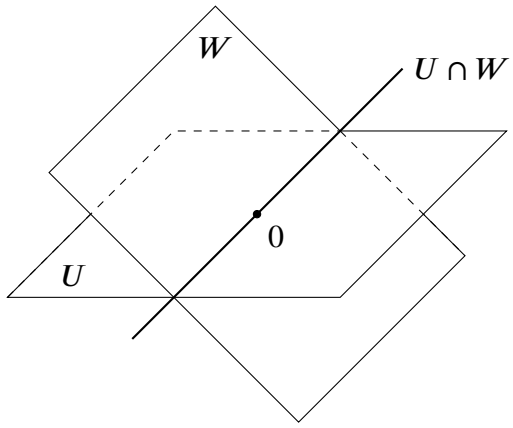


Figure 4.1: Intersection of two subspaces is a subspace

We saw above that the intersection of hyperplanes is a subspace. We can generalise the idea and look at the intersection of two or more subspaces of an arbitrary vector space.

Proposition 4.7. *Let V be a vector space over $\mathbb{F}$ and let U, W subspaces of V. Then $U \cap W$ is a subspace of V as well.*

Proof. As U and W both contain the zero vector, $U \cap W \neq \emptyset$. We verify the two conditions from Proposition 4.5. Let $u, v \in U \cap W$. Then $u, v \in U$ and $u, v \in W$ by definition of an intersection. As U is a subspace, $u + v \in U$. Similarly, as W is a subspace, $u + v \in W$. Hence by definition of an intersection $u + v \in U \cap W$. For the second condition assume that $u \in U \cap W$ and that $\lambda \in \mathbb{F}$. Then $u \in U$ and $u \in W$. As U is a subspace $\lambda u \in U$, and as W is a subspace $\lambda u \in W$ as well. Hence $\lambda u \in U \cap W$, proving that $U \cap W$ is a subspace of V. $\square$

Note however that the union $U \cup W$ is not in general a subspace since the sum of a vector in U and a vector in W does not need to be in $U \cup W$. For instance, two crossing lines in the plane passing through the origin represent two subspaces, but their union is not a subspace as illustrated in Figure 4.2. However, the picture suggests that if we

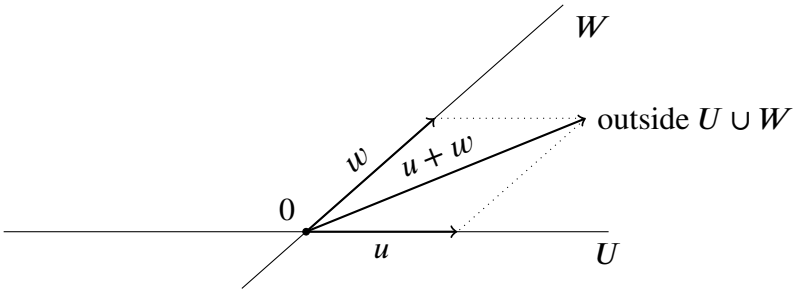


Figure 4.2: Union of two subspaces is not a subspace.

take the sum $u + w$ for $u \in U$ and $w \in W$ we get a subspace, namely the sum of the two. We define

$$U + W := \{u + w : u \in U,\ w \in W\}.$$

In fact we have the following proposition.

Proposition 4.8. *Let V be a vector space over $\mathbb{F}$ and let U, W subspaces of V. Then $U + W$ is a subspace of V.*

Proof. We again check the two conditions from Proposition 4.5. First let $v_1, v_2 \in U + W$. By definition of $U + W$ these vectors are of the form $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ with $u_1, u_2 \in U$ and $w_1, w_2 \in W$. Hence

$$v_1 + v_2 = (u_1 + w_1) + (u_2 + w_2) = (u_1 + u_2) + (w_1 + w_2) \in U + W$$

since U is a subspace and so $u_1 + u_2 \in U$ and similarly W is a subspace and so $w_1 + w_2 \in W$. For the second condition suppose that $v = u + w$ with $u \in U$ and $w \in W$. If $\lambda \in \mathbb{F}$, then $\lambda u \in U$ because U is a subspace and $\lambda w \in W$ because W is a subspace. Hence

$$\lambda v = \lambda(u + w) = \lambda u + \lambda w \in U + W$$

as claimed. Hence $U + W$ is a subspace. □

Intersections and sums of subspaces can be defined for any finite number of subspaces. In case of the intersection we can take any number of subspaces, but for sums we need to be careful as only finite sums are well defined. See the tutorials for details.

Chapter II

The structure of abstract vector spaces

The aim of this chapter is to prove that every finitely generated vector space has a basis. For this we need to introduce the notion of linear combinations, linear independence and bases. A basis is the generalisation of the standard basis in $\mathbb{R}^N$ introduced in first year linear algebra to abstract vector spaces. The idea is that every element of the vector space can be written as a "linear combination" of the basis elements in a unique manner. In the abstract setting the existence of a "basis" and the notion of "dimension" is not at all obvious, and we carefully justify them.

5 Linear combinations

We first introduce some terminology. Some of this might seem a bit pedantic, but it is useful for efficiently and precisely formulating results. Let X, I be sets. For each $i \in I$ let x_i be an element of X. We call

$$(x_i)_{i\in I}$$

a *family* in a set X indexed by the set I.

Remarks 5.1. (a) In effect $(x_i)_{i\in I}$ is just an different way of saying that $i \mapsto x_i$ is a function from I to X.

(b) If $I = \{1, \dots, n\}$ is a finite set a family is just a finite sequence $(x_1, x_2, \dots, x_n)$ in X. We call such a family a *finite family* in X. For a finite family we also write

$$(x_i)_{i=1,\dots,n}$$

(c) A family is not a set since, in contrary to sets,

- repetition of elements is allowed;
- the order of elements is important.

For instance, $(2, 3, 7, 7)$ is a family but not a set since one element is repeated. Moreover, $(2, 7, 3, 7)$ is a different family from the above since the order of elements is different.

(d) We sometimes look at *subfamilies* of $(x_i)_{i\in I}$. By this we mean a family $(x_j)_{j\in J}$, where $J \subseteq I$. We call it a *proper subfamily* if $J \neq I$, and a *finite subfamily* if J is finite.

From now on let V be a vector space over an arbitrary field $\mathbb{F}$. Recall that we call any element of v a "vector" even if it is not a vector in the usual sense, and any element of $\mathbb{F}$ a "scalar".

Definition 5.2 (linear combination, span)**.** (a) Let $(v_1, \dots, v_n)$ be a finite family of vectors in V. A *linear combination* of $(v_i)_{i=1,\dots,n}$ is an expression of the form

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n$$

where $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{F}$.
(b) A linear combination of an arbitrary family of vectors $(v_i)_{i\in I}$ is a linear combination of any finite subfamily.
(c) If $(v_i)_{i\in I}$ is an arbitrary family of vectors we let

$$\operatorname{span}(v_i)_{i\in I}$$

be the set of linear combinations of $(v_i)_{i\in I}$. We call it the *span* of the family $(v_i)_{i\in I}$. If $I = \emptyset$ we define the span to be $\{0\}$ by convention.

Remark 5.3. Even if $(v_i)_{i\in I}$ is an infinite family we often write

$$\sum_{i\in I} \lambda_i v_i \tag{5.1}$$

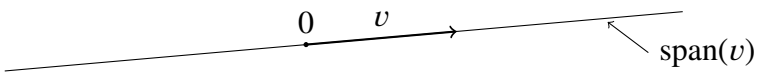
for a linear combination with the convention that at most finitely many $\lambda_i \in \mathbb{F}$ are non-zero.

It is important to remember that linear combinations are always *finite sums*, even if the family is infinite! Hence $\operatorname{span}(v_i)_{i\in I}$ consists of all sums of the form (5.1) with at most finitely many $\lambda_i \in \mathbb{F}$ non-zero.

Examples 5.4. (a) If (v) is a "family" of just one vector a linear combination is of the form λv with $\lambda \in \mathbb{F}$ and

$$\operatorname{span}(v) = \{\lambda v : \lambda \in \mathbb{F}\}$$

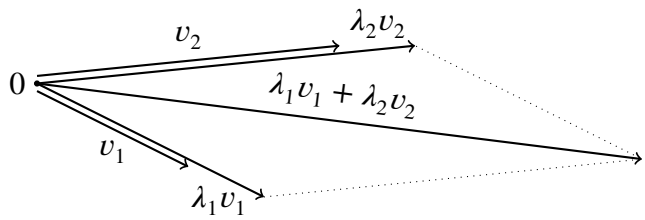
This is a "line" in V. If v is a "real" vector we have the picture as below.



(b) If (v_1, v_2) is a family of two vectors a linear combination is $\lambda_1 v_1 + \lambda_2 v_2$. The span of the family is

$$\operatorname{span}(v_1, v_2) = \{\lambda_1 v_1 + \lambda_2 v_2 : \lambda_1, \lambda_2 \in \mathbb{F}\}$$

This describes a "plane" in V spanned by the vectors v_1 and v_2 as shown in the picture below.



(c) The differential equation $\ddot{x} = -\omega^2 x$ for simple harmonic motion of frequency $\omega > 0$ has solutions $x_1(t) = \cos \omega t$ and $x_2(t) = \sin \omega t$. The "general solution" is a linear combination of the two, namely

$$x(t) = A \cos \omega t + B \sin \omega t = A x_1(t) + B x_2(t)$$

with $A, B \in \mathbb{R}$.

(d) Find an equation for the plane representing

$$\operatorname{span}\left\{\begin{bmatrix}1\\5\\3\end{bmatrix}, \begin{bmatrix}-1\\2\\4\end{bmatrix}\right\} \subseteq \mathbb{R}^3.$$

The plane is given by all vectors that can be written in the form

$$s\begin{bmatrix}1\\5\\3\end{bmatrix} + t\begin{bmatrix}-1\\2\\4\end{bmatrix} = \begin{bmatrix}x\\y\\z\end{bmatrix},$$

where $s, t \in \mathbb{R}$. We can find the plane by row reducing the augmented matrix

$$\left[\begin{array}{cc|c}1 & -1 & x\\5 & 2 & y\\3 & 4 & z\end{array}\right] \xrightarrow[R_3-3R_1]{R_2-5R_1} \left[\begin{array}{cc|c}1 & -1 & x\\0 & 7 & y-5x\\0 & 7 & z-3x\end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{cc|c}1 & -1 & x\\0 & 7 & y-5x\\0 & 0 & 2x-y+z\end{array}\right].$$

The last row means that a vector $\begin{bmatrix}x\\y\\z\end{bmatrix}$ is in the span of the two given vectors if and only if $0 = 2x - y + z$. That is the equation of a plane. We can easily check that the given vectors are in that plane.

Remark 5.5. If $(v_i)_{i\in I}$ is a family of vectors in V, then

(i) $\operatorname{span}(v_i)_{i\in I}$ is a subspace of V;

(ii) if W is a subspace of V and $v_i \in W$ for all $i \in I$, then $\operatorname{span}(v_i)_{i\in I} \subseteq W$.

This means that $\operatorname{span}(v_i)_{i\in I}$ is the smallest subspace of V containing all v_i, $i \in I$.

Proof. (i) Let $v = \sum_{i\in I} \lambda_i v_i$ and $w = \sum_{i\in I} \mu_i v_i$ be in $\operatorname{span}(v_i)_{i\in I}$. Recall that only finitely many λ_i, μ_i are non-zero, so we have for $\lambda \in \mathbb{F}$

$$v + w = \sum_{i\in I} (\lambda_i + \mu_i) v_i,$$
$$\lambda v = \sum_{i\in I} \lambda(\lambda_i v_i) = \sum_{i\in I} (\lambda\lambda_i) v_i.$$

Also in these sums at most finitely many $(\lambda_i + \mu_i)$ and $\lambda\lambda_i$ are non-zero, so $v + w$ and λv are in $\operatorname{span}(v_i)_{i\in I}$. Hence $\operatorname{span}(v_i)_{i\in I}$ is a subspace of V.

(ii) Since W is a subspace every linear combination of vectors in W is in W. In particular, if $v_i \in W$ for all $i \in I$, then every linear combination of $(v_i)_{i\in I}$ is in W, that is, $\operatorname{span}(v_i)_{i\in I} \subseteq W$. □

Remark 5.6. In Proposition 4.8 we proved that the sum of two subspaces U and W of a vector space V is a subspace as well. We have in fact

$$U + W = \operatorname{span}(U \cup W).$$

Proof. If $v \in U + W$, then $v = u + w$ for some $u \in U$ and $w \in W$. In particular, $u, w \in U \cup W$ and so $v = u + w \in \operatorname{span}(U \cup W)$. Hence $U + V \subseteq \operatorname{span}(U \cup W)$. If $u \in U$, then $u = u + 0 \in U + W$. Similarly, if $w \in W$, then $w = 0 + w \in U + W$. Therefore $U \cup W \subseteq U + W$. As $U + W$ is a subspace it follows that $\operatorname{span}(U \cup W) \subseteq U + W$. □

We will be interested in "minimal" spanning families. As it turns out they have special properties we discuss in the next section.

6 Linear independence and bases

We want to consider families of vectors in a vector space so that none of them can be written as a linear combination of the others. As usual V is a vector space over the field $\mathbb{F}$.

Definition 6.1 (linear independence)**.** (a) Let $(v_1, \dots, v_n)$ be a family of vectors in V. We say the family is *linearly independent* if

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n = 0$$

implies that $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$.
(b) An arbitrary family $(v_i)_{i\in I}$ is called *linearly independent* if every finite subfamily is linearly independent.
(c) A family $(v_i)_{i\in I}$ is called *linearly dependent* if there exists a finite subfamily $(v_{i_1}, \dots, v_{i_m})$ and λ_{i_k}, not all zero, such that

$$\lambda_{i_1} v_{i_1} + \lambda_{i_2} v_{i_2} + \cdots + \lambda_{i_m} v_{i_m} = 0.$$

(This means that the family is not linearly independent.)

We collect some simple consequences of the above definitions.

Remarks 6.2. (a) If $(v_i)_{i\in I}$ is linearly independent and $(v_{i_1}, v_{i_2}, \dots, v_{i_m})$ a subfamily, then that subfamily is linearly independent as well.

Proof. Every linear combination of the subfamily is a linear combination of the whole family since we can choose $\lambda_i = 0$ if $i \neq i_k$ for $k = 1, \dots, m$. Hence, if

$$\lambda_{i_1} v_{i_1} + \lambda_{i_2} v_{i_2} + \cdots + \lambda_{i_m} v_{i_m} = 0,$$

then also

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n = 0.$$

and by the linear independence of $(v_1, \dots, v_n)$ we have $\lambda_{i_1} = \lambda_{i_2} = \cdots = \lambda_{i_m} = 0$. Hence the subfamily is linearly independent. □

(b) If $(v_i)_{i \in I}$ is linearly independent, then every subfamily is linearly independent. This follows from the definition and (a).

(c) If $v_j = 0$ for some $j \in I$, then $(v_i)_{i \in I}$ is linearly dependent because we can write $0 = \lambda v_j$ for all $\lambda \neq 0$, and so there is a linearly dependent subfamily.

(d) If there is a repeated vector in $(v_i)_{i \in I}$, then the family is linearly dependent. Indeed, suppose that $v_j = v_k$ for some $j, k \in I$ with $j \neq k$. Then we can write $0 = v_j + (-1)v_k$, so there is a linearly dependent subfamily.

(e) The family with one single element (v) is linearly independent if and only if $v \neq 0$. This follows since $\lambda v = 0$ implies that $\lambda = 0$ or $v = 0$ in any vector space (see Proposition 3.1(vii)).

We defined linear independence with the idea in mind that no vector can be written as a linear combination of the others. We show that this is the case.

Lemma 6.3. *A family $(v_1, \dots, v_n)$ in the vector space V is linearly independent if and only if none of its elements can be written as a linear combination of the others. Equivalently $(v_1, \dots, v_n)$ is linearly dependent if and only if one of its elements can be written as a linear combination of the others.*

Proof. If $(v_1, \dots, v_n)$ is linearly dependent, then there exists a linear combination

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n = 0.$$

with $\lambda_k \neq 0$ for some $k \in \{1, \dots, n\}$. Hence

$$v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} -\frac{\lambda_i}{\lambda_k} v_i,$$

so v_k is a linear combination of the others. If on the other hand v_k is a linear combination of the others, that is,

$$v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} \lambda_i v_i,$$

then

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n = 0$$

with $\lambda_k := -1 \neq 0$. Hence the family is linearly dependent. □

We next look at some examples of linearly independent sets.

Examples 6.4. (a) The vectors $e_1 = (0, 1)$ and $e_2 = (0, 1)$ are linearly independent in $\mathbb{R}^2$. Indeed, if

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \lambda \\ \mu \end{bmatrix}$$

then $\lambda = \mu = 0$.

(b)Similarly the vectors $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$ and $e_3 = (0, 0, 1)$ are linearly independent in $\mathbb{R}^3$.

(c) The above examples can be generalised. We let $V = \mathbb{F}^n$ and consider the family of vectors

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}.$$

Then the family $(e_1, \dots, e_n)$ is linearly independent. Indeed, note that if

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \lambda_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \cdots + \lambda_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \lambda_1 \\ \lambda_1 \\ \lambda_3 \\ \vdots \\ \lambda_n \end{bmatrix},$$

then $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$.

(d) In the vector space $\mathbb{F}[X]$ of polynomials over $\mathbb{F}$ the family $(1, X, X^2, X^3, \dots)$ is linearly independent because any linear combination of this family is a polynomial. More precisely,

$$a_0 + a_1 X + a_2 X^2 + \cdots + a_n X^n = 0$$

means that the above is the zero polynomial and so $a_0 = a_1 = \cdots = a_n = 0$. Recall that we only need to check that any finite subfamily is linearly independent, so it is sufficient to check that for each $n \in \mathbb{N}$ the family $(1, X, X^2, \dots, X^n)$ is linearly independent.

(e) The solutions $x_1(t) = \cos \omega t$ and $x_2 = \sin \omega t$ are linearly independent solutions of the equation of simple harmonic motion $\ddot{x} = -\omega^2 x$. Indeed, suppose that

$$0 = A \cos \omega t + B \sin \omega t$$

for all $t \in \mathbb{R}$, that is, the linear combination is the zero function. Hence in particular for $t = 0$

$$0 = A \cos 0 + B \sin 0 = A + 0 = A$$

and for $t = \pi/2\omega$

$$0 = A \cos(\pi/2) + B \sin(\pi/2) = 0 + B = B.$$

Hence $A = B = 0$, showing that the two solutions are linearly independent.

Lemma 6.5. *Suppose that $(v_i)_{i\in I}$ is is a family of vectors in V. Then the following statements are equivalent.*

(i) *$(v_i)_{i\in I}$ is linearly independent.*

(ii) *Every $v \in \operatorname{span}(v_i)_{i\in I}$ can be written in a unique way as a linear combination $v = \sum_{i\in I} \lambda_i v_i$ of $(v_i)_{i\in I}$.*

Proof. Suppose that $(v_i)_{i\in I}$ is a linearly independent family of vectors in V and that $v \in \operatorname{span}(v_i)_{i\in I}$ can be written as

$$v = \sum_{i\in I} \lambda_i v_i = \sum_{i\in I} \mu_i v_i$$

(recall that at most finitely many λ_i, μ_i are non-zero). Then

$$0 = v - v = \sum_{i\in I} (\lambda_i - \mu_i) v_i.$$

By the linear independence of $(v_i)_{i\in I}$ it follows that $\lambda_i - \mu_i = 0$, and so $\lambda_i = \mu_i$ for all $i \in I$. Hence v can be written in a unique way as a linear combination of $(v_i)_{i\in I}$.

To prove the converse we assume that $\lambda_i \in \mathbb{F}$ are such that $0 = \sum_{i\in I} \lambda_i v_i$. We also have $0 = \sum_{i\in I} 0 v_i$. By assumption the representation of zero is unique and hence $\lambda_i = 0$ for all $i \in I$. □

We are now interested in "minimal" families spanning the whole vector space, motivating the following definition.

Definition 6.6 (spanning set, basis)**.** (a) A family $(v_i)_{i\in I}$ is called a *spanning family* in V if

$$V = \operatorname{span}(v_i)_{i\in I}.$$

We call V *finitely generated* if V has a finite spanning family of vectors.
(b) A family $(v_i)_{i\in I}$ is called a *basis* if

- $(v_i)_{i\in I}$ is a spanning family in V

and

- $(v_i)_{i\in I}$ is linearly independent.

Examples 6.7. (a) The family vectors $(e_1, \dots, e_n)$ as defined in Example 6.4(c) form a basis of $\mathbb{F}^n$. We showed it is linearly independent, and it is quite obvious that it is spanning as well. It is called the *standard basis* of $\mathbb{F}^n$.

(b) The family $(1, X, X^2, \dots)$ is a basis of the vector space of polynomials $\mathbb{F}[X]$. We showed in Example 6.4(d) that the family is linearly independent. Clearly it is also spanning by definition of polynomials.

(c) The solutions of the equation of simple harmonic motion $\ddot{x} = -\omega^2 x$ form a vector space. We have seen in Example 6.4(e) that the solutions $x_1(t) = \cos \omega t$ and $x_2 = \sin \omega t$ are linearly independent and that the general solution can be written as a linear combination of these two solutions. Hence $(\cos \omega t, \sin \omega t)$ is a basis for the vector space of solutions.

We next give several characterisations of bases.

Theorem 6.8 (Characterisation of bases)**.** *Let $(v_i)_{i\in I}$ be a family of vectors in $V \neq \{0\}$. Then the following assertions are equivalent.*

(i) *$(v_i)_{i\in I}$ is a basis of V.*

(ii) *$(v_i)_{i\in I}$ is a minimal spanning family, that is, every proper subfamily of $(v_i)_{i\in I}$ does not span the vector space V.*

(iii) *$(v_i)_{i\in I}$ is a maximal linearly independent family, that is, any family containing $(v_i)_{i\in I}$ as a proper subfamily is linearly dependent.*

(iv) *Every $v \in V$ can be written in a unique way as a linear combination of $(v_i)_{i\in I}$.*

Proof. (i) $\Rightarrow$ (ii) By definition a basis $(v_i)_{i\in I}$ is spanning and linearly independent. It is not possible to remove any vector from the basis without loosing the spanning property because, as proved in Lemma 6.3, none of the elements in a linearly independent family can be written as a linear combination of the others.

(ii) $\Rightarrow$ (iii) If $(v_i)_{i\in I}$ is a spanning family, then by Lemma 6.3 any larger family is linearly dependent, since every new element can be written as a linear combination of (v_i). To show linear independence we do a proof by contrapositive, assuming that $(v_i)_{i\in I}$ is linear dependent. Then there exists $k \in I$ such that

$$v_k = \sum_{\substack{i\in I \\ i\neq k}} \lambda_i v_i,$$

so $(v_i)_{i\in I\setminus\{k\}}$ is still spanning. Hence a minimal spanning family is linearly independent.

(iii) $\Rightarrow$ (i) Suppose $(v_i)_{i\in I}$ is a maximal linearly independent set. To show it is spanning we let $v \in V$ be arbitrary. Since $(v_i)_{i\in I}$ is a maximal linearlty independent set, the family $(v_i)_{i\in I}$ augmented by v is linearly dependent. Hence there exist coefficients $\lambda, \lambda_i \in \mathbb{F}$, not all equal to zero, such that

$$0 = \lambda v + \sum_{i\in I} \lambda v_i.$$

We have $\lambda \neq 0$ since otherwise all coefficients are zero. Indeed, if $\lambda = 0$, then $\sum_{i\in I} \lambda v_i = 0$ and by the linear independence of $(v_i)_{i\in I}$, also $\lambda_i = 0$ for all $i \in I$. We can therefore write

$$v = \sum_{i\in I} \Big(-\frac{\lambda_i}{\lambda}\Big) v_i.$$

As the above arguments apply to any choice of $v \in V$, the family $(v_i)_{i\in I}$ is spanning and hence $(v_i)_{i\in I}$ is a basis.

(iv) $\Leftrightarrow$ (i). Saying that every $v \in V$ can be written as a linear combination of $(v_i)_{i\in I}$ is equivalent to saying that $(v_i)_{i\in I}$ spans V. The uniqueness of the representation is equivalent to the linear independence by Lemma 6.5. □

So far we discussed properties of bases, and we have seen examples of bases, but what we have not proved is that they exist in an arbitrary vector space. Also, it is not obvious that every basis has the same length, that is, number of elements. These questions are addressed in the next section.

7 Bases of finitely generated vector spaces

In this section we mainly consider finitely generated vector spaces. By this we mean that the vector space is spanned by a finite set. One consequence of this is that such vector spaces have a basis.

Proposition 7.1. *Suppose that V is a finitely generated vector space over the field $\mathbb{F}$. Then V has a basis.*

Proof. Suppose that $(v_1, \dots, v_m)$ is a finite spanning family. If is linearly independent it is a basis (by definition of a basis). If $(v_1, \dots, v_m)$ is not linearly independent, then some vector can be written as a linear combination of the others. Hence if we remove that vector from the family we still have a spanning family. We repeat that process until the family becomes a minimal spanning set. Then it follows from Theorem 6.8 that it is a basis. □

A question arising now is: if we have two or more bases, do all of them have the same number of elements? The answer is yes, but the proof of this is more difficult than what we have done so far. As a first step we prove a replacement lemma.

Lemma 7.2 (replacement lemma)**.** *Suppose that $(v_1, \dots, v_n)$ is a basis of the vector space V and that*

$$w = \lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n \tag{7.1}$$

for some $\lambda_1, \dots, \lambda_n \in \mathbb{F}$. If $\lambda_k \neq 0$, then $(v_1, \dots, v_{k-1}, w, v_{k+1}, \dots, v_n)$ is a basis of V.

Proof. If we renumber the basis elements we can assume without loss of generality that $k = 1$. Hence $\lambda_1 \neq 0$ and the claim is that $(w, v_2, \dots, v_n)$ is a basis. We have to show (i) that the family is spanning and (ii) that it is linearly independent.

(i) To show that $(w, v_2, \dots, v_n)$ is a spanning family we let $v \in V$. Since $(v_1, \dots, v_n)$ is a basis we can write

$$v = \mu_1 v_1 + \mu_2 v_2 + \cdots + \mu_n v_n.$$

By (7.1) and the assumption that $\lambda_1 \neq 0$

$$v_1 = \frac{1}{\lambda_1} w - \frac{\lambda_2}{\lambda_1} v_2 + \frac{\lambda_3}{\lambda_1} v_3 + \cdots + \frac{\lambda_n}{\lambda_1} v_n$$

and therefore

$$\begin{aligned} v &= \mu_1\Big(\frac{1}{\lambda_1}w - \frac{\lambda_2}{\lambda_1}v_2 + \frac{\lambda_3}{\lambda_1}v_3 + \cdots + \frac{\lambda_n}{\lambda_1}v_n\Big) + \mu_2 v_2 + \cdots + \mu_n v_n \\ &= \frac{\mu_1}{\lambda_1}w + \Big(\mu_2 - \frac{\lambda_2}{\lambda_1}\mu_1\Big)v_2 + \Big(\mu_3 - \frac{\lambda_3}{\lambda_1}\mu_1\Big)v_3 + \cdots + \Big(\mu_n - \frac{\lambda_n}{\lambda_1}\mu_1\Big)v_n. \end{aligned}$$

Since the above argument applies to every choice of $v \in V$ we conclude that

$$\operatorname{span}(w, v_2, \dots, v_n) = V.$$

(ii) To show that $(w, v_2, \dots, v_n)$ is linearly independent we assume that

$$\mu w + \mu_1 v_2 + \mu_3 v_3 + \cdots + \mu_n v_n = 0 \tag{7.2}$$

and show that necessarily $\mu = \mu_2 = \cdots = \mu_n = 0$. Using (7.1) we get

$$\begin{aligned} 0 &= \mu(\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n) + \mu_2 v_2 + \mu_3 v_3 + \cdots + \mu_n v_n \\ &= \mu\lambda_1 v_1 + (\mu\lambda_2 + \mu_2)v_2 + (\mu\lambda_3 + \mu_3)v_3 + \cdots + (\mu\lambda_n + \mu_n)v_n. \end{aligned}$$

As $(v_1, \dots, v_n)$ is linearly independent

$$\mu\lambda_1 = \mu\lambda_2 + \mu_2 = \mu\lambda_3 + \mu_3 = \cdots = \mu\lambda_n + \mu_n = 0.$$

By assumption $\lambda_1 \neq 0$ and therefore $\mu = 0$ since from the above $\mu\lambda_1 = 0$. Hence the above reduces to $\mu_2 = \mu_3 = \cdots = \mu_n = 0$. This proves that $(w, v_2, \dots, v_n)$ is linearly independent. □

The above lemma provides the essential step in proving the following theorem often referred to as the *replacement theorem*. The idea is not just to replace one vector in a basis, but a whole family of linearly independent vectors. We do that by induction, replacing one after the other using the replacement lemma.

Theorem 7.3 (replacement theorem)**.** *Suppose that $(v_1, \dots, v_n)$ is a basis of V and that $(w_1, \dots, w_m)$ is linearly independent. Then*

(i) *$m \leq n$;*

(ii) *There exist indices $i_1, \dots, i_m \in \{1, \dots, n\}$ so that if we replace v_{i_k} by w_k for all $k = 1, \dots, m$, then we still have a basis of V.*

If we renumber the basis elements in $(v_1, \dots, v_n)$ so that $i_k = k$ for $k = 1, \dots, m$, then (ii) *means that $(w_1, \dots, w_m, v_{m+1}, \dots, v_n)$ is a basis of V.*

Proof. We give a proof by induction on m. If $m = 0$, then no vector in $(v_1, \dots, v_n)$ is replaced, so (i) and (ii) are obvious. For the induction step assume that we know that

(ii) hold for some $0 \leq m = k < n$, that is, $(w_1, \dots, w_k, v_{k+1}, \dots, v_n)$ is a basis for V. In particular, the family is spanning and so there exist $\lambda_i \in \mathbb{F}$ such that

$$w_{k+1} = \lambda_1 w_1 + \cdots + \lambda_k w_k + \lambda_{k+1} v_{k+1} + \cdots + \lambda_n v_n.$$

We claim that $\lambda_i \neq 0$ for some $i \in \{k+1, \dots, n\}$. If not, then

$$w_{k+1} = \lambda_1 w_1 + \cdots + \lambda_k w_k.$$

But this means that one vector of the linearly independent family $(w_1, \dots, w_k, w_{k+1})$ is a linear combination of the others. However, we know that this is not possible (see Lemma 6.3). Hence $\lambda_i \neq 0$ for some $i \in \{k+1, \dots, n\}$. Reordering the vectors $v_{k+1}, \dots, v_n$ we can assume that $\lambda_{k+1} \neq 0$. Applying Lemma 7.2 we conclude that

$$(w_1, \dots, w_k, w_{k+1}, v_{k+2} \dots, v_n)$$

is a basis of V. This proves the statement for $k+1$, and we stop the induction when we reach $k+1 = m \leq n$. If $m \geq n$ the above induction argument shows in particular that $(w_1, \dots, w_n)$ is a basis of V. Now Theorem 6.8(iii) shows that $m \leq n$ as otherwise $(w_1, \dots, w_m)$ is linearly dependent. □

Remark 7.4. It is worth noting that $m \leq n$ is a conclusion in the above theorem, and not an assumption!

The replacement theorem is one of the key theorems in the theory of finitely generated vector spaces. It was first proved by Grassmann around 1850. It was rediscovered by Steinitz in 1913 and is often called the "Steinitz replacement lemma". An important consequence of the above theorem is the following fact on finitely generated vector spaces.

Theorem 7.5. *Suppose that V is a finitely generated vector space. Then V has a basis. Moreover, every basis has the same length (number of elements).*

Proof. From Proposition 7.1 we know that every finitely generated vector space has a basis. Assume that there are bases of n and m elements, respectively . In particular we have a basis of n elements and a linearly independent family of m elements. Hence by the replacement theorem $m \leq n$. Interchanging the roles of the two families we conclude that $n \leq m$. Hence $m = n$ as claimed. □

The above justifies the following definition.

Definition 7.6 (dimension)**.** For a vector space V over $\mathbb{F}$ we define its *dimension* to be

$$\dim(V) = \begin{cases} n & \text{if } V \text{ has a basis of } n \text{ elements} \\ \infty & \text{if } V \text{ is not finitely generated} \end{cases}$$

The above definition makes sense since in every finitely generated vector space every basis has the same number of elements.

Remarks 7.7. (a) The dimension of $\mathbb{F}^n$ as a vector space over $\mathbb{F}$ is n since the standard basis and therefore every basis has precisely n elements.

(b) The space $\mathbb{F}[X]$ of polynomials over $\mathbb{F}$ is infinite dimensional since there is a basis $(1, X, X^2, \dots)$ of infinitely many elements. The subspace $\mathbb{F}_n[X]$ of polynomials of degree at most n has dimension $n+1$ since $(1, X, X^2, \dots, X^n)$ is a basis.

(e) The space of solutions of $\ddot{x} = -\omega^2 x$ is two dimensional since $(\cos \omega t, \sin \omega t)$ is a basis.

(c) Note that the dimension of a vector space depends on the choice of $\mathbb{F}$. For instance we can view $\mathbb{C}$ as a vector space over $\mathbb{C}$. Then it has dimension one with (1) being a basis. On the other hand we can view $\mathbb{C}$ as a vector space over $\mathbb{R}$. Then its dimension is two with $(1, i)$ being a basis.

We next derive consequences of the fact that every basis has the same length.

Corollary 7.8. *If* $\dim V = n$*, then the following assertions are equivalent.*

(i) $(v_1, \dots, v_n)$ *is a basis;*

(ii) $(v_1, \dots, v_n)$ *is linearly independent;*

(iii) $\operatorname{span}(v_1, \dots, v_n) = V$*;*

Proof. This follows from Theorem 6.8 and Theorem 7.5 since every basis has precisely n vectors. □

Corollary 7.9. *If* $W \subseteq V$ *is a subspace of* V*, then* $\dim W \leq \dim V$.

Proof. If $\dim V = \infty$ this is obvious. If $\dim V < \infty$, then there is no linearly independent family in W of more than n elements, since they are also linearly independent in V. Hence $\dim W \leq \dim V$. □

Remark 7.10. Theorem 7.5 shows that every finitely generated vector space has a basis. One can show, using (the somewhat controversial) Zorn lemma from set theory to show that *every* vector space has a basis. The idea is to start with a linearly independent family and keep adding more and more vectors, keeping it linearly independent. That process is then repeated until getting a maximal linearly independent family which is a basis according to Theorem 6.8. The only problem is that a basis in general is not countable, so an ordinary induction proof does not work. The solution is what is often referred to as "transfinite induction", namely Zorn's lemma.

One consequence of the replacement theorem is also that every linearly independent family can be extended to a basis of a vector space.

Theorem 7.11. *Suppose* V *is a vector space over* $\mathbb{F}$ *with* $\dim(V) = n$*, and that the family* $(w_1, \dots, w_m)$ *is a linearly independent family in* V*. Then there exist vectors* $v_{m+1}, \dots, v_n$ *such that* $(w_1, \dots, w_m, v_{m+1}, \dots, v_n)$ *is a basis of* V*.*

Proof. From Theorem 7.5 we know that V has a basis $(v_1, \dots, v_n)$. By the replacement theorem (Theorem 7.3) we have $m \leq n$ and, after possibly renumbering the elements of $(v_1, \dots, v_n)$, the family $(w_1, \dots, w_m, v_{m+1}, \dots, v_n)$ is a basis of V. □

Suppose now that V is a vector space over $\mathbb{F}$ with $\dim(V) = n$. If we choose a basis $(v_1, \dots, v_n)$ in V we know that from Theorem 6.8(iv) that for every vector $v \in V$ there exists a unique column vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{F}^n$$

such that

$$v = x_1 v_1 + x_2 v_2 + \cdots + x_n v_n. \tag{7.3}$$

This leads to the following definition.

Definition 7.12 (components of a vector)**.** The unique numbers $x_1, \dots, x_n \in \mathbb{F}$ such that (7.3) holds are called the *components* of V with respect to the basis $(v_1, \dots, v_n)$. The vector

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{F}^n$$

is called the *coordinate vector* of v with respect to the basis $(v_1, \dots, v_n)$.

Remarks 7.13. (a) It is important to remember that the components of a vector depend on the choice of the basis. For instance consider $V = \mathbb{R}^2$. Let (e_1, e_2) be the standard basis given by $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Then the coordinate vector of $(3, 4)$ is $(3, 4)$ as expected. However, if we change the order of the basis elements and the basis is (e_2, e_1) then the coordinate vector is $(4, 3)$. We can also look at a completely different basis. The vectors $v_1 = e_1 = (1, 0)$ and $v_2 = (1, 1)$ are linearly independent and therefore (v_1, v_2) is a basis. Then

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = (-1) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (-1)v_1 + 4v_2.$$

Hence the coordinate vector of $(3, 4)$ with respect to the basis (v_1, v_2) is $(-1, 4)$. We will study changes of bases later when we know more about linear operators.

(b) Let V be a vector space over $\mathbb{F}$ with $\dim V = n$, and let $(v_1, \dots, v_n)$ be a basis of V. Define a map $T : \mathbb{F}^n \to V$ by setting

$$Tx := x_1 v_1 + x_2 v_2 + \cdots + x_n v_n$$

for all $x = (x_1, \dots, x_n) \in \mathbb{F}^n$. Our discussion above shows that this map is bijective. Moreover, as an immediate consequence of its definition it satisfies the properties

$$\begin{aligned} T(x+y) &= Tx + Ty \quad && \text{for all } x, y \in \mathbb{F}^n; \\ T(\lambda x) &= \lambda T x && \text{for all } x \in \mathbb{F}^n,\ \lambda \in \mathbb{F}. \end{aligned}$$

Such a map is called a *linear operator*, and because it is bijective an *isomorphism*. The fact that T is bijective shows that, by choosing a basis, every n-dimensional vector space over $\mathbb{F}$ can be identified with $\mathbb{F}^n$.

In the next chapter we give a precise definition of linear operators and discuss their basic properties.

Chapter III

Matrices and Linear Operators

Linear operators are functions between vector spaces which are in some sense compatible with the structure of the vector space, in particular with addition and multiplication by scalars. In this chapter we discuss basic properties and the relationship to matrices. A highlight of this section is the discussion of eigenvalues and eigenvectors of linear operators in finite dimensional spaces, and the associated decompositions of vector spaces into invariant subspaces.

8 Definition and examples of linear operators

Linear maps like $x \mapsto ax$ are the simplest maps. In a generalised form they appear in many different contexts. Even in nonlinear problems a common method is to "linearise", and then try and draw conclusions on the non-linear problem from the knowledge gained by looking at an appropriate linear problem. From an abstract, axiomatic point of view, linear operators are functions (maps) compatible with the vector space structure.

Definition 8.1 (linear operator)**.** (a) Let V, W be vector spaces over $\mathbb{F}$. A *linear operator* is a function $T : V \to W$ with the following properties:

- $T(u + v) = Tu + Tv$ for all $u, v \in V$;
- $T(\lambda u) = \lambda Tu$ for all $u \in V$, $\lambda \in \mathbb{F}$.

(b) We call

$$L(V, W) := \{A : V \to W \mid A \text{ linear}\}$$

the space of linear operators from V into W.
(c) We call $T \in L(V, W)$ an *isomorphism* if it is bijective.
(d) A linear operator from V into V is called an *endomorphism*. We set $L(V) := L(V, V)$.

Remarks 8.2. (a) The space $L(V, W)$ of linear operators is a vector space. It is a subspace of the space $F(V, W)$ of all functions from V into W.

(b) Note that $T0 = 0$ for every linear operator since $T0 = 0T0 = 0$.

There are many examples of linear operators.

Examples 8.3. (a) A set of m linear equations of n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12} + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22} + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2} + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

involves a linear operator $x \mapsto Ax$, where A is the matrix $A = [a_{ij}]_{1\le i\le m, 1\le j\le n}$. Linear algebra helps to understand the structure of solution. We can also see this geometrically. Every equation describes an $(n-1)$-dimensional plane in $\mathbb{R}^n$. The set of m equations describes the intersection of m such planes. Hence there are the following possibilities:

- If $m < n$ we expect many solutions as the intersection is not just one point. For instance, the intersection of two planes in $\mathbb{R}^3$ is a line, and all points on the line form the solution.
- If $m = n$ the intersection is expected to be exactly one point, unless two or more planes coincide.
- If $m > n$, we need to be lucky to get a solution since the planes do not need to intersect in one point. For instance, three lines in $\mathbb{R}^2$ or four planes in $\mathbb{R}^3$ have an empty intersection in most cases.

These cases essentially mirror more general properties of abstract linear operators in finite dimensional spaces.

(b) Many geometric transformation in the plane and space can be represented by linear transformations. For instance, consider a rotation R_φ in the plane by the angle φ about the origin O. Then for points x, y in the plane clearly $R_\varphi(x + y) = R_\varphi x + R_\varphi y$ (rotation of a triangle leads to a congruent triangle) and $R_\varphi(\lambda x) = \lambda R_\varphi x$ (similar vectors are similar after rotation) as illustrated in Figure 8.1.

(c) Differentiation of functions is a linear operation since for functions u, v

$$(u + v)' = u' + v' \qquad \text{and} \qquad (\lambda u)' = \lambda u'.$$

The same is true for higher derivatives and also partial derivatives. This leads to what is often referred to as *linear differential operators*. Examples appear in ordinary differential equations with equations of second order

$$ay'' + by' + cy = 0$$

studied in first year calculus. Other examples include the operators from vector calculus including the divergence, nabla and Laplace operators. They play an important role in the theory and application of partial differential equations.

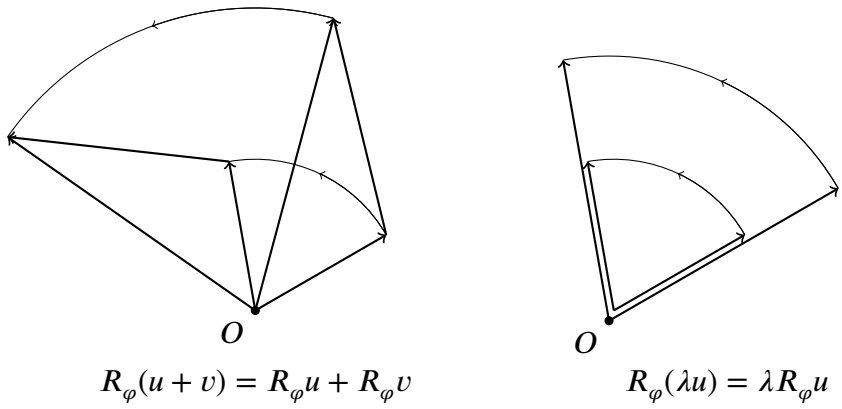


Figure 8.1: Linearity of rotation

(d) Integration is a linear operation since for functions u, v

$$\int_a^b u + v\,dx = \int_a^b u\,dx + \int_a^b v\,dx \qquad \text{and} \qquad \int_a^b \lambda u\,dx = \lambda \int_a^b u\,dx.$$

Integral operators are extensively used in partial differential equations and harmonic analysis.

Remarks 8.4. (a) A map $T : V \to W$ is linear if and only if

$$T(\lambda u + \mu v) = \lambda T u + \mu T v$$

for all $u, v \in V$ and $\lambda, \mu \in \mathbb{F}$. To show that the above implies linearity look at the cases $\lambda = \mu = 1$ and $v = 0$. Conversely apply the conditions of linearity first for addition, then for multiplication by scalars.

(b) By induction it follows that for a linear map $T : V \to W$

$$T(\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n) = \lambda_1 T v_1 + \lambda_2 T v_2 + \cdots + \lambda_n T v_n.$$

(c) If $T : V \to W$ is an isomorphism, then T^{-1} is also linear. To see this we note that by the linearity of T and the definition of an inverse function

$$T(\lambda T^{-1} u + \mu T^{-1} v) = \lambda T T^{-1} u + \mu T T^{-1} v = \lambda u + \mu v = T\big(T^{-1}(\lambda u + \mu v)\big).$$

As T is injective we conclude that $T^{-1}(\lambda u + \mu v) = \lambda T^{-1} u + \mu T^{-1} v$, which implies the linearity of T^{-1}.

We have met one isomorphism already, namely the one associating a vector with its coordinate vector.

Example 8.5. If $(v_1, \ldots, v_n)$ is a basis of V, then the function taking $v \in V$ to its coordinate vector $x \in \mathbb{F}^n$ is an isomorphism as seen at the end of the previous section.

9 Image and kernel

As with any other function we are interested in conditions that ensure it is surjective or injective. Let V, W be vector spaces and $T \in L(V, W)$ a linear operator. Then T is injective if and only if $Tu = Tv$ implies $u = v$. By the linearity of T this means T is injective if and only if

$$0 = Tu - Tv = T(u - v)$$

implies that $u - v = 0$. Since we do not know what u, v are the above is equivalent to $Tw = 0$ implies $w = 0$. This motivates the following definition.

Definition 9.1 (kernel and image)**.** Let V, W be vector spaces and $T \in L(V, W)$ a linear operator. We call

$$\ker T := \{u \in V : Tu = 0\} \subseteq V$$

the *kernel* (or *null space*) of T. Moreover,

$$\operatorname{im} T := \{Tu : u \in V\} \subseteq W$$

is called the *image* (or *range*) of T.

The discussion above proves last part of the following proposition.

Proposition 9.2. *Let V, W be vector spaces and $T \in L(V, W)$. Then*

(i) $\ker T$ *is a subspace of V;*

(ii) $\operatorname{im} T$ *is a subspace of W;*

(iii) T *is injective if and only if* $\ker T = \{0\}$.

Proof. (i) Suppose that $u, v \in \ker T$, that is, $Tu = Tv = 0$. Then by the linearity $T(u + v) = Tu + Tv = 0 + 0 = 0$, so $u + v \in \ker T$. Similarly $T(\lambda u) = \lambda Tu = \lambda 0 = 0$, so also $\lambda u \in \ker T$ for all $\lambda \in F$. Hence $\ker T$ is a subspace of V by Proposition 4.5.

(ii) Suppose that $w_1, w_2 \in \operatorname{im} T$, that is, there exist $v_1, v_2 \in V$ such that $Tv_1 = w_1$ and $Tv_2 = w_2$. Then by the linearity $w_1 + w_2 = Tv_1 + Tv_2 = T(v_1 + v_2)$, so $v_1 + v_2 \in \operatorname{im} T$. Similarly $\lambda w_1 = \lambda Tv_1 = T(\lambda v_1)$, so also $\lambda w_1 \in \operatorname{im} T$ for all $\lambda \in F$. Hence $\operatorname{im} T$ is a subspace of W.

(iii) This is proved in the discussion at the start of this section. □

We next characterise linear operators in terms of bases.

Theorem 9.3. *Suppose that $(v_i)_{i\in I}$ is a basis of the vector space V, and that $T \in L(V, W)$ is linear. Then*

(i) *T is uniquely determined by the vectors Tv_i, $i \in I$;*

(ii) *The family $(Tv_i)_{i\in I}$ is linearly independent if and only if T is injective.*

(iii) *T is an isomorphism (bijective) if and only if $(Tv_i)_{i\in I}$ is a basis in W.*

(iv) $\dim(\operatorname{im} T) \leq \dim V$.

Proof. (i) Suppose that $v \in V$. Since $(v_i)_{i\in I}$ is a basis there exist unique $\lambda_i \in \mathbb{F}$, zero except for at most finitely many $i \in I$, so that $v = \sum_{i\in I} \lambda_i v_i$. If Tv_i are know for every $i \in I$, then by the linearity

$$Tv = T\Big(\sum_{i\in I} \lambda_i v_i\Big) = \sum_{i\in I} \lambda_i T v_i$$

and so Tv is uniquely determined.

(ii) First suppose that $(Tv_i)_{i\in I}$ is linearly independent. Let $v \in \ker T$, that is, $Tv = 0$. Since $(v_i)_{i\in I}$ is a basis we can write $v = \sum_{i\in I} \lambda_i v_i$ with at most finitely many $\lambda \in \mathbb{F}$ non-zero. Hence

$$0 = Tv = T\Big(\sum_{i\in I} \lambda_i v_i\Big) = \sum_{i\in I} \lambda_i T v_i$$

By assumption $(Tv_i)_{i\in I}$ is linearly independent, and so $\lambda_i = 0$ for all $i \in I$. Hence $v = \sum_{i\in I} \lambda_i v_i = 0$, showing that $\ker T = 0$. By Proposition 9.2 T is injective. Now suppose that T is injective and that

$$0 = \sum_{i\in I} \lambda_i T v_i.$$

Then by the linearity of T we have $0 = T\big(\sum_{i\in I} \lambda_i v_i\big)$ and so $\sum_{i\in I} \lambda_i v_i \in \ker T$. Since T is injective $\sum_{i\in I} \lambda_i v_i = 0$, and by the linear independence of $(v_i)_{i\in I}$ we get $\lambda_i = 0$ for all $i \in I$.

(iii) From (ii) we know that T is injective if and only if $(Tv_i)_{i\in I}$ is linearly independent. Moreover, it is spanning if and only if $\operatorname{im} T = W$. Hence (iii) follows.

(iv) This follows from (i) and Theorem 6.8 which shows that the cardinality of a basis of $\operatorname{im} T$ cannot exceed that of a basis of V. □

Corollary 9.4. *Let V, W be finite dimensional spaces with* $\dim V = n$ *and* $\dim W = m$. *Then the following assertions hold.*

(i) *There exists an isomorphism between V and W if and only if $m = n$;*

(ii) $\dim(\operatorname{im} T) \leq \min\{m, n\}$ *for any $T \in L(V, W)$;*

Proof. (i) If $T \in L(V, W)$ is an isomorphism, then by Theorem 9.3(iii) we must have $\dim(W) = n$. On the other hand, if $\dim V = \dim W = n$, then choose a basis $(v_1, \dots, v_n)$ in V and a basis $(w_1, \dots, w_n)$ in W. According to (i) the linear map determined by $Tv_i := w_i$ defines a linear operator by Theorem 9.3(i). That linear operator is an isomorphism by Theorem 9.3(iii)

(ii) By Theorem 9.3(ii) the image of T can have at most dimension n. If $m \leq n$ it can have at most dimension m. □

In the above theorem we did not assume that V is finite dimensional. The following result is an important property of linear operators on a finite dimensional space.

Theorem 9.5. *Suppose that V, W are vector spaces and $T \in L(V, W)$. If V is finite dimensional, then*

$$\dim V = \dim(\ker T) + \dim(\operatorname{im} T).$$

More precisely, suppose that $(w_1, \dots, w_m)$ is a basis of $\operatorname{im} T$ *and that $v_1, \dots, v_m \in V$ are such that $Tv_k = w_k$ for $k = 1, \dots, m$. If $(u_1, \dots, u_p)$ is a basis of* $\ker T$*, then $(u_1, \dots, u_p, v_1, \dots, v_m)$ is a basis of V.*

Proof. As shown above $m := \dim(\operatorname{im} T) \leq \dim V$, and $p := \dim(\ker T) \leq \dim V$. We choose a basis $(w_1, \dots, w_m)$ of $\operatorname{im} T$ and vectors $v_1, \dots, v_m \in V$ such that $Tv_k = w_k$ for $k = 1, \dots, m$. Further let $(u_1, \dots, u_p)$ be a basis of $\ker T$. To show that $\mathcal{B} := (v_1, \dots, v_m, u_1, \dots, u_p)$ is a basis of V we need to show that it is spanning and linearly independent in V. Then $\dim V = m + p$ as claimed.

(i) We show that $\mathcal{B}$ is a spanning family for V. Suppose that $v \in V$. Since $(w_1, \dots, w_m)$ is a basis of $\operatorname{im} T$ there exist $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ such that

$$Tv = \lambda_1 w_1 + \cdots + \lambda_m w_m.$$

Using the linearity of T and the definition of v_i we get

$$\begin{aligned} 0 = \lambda_1 w_1 + \cdots + \lambda_m w_m - Tv = \lambda_1 Tv_1 + \cdots + \lambda_m Tv_m - Tv \\ = T(\lambda_1 v_1 + \cdots + \lambda_m v_m - v). \end{aligned}$$

Hence

$$\lambda_1 v_1 + \cdots + \lambda_m v_m - v \in \ker T$$

and since $(u_1, \dots, u_p)$ is a basis for $\ker T$ there exist $\mu_1, \dots, \mu_p \in \mathbb{F}$ such that

$$\lambda_1 v_1 + \cdots + \lambda_m v_m - v = \mu_1 u_1 + \cdots + \mu_p u_p.$$

In particular

$$v = \lambda_1 v_1 + \cdots + \lambda_m v_m - \mu_1 u_1 - \cdots - \mu_p u_p.$$

As the above arguments work for every choice of $v \in V$ we conclude that $\operatorname{span} \mathcal{B} = V$.

(ii) We next show that $\mathcal{B}$ is linearly independent. Hence assume that $\lambda_1, \dots, \lambda_m$ and $\mu_1, \dots, \mu_p$ are scalars such that

$$0 = \lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_p u_p \tag{9.1}$$

Using the linearity of T and that $Tu_k = 0$ for $k = 1, \dots, p$ we get

$$\begin{aligned} 0 &= T(\lambda_1 v_1 + \cdots + \lambda_m v_m + \mu_1 u_1 + \cdots + \mu_p u_p) \\ &= \lambda_1 T v_1 + \cdots + \lambda_m T v_m + \mu_1 T u_1 + \cdots + \mu_p T u_p = \lambda_1 w_1 + \cdots + \lambda_m w_m. \end{aligned}$$

By the linear independence of $(w_1, \dots, w_m)$ we see that $\lambda_1 = \lambda_2 = \cdots = \lambda_m = 0$. Hence from (9.1) we conclude that

$$\mu_1 u_1 + \cdots + \mu_p u_p = 0.$$

As $(u_1, \dots, u_p)$ is linearly independent we also have $\mu_1 = \cdots = \mu_p = 0$. This shows that $\mathcal{B}$ is linearly independent as claimed. □

From the above we get some important facts on linear operators on finite dimensional spaces, not true for arbitrary functions.

Corollary 9.6. *Let V, W be finite dimensional spaces with $\dim V = \dim W = n$ and $T \in L(V, W)$. Then the following assertions are equivalent.*

(i) *T is bijective, that is, an isomorphism;*

(ii) *T is surjective;*

(iii) *T is injective.*

Proof. By Theorem 9.5 we have $\dim V = n = \dim(\operatorname{im} T) + \dim(\ker T)$. If T is injective, then $\ker T = \{0\}$, so $\dim(\operatorname{im} T) = \dim W = n$ and so $\operatorname{im} T = V$. Similarly, if T is surjective, then $\dim(\operatorname{im} T) = \dim W = n$, and so $\ker T = \{0\}$. Hence T is injective. □

10 Direct sums and projections

Suppose that V is a vector space over $\mathbb{F}$ and U, W subspaces of V. We know from Proposition 4.7 and 4.8 that $U \cap W$ and $U + W$ are a subspaces of V.

Lemma 10.1. *Suppose that U and W are subspaces of the vector space V. If U and W are finite dimensional, then*

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. We define a map $U \times W \to V$ by $T(u, w) := u + w$. The map T is linear since for (u_1, w_1), $(u_2, w_2) \in U \times W$ and $\lambda, \mu \in \mathbb{F}$

$$\begin{aligned} &T\big(\lambda(u_1, w_1) + \mu(u_2, w_2)\big) = T(\lambda u_1 + \mu u_2, \lambda w_1 + \mu w_2) \\ &= (\lambda u_1 + \mu u_2) + (\lambda w_1 + \mu w_2) = \lambda(u_1 + u_2) + \mu(w_1 + w_2) = \lambda T(u_1, u_2) + \mu T(w_1, w_2). \end{aligned}$$

To determine the kernel of T assume that $T(u, w) = u + w = 0$. Then $w = -u$ and since U, W are subspaces we conclude that $-u$ and therefore u is in W. Hence $(u, w) \in \ker T$ if and only if $w = -u$ and $u \in U \cap W$, that is,

$$\ker T = \{(u, -u) : u \in U \cap W\} \subseteq U \times W.$$

We next show that $\ker T$ is isomorphic to $U \cap W$. We have to show that there is a bijection from $\ker T$ to $U \cap W$. We define $S(u, -u) := u$. Clearly S is linear. The map $S : \ker T \to U \cap W$ is surjective because for evey $u \in U \cap W$ we have $(u, -u) \in \ker T$ and $S(u, -u) = u$. It is injective, since $S(u, -u) = 0$ implies that $u = 0$ and therefore $(u, -u) = (0, 0)$. Hence S is injective and so $\ker T$ is isomorphic to $U \cap V$.

Suppose that U and W are finite dimensional and that $(u_1, \dots, u_n)$ is a basis of U and $(w_1, \dots, w_m)$ is a basis of W. We claim that

$$\big((u_1, 0), \dots, (u_n, 0), (0, w_1), \dots (0, w_m)\big)$$

is a basis of $U \times W$. The family is clearly spanning since every $(u, w) = (u, 0) + (0, w)$ can clearly be written as a linear combination. It is also linearly independent since

$$\begin{aligned} (0, 0) = \lambda_1(u_1, 0) + \cdots + \lambda_n(u_n, 0) + \mu_1(0, w_1) + \cdots + \mu_m(0, w_m) \\ = (\lambda_1 u_1 + \cdots + \lambda_n u_n, \mu_1 w_1 + \cdots + \mu_m w_m) \end{aligned}$$

Hence $\lambda_1 u_1 + \cdots + \lambda_n u_n = 0$ and $\mu_1 w_1 + \cdots + \mu_m w_m = 0$. By the linear independence of $(u_1, \dots, u_n)$ and $(w_1, \dots, w_m)$ all coefficients λ_i, μ_i are zero. Hence

$$\dim(U \times W) = n + m = \dim U + \dim W.$$

By the dimension formula from 9.5 and from what we proved above

$$\begin{aligned} \dim U + \dim W &= \dim V \times W \\ &= \dim(\operatorname{im} T) + \dim(\ker T) = \dim(U + W) + \dim(U \cap W). \end{aligned}$$

Rearranging the above the required identity follows. □

We next look at some special sums.

Definition 10.2 (direct sum, complement)**.** Let U, W be subspaces of the vector space V. We call $U + W$ a *direct sum* if $U \cap W = \{0\}$ and write

$$U \oplus W.$$

If $U \oplus W = V$ we call W a *complement* of U in V.

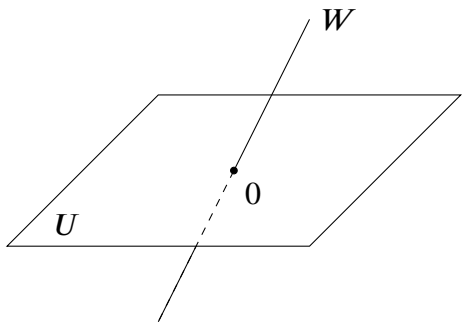


Figure 10.1: Complemented subspaces in $\mathbb{R}^3$.

In $\mathbb{R}^3$, complemented subspaces are for instance a plane and a line passing through the origin and intersecting transversally (so the line is not in the plane) as shown in Figure 10.1. Note that complements are never unique. In the above example any line transversal to the plane is a complement to the plane, and there are clearly infinitely many of those. In a finite dimensional vector space every subspace has a complement.

Proposition 10.3. *If V is finite dimensional and U is a subspace of V, then there exists a complement W such that $V = U \oplus W$.*

Proof. Assume that $\dim V = n$ and $\dim U = \ell$. Let $(u_1, \dots, u_\ell)$ be a basis for U. By Theorem 7.11 we can complement this to a basis $(u_1, \dots, u_\ell, w_{\ell+1}, \dots, w_n)$ of V. Then $W := \operatorname{span}(w_{\ell+1}, \dots, w_n)$ clearly is a complement of U in V. □

One can show that a subspace of any vector space is complemented, but to show this is beyond the scope of this course.

Remark 10.4. Note that $U \oplus W$ is a direct sum if and only if for every $v \in U + W$ there exist unique $u \in U$ and $w \in W$ such that $u + w = v$. To see this assume that $u_1 + w_1 = u_2 + w_2 = v$ with $u_1, u_2 \in U$ and $w_1, w_2 \in W$. Then $0 = (u_1 - u_2) + (w_1 - w_2)$ and so $u_1 - u_2 = -(w_1 - w_2)$. The left hand side is in U and the right hand side in W. Hence $u_1 - u_2, w_1 - w_2 \in U \cap W$. Hence the representation is unique if and only if $U \cap W = \{0\}$ as otherwise we can let $u_2 = u_1 + h$ and $w_2 = w_1 + h$ for $h \in U \cap W$ to get different representations.

If $V = U \oplus W$, then every v has a unique representation in the form $v = u + w$ with $u \in U$ and $w \in W$. Hence we can define maps $P, Q : V \to V$ by $Pv := u$ and $Qv := w$. The map P is called the projection of v onto u parallel to W. The idea is illustrated in Figure 10.2.

If $u \in U$, then clearly $u = u + 0$, and so by definition of P we have $Pu = u$. Hence in particular $P^2 := P \circ P = P$ and similarly $Q^2 = Q$. These projections are linear operators. Indeed, let $v_1 = u_1 + w_1$ and $v_2 = u_2 + w_2$ the unique decompositions of $v_1, v_2 \in V$ with $u_1, u_2 \in U$ and $w_1, w_2 \in W$. Then by definition of P

$$\begin{aligned} P(\lambda_1 v_1 + \lambda_2 v_2) &= P\big((\lambda_1 u_1 + \lambda_2 u_2) + (\lambda_1 w_1 + \lambda_2 w_2)\big) \\ &= \lambda_1 u_1 + \lambda_2 u_2 = \lambda_1 P v_1 + \lambda_2 P v_2 \end{aligned}$$

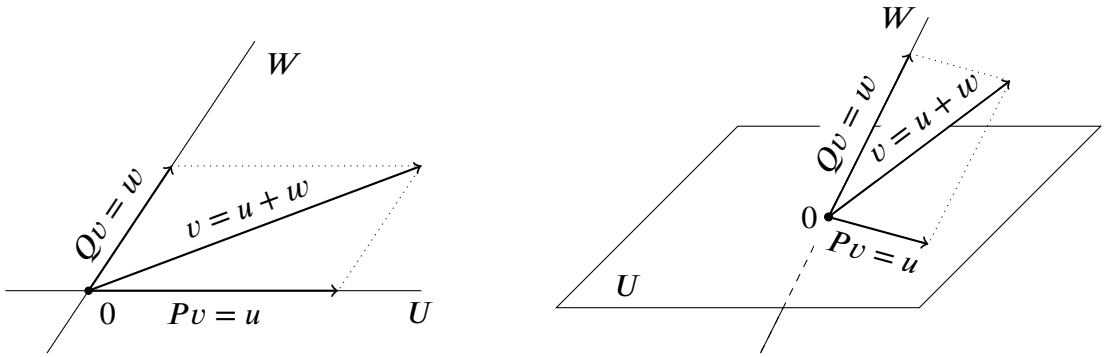


Figure 10.2: Projections onto complemented subspaces

Hence P is linear. Similarly, Q is linear as well. Finally, also by definition of P we have $Pw = 0$ for all $w \in W$, so $\ker P = \operatorname{im} Q = W$. Similarly $\ker Q = \operatorname{im} P = U$. We show now that there is a one-to-one relationship between projections and complemented subspaces.

Definition 10.5 (projection)**.** A linear operator $P : V \to V$ is called a *projection* if $P^2 = P$.

The next proposition tells us that there is a one-to-one correspondence between projections and direct sums of subspaces.

Proposition 10.6. *Let V be a vector space over $\mathbb{F}$.*

(i) *If $V = U \oplus W$, then there exists a projection P such that* $\operatorname{im} P = U$ *and* $\ker P = W$.

(ii) *Suppose that $P : V \to V$ is a projection. Then $V = \operatorname{im} P \oplus \ker P$. Moreover, $Q := I - P$ is a projection with* $\operatorname{im} Q = \ker P$.

Proof. (i) follows from the discussion above.

(ii) If $v \in V$ we have $v = (v - Pv) + Pv$. We set $u = Pv$ and $w = v - Pv$ so that $v = u + w$. Clearly $u = Pv \in \operatorname{im} P$. Now since $P^2 = P$

$$Pw = P(v - Pv) = Pv - P^2 v = Pv - Pv = 0,$$

so $w \in \ker P$. Hence $V = \operatorname{im} P + \ker P$. If $v \in \operatorname{im} P \cap \ker P$, then $v = Px$ for some $x \in V$ since $v \in \operatorname{im} P$ and $Pv = 0$ since $v \in \ker P$. In particular $v = Px = P^2 x = Px = 0$, so $v = 0$. Therefore $V = \operatorname{im} P \oplus \ker P$. □

11 Matrix representation of linear operators

So far we have developed in a coordinate free fashion. Ultimately, when solving problems explicitly or numerically we have to introduce coordinates. By this we mean we

choose a basis in the vector spaces involved. As we shall see problems involving linear operators reduce to problems involving matrices. Hence let V, W be finite dimensional vector spaces and $T \in L(V, W)$ a linear operator. We choose a basis $(v_1, \dots, v_n)$ of V and a basis $(w_1, \dots, w_m)$ of W. Given $v \in V$ we can uniquely express V as a linear combination of the basis in V, that is, there is a unique vector $(x_1, \dots, x_n) \in \mathbb{F}^n$ such that

$$v = x_1 v_1 + x_2 v_2 + \cdots + x_n v_n.$$

Applying the linear operator T we get

$$Tv = x_1 T v_1 + x_2 T v_2 + \cdots + x_n T v_n.$$

Now for every $j = 1, \dots, n$ we can express Tv_j in a unique fashion as a linear combination of the basis in W, that is,

$$Tv_j = b_{1j} w_1 + b_{2j} w_2 + \cdots + b_{mj} w_m \tag{11.1}$$

with $b_{ij} \in \mathbb{F}$ uniquely determined, where $1 \le i \le n$ and $1 \le j \le m$. Hence we get

$$Tv = \sum_{j=1}^{n} x_j \Big(\sum_{i=1}^{m} b_{ij} w_i \Big) = \sum_{i=1}^{m} \Big(\sum_{j=1}^{n} b_{ij} x_j \Big) w_i. \tag{11.2}$$

Hence the coordinate vector of Tv with respect to the basis $(w_1, \dots, w_m)$ is

$$\begin{bmatrix} \sum_{j=1}^{n} b_{1j} x_j \\ \sum_{j=1}^{n} b_{2j} x_j \\ \vdots \\ \sum_{i=1}^{n} b_{mj} x_i \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & b_{m3} & \dots & b_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Hence the matrix $B := [b_{ij}]_{1 \le i \le n, 1 \le j \le m}$ represents the linear operator T: If $x = (x_1, \dots, x_n)$ is the coordinate vector of v with respect to the basis $(v_1, \dots, v_n)$, then Bx is the coordinate vector of Tv with respect to the basis $(w_1, \dots, w_n)$. This leads to the following definition.

Definition 11.1 (matrix representation)**.** We call the matrix B defined above the *matrix representation* of the linear operator $T : V \to W$ with respect to the bases $\boldsymbol{v} = (v_1, \dots, v_n)$ in V and $\boldsymbol{w} = (w_1, \dots, w_n)$ in W.

Remarks 11.2. (a) We best capture the situation above in the following diagram:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "S_{\boldsymbol{v}}"] & W \arrow[d, "S_{\boldsymbol{w}}"] \\ \mathbb{F}^n \arrow[r, "B"] & \mathbb{F}^m \end{tikzcd} \tag{11.3}$$

The map $S_{\boldsymbol{v}}$ is the isomorphism mapping a vector in V onto its coordinate coordinate vector $x \in \mathbb{F}^n$ with respect to the basis $\boldsymbol{v} = (v_1, \dots, v_n)$. The linear map $S_{\boldsymbol{w}}$ is defined similarly. We then have

$$BS_{\boldsymbol{v}} = S_{\boldsymbol{w}}T.$$

We say that the diagram is *commutative*.

(b) From the above considerations the columns of the matrix B are the coordinate vectors of $T\boldsymbol{v}_j$ with respect to the basis $\boldsymbol{w}$ in W, that is,

$$\begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{mj} \end{bmatrix} = S_{\boldsymbol{w}}Tv_j.$$

(c) As a special case consider the identity map id_V defined by $\mathrm{id}_V(v) = v$ for all $v \in V$. We then have the diagram

$$\begin{tikzcd} V \arrow[r, "\mathrm{id}_V"] \arrow[d, "S_{\boldsymbol{v}}"] & V \arrow[d, "S_{\boldsymbol{v}}"] \\ \mathbb{F}^n \arrow[r, "I_n"] & \mathbb{F}^n \end{tikzcd}$$

According to (b) the i-th column of the matrix is $S_{\boldsymbol{v}}\,\mathrm{id}_V(v_i) = S_{\boldsymbol{v}}(v_i) = e_i$, where e_i is the i-th standard basis vector. Hence the matrix representation of id_V is the $n \times n$ *identity matrix*

$$I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

with 1 in the diagonal and zeros elsewhere. We also write

$$I_n = [\delta_{ij}]_{1 \le i,j \le n}$$

where δ_{ij} is the *Kronecker delta*. By definition $\delta_{ij} = 1$ if $i = j$ and zero otherwise.

(d) The above shows that there is a one-to-one correspondence between linear operators from V to W and the $m \times n$ matrices. Indeed, if $T \in L(V, W)$ is given, then

$$B = S_{\boldsymbol{w}}TS_{\boldsymbol{v}}^{-1}.$$

On the other hand, if $B \in \mathrm{Mat}(m \times n, \mathbb{F})$ is given, then

$$T = S_{\boldsymbol{w}}^{-1}BS_{\boldsymbol{v}}.$$

We can easily read this off from the commutative diagram (11.3).

Examples 11.3. (a) We want to find a linear operator that maps the unit square $(0,1) \times (0,1)$ onto the parallelogram defined by the vectors $(2,5)$ and $(-1,3)$. According to the above we need to find the image of the standard basis vectors in $\mathbb{R}^2$. The square is defined by the two standard basis vectors, so we require that

$$T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}2\\5\end{bmatrix} \qquad \text{and} \qquad T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}-1\\3\end{bmatrix}.$$

The matrix has as its column the image of the standard basis elements, so the matrix representing T with respect to the standard basis is

$$A = \begin{bmatrix}2 & -1\\5 & 3\end{bmatrix}.$$

(b) Find the matrix that describes a rotation of θ about the origin in $\mathbb{R}^2$. We have shown in Example 8.3(b) that such a rotation is a linear map. To get the matrix representation with respect to the standard basis we again have to find the images of the standard basis elements as shown in Figure 11.1. Hence we see that

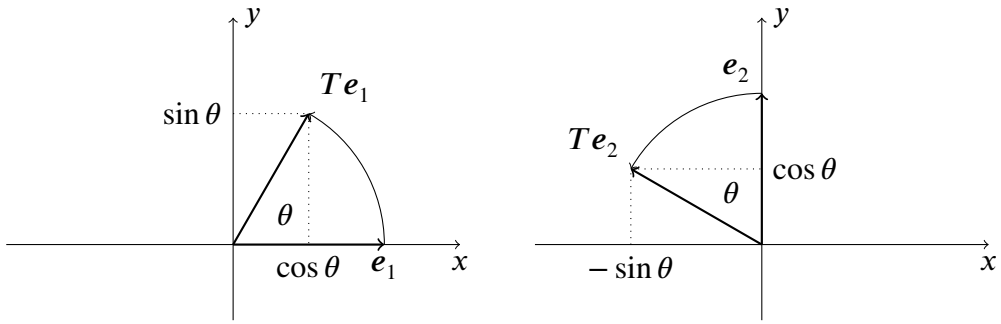


Figure 11.1: Image of standard basis vectors under a rotation

$$Te_1 = T\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}\cos\theta\\\sin\theta\end{bmatrix} \qquad \text{and} \qquad Te_2 = T\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}-\sin\theta\\\cos\theta\end{bmatrix}.$$

Therefore the matrix representing a rotation by the angle θ about the origin in the counterclockwise direction is given by

$$A = \begin{bmatrix}\cos\theta & -\sin\theta\\\sin\theta & \cos\theta\end{bmatrix}.$$

Again we use that the columns of the matrix are the images of the standard basis vectors under the linear transformation.

(c) We know that the subspace $\mathbb{R}_n[X]$ of $\mathbb{R}[X]$ of polynomials of degrees at most n is a finite dimensional space of dimension $n+1$ with basis $(1, X, X^2, \dots, X^n)$. We define a linear operator by (formal) differentiation:

$$T(a_0 + a_1X + a_2X^2 + \cdots + a_nX^n) := a_1 + 2a_2X + \cdots + na_nX^{n-1}.$$

Note that the coordinate vector of $a_0 + a_1X + a_2X^2 + \cdots + a_nX^n$ is given by

$$\begin{bmatrix} a_0 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^{n+1}.$$

We determine the matrix representation of $T : \mathbb{R}_n[X] \to \mathbb{R}_n[X]$ with respect to the above basis. Again we need to determine the image of each basis element, namely X^k. If $k = 0$, then $TX^0 = T1 = 0$. If $k \geq 1$, then

$$TX^k = kX^{k-1},$$

so we need to find the coordinate vector of kX^{k-1} with respect to the basis $(1, X, \dots, X^n)$. There is only one non-zero coefficient if we view kX^{k-1} as a linear combination of $(1, X, \dots, X^n)$, and that is the k-th (remember $k = 0, 1, \dots, n$). Hence the coordinate vector of kX^{k-1} is given by

$$k\text{-th component} \to \begin{bmatrix} 0 \\ \vdots \\ 0 \\ k \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{n+1}$$

with the k at the k-th position since k is the coefficient of X^{k-1} which is the k-th element in the basis $(1, X, \dots, X^n)$. Hence the matrix is

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots\dots\dots\dots \\ . & 0 & 2 & 0 & \dots\dots\dots \\ \dots\dots & 0 & 3 & 0 & \dots\dots \\ \dots\dots\dots\dots\dots\dots\dots\dots \\ \dots\dots\dots\dots & & 0 & n-1 \\ 0 & \dots\dots\dots\dots & & 0 \end{bmatrix}.$$

The last row is zero since the derivative of a polynomial of degree n is of degree $n - 1$.

We next look at the composition of two linear operators $T : V \to W$ and $S : W \to U$. The aim is to find the corresponding matrix representation. We again assume we have bases $\boldsymbol{v} = (v_1, \dots, v_n)$ in V, $\boldsymbol{w} = (w_1, \dots, w_m)$ in W and $\boldsymbol{u} = (u_1, \dots, u_p)$ in U. Let $B = [b_{ij}]_{1 \leq i \leq m, 1 \leq j \leq n}$ be the matrix representing T and $A = [a_{ki}]_{1 \leq k \leq p, 1 \leq i \leq m}$ be the matrix representing S. To find the matrix representing the composition $S \circ T$ we need to find the coordinate vectors of $(S \circ T)v_j$ with respect to the basis $(u_1, \dots, u_p)$. We know from Remark 11.2(b) that

$$Tv_j = \sum_{i=1}^{m} b_{ij} w_i \qquad \text{and} \qquad Sw_i = \sum_{k=1}^{p} a_{ki} u_k.$$

Hence

$$(S\circ T)v_j = S(Tv_j) = S\Big(\sum_{j=1}^{m} b_{ij}w_i\Big) = \sum_{j=1}^{m} b_{ij}Sw_i$$
$$= \sum_{j=1}^{m} b_{ij}\Big(\sum_{k=1}^{p} a_{ki}u_k\Big) = \sum_{k=1}^{p}\Big(\sum_{j=1}^{m} a_{ki}b_{ij}\Big)u_k.$$

Therefore the coefficients of the $m \times p$ matrix C representing $S\circ T$ with respect to the bases $(v_1, \dots, v_n)$ in V and $(u_1, \dots, u_p)$ in U are given by

$$c_{kj} := \sum_{i=1}^{m} a_{ki}b_{ij}$$

for $1 \le k \le p$ and $1 \le j \le n$. This corresponds to matrix multiplication $C = AB$ as defined in first year. The situation is captured in the following commutative diagram.

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "S_v"] \arrow[rr, bend left, "S\circ T"] & W \arrow[r, "S"] \arrow[d, "S_w"] & U \arrow[d, "S_u"] \\
\mathbb{F}^n \arrow[r, "B"] \arrow[rr, bend right, "C = AB"'] & \mathbb{F}^m \arrow[r, "A"] & \mathbb{F}^p
\end{tikzcd} \tag{11.4}$$

For linear operators, in analogy to matrix multiplication, we usually omit the composition sign and define $ST := S\circ T$.

Definition 11.4 (matrix multiplication)**.** Suppose that $A \in \mathrm{Mat}(p \times m; \mathbb{F})$ and $B \in \mathrm{Mat}(m \times n; \mathbb{F})$. Then the product $C = AB$ is the $n \times p$ matrix with coefficients

$$c_{kj} := \sum_{i=1}^{m} a_{ki}b_{ij},$$

where $1 \le k \le p$ and $1 \le j \le n$

From the above considerations we have the following theorem.

Theorem 11.5. *Let U, V, W be vector spaces and $T : V \to W$ and $S : W \to U$. If A and B are the basis representations of the linear operators T and S with respect to some basis in U, V, W, then the matrix product AB is the matrix representation of the composition $S\circ T : V \to U$ (see also the diagram* (11.4)*).*

Remarks 11.6. (a) It is important to note that not all matrices can be multiplied. We can only multiply an $p \times m$ matrix with an $m \times n$ matrix. The result is a $p \times n$ matrix. Graphically this is illustrated in Figure 11.2. Also note that the coefficient c_{kj} is like the product of the k-th row of the matrix A and the j-th column of the matrix B. Again this is illustrated in Figure 11.2.

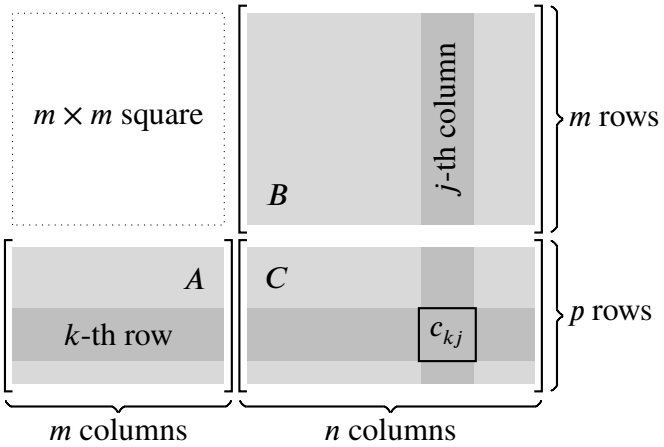


Figure 11.2: Product $C = AB$ of compatible matrices A and B

(b) Even if AB and BA exist we do not in general expect $AB = BA$. In particular, if $A \in \mathrm{Mat}(m{\times}n; \mathbb{F})$ and $B \in \mathrm{Mat}(n{\times}m; \mathbb{F})$, then $AB \in \mathrm{Mat}(m{\times}m)$ and $BA \in \mathrm{Mat}(n{\times}n)$, and therefore they cannot be equal if $n \neq m$. Even if $n = m$, then matrix multiplication is not commutative as the example below shows.

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

We next look at the special case $\dim V = \dim W = n$ and T is an isomorphism. Then we have the following commutative diagrams:

$$\begin{tikzcd}
V \arrow[r, "T"] \arrow[d, "S_v"] \arrow[rr, bend left, "\mathrm{id}_V = T^{-1} \circ T"] & W \arrow[r, "T^{-1}"] \arrow[d, "S_w"] & V \arrow[d, "S_v"] \\
\mathbb{F}^n \arrow[r, "B"] \arrow[rr, bend right, "I_n = AB"'] & \mathbb{F}^n \arrow[r, "A"] & \mathbb{F}^n
\end{tikzcd}
\qquad
\begin{tikzcd}
W \arrow[r, "T^{-1}"] \arrow[d, "S_w"] \arrow[rr, bend left, "\mathrm{id}_W = T \circ T^{-1}"] & V \arrow[r, "T"] \arrow[d, "S_v"] & W \arrow[d, "S_w"] \\
\mathbb{F}^n \arrow[r, "A"] \arrow[rr, bend right, "I_n = BA"'] & \mathbb{F}^n \arrow[r, "B"] & \mathbb{F}^n
\end{tikzcd}$$

We use Remark 11.6(c) for the fact that I_n is the matrix representation of id_V and id_W, respectively. The matrix B is therefore the *inverse matrix* of A. Recall that an $n \times n$ matrix A has an inverse if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n$$

Such an inverse matrix is unique if it exists since if $AC = CA = I_n$, then

$$B = BI_n = B(AC) = (BA)C = I_nC = C.$$

If it exists, we denote the inverse matrix by A^{-1}. We furthermore deduce the following fact.

Proposition 11.7. *Suppose that $A \in \mathrm{Mat}(n \times n; \mathbb{F})$. Then the following assertions are equivalent.*

(i) *A is invertible;*

(ii) *There exists an $n \times n$ matrix B such $AB = I_n$;*

(iii) *There exists an $n \times n$ matrix B such $BA = I_n$.*

Proof. It is clear that (i) implies the weaker statements (ii) and (iii). We prove that (ii) implies (1). Since $ABx = x$ for all $x \in \mathbb{F}$ it follows that A is surjective. As A is a linear operator between spaces of the same dimension it follows from Corollary 9.6 that A is bijective, so A^{-1} exists. Likewise, if (iii) holds, then B is surjective and hence by the same argument as before B^{-1} exists. Moreover, $Ax = (B^{-1}B)Ax = B^{-1}BAx = B^{-1}x$, so A is invertible as claimed. □

In general $AB \neq BA$ as we know, so it is surprising that if $AB = I_n$, then the matrices A and B commute. The reason for that is that matrix multiplication represents composition of linear operators which is what we exploited to prove the above proposition.

We next collect some more elementary properties of matrices. We assume throughout that the matrices A, B and C involved are of appropriate dimensions for the operations considered, and that $\lambda, \mu \in \mathbb{F}$.

(1) We know already that $\mathrm{Mat}(m \times n; \mathbb{F})$ is a vector space over $\mathbb{F}$.

(2) We have the associative law

$$(AB)C = A(BC).$$

This is easily checked directly from the definition of matrix multiplication, or follows if we interpret them as a composition of linear operators which is clearly associative.

(3) Matrix multiplication is left and right distributive, that is,

$$A(B + C) = AB + AC \qquad \text{and} \qquad (A + B)C = AC + BC.$$

(4) Scalars can be moved freely:

$$\lambda(AB) = (\lambda A)B = A(\lambda B).$$

(5) If A is an $m \times n$ matrix, then

$$I_m A = A I_n = A.$$

(6) If A, B are invertible $n \times n$ matrices, then AB is invertible and

$$(AB)^{-1} = B^{-1} A^{-1}.$$

This easily follows from properties of inverse functions if we interpret the matrices as linear operators.

12 Change of basis and transition matrices

The choice of basis is arbitrary. To simplify problems we often choose a basis adapted to the problem under consideration. Hence we need to know how to change from one basis to another. We first look at how the coordinate vectors change when the basis is changed. Hence assume that $\boldsymbol{u} = (u_1, \dots, u_n)$ and $\boldsymbol{v} = (v_1, \dots, v_i)$ are bases of the vector space V. Then we have two isomorphisms $S_{\boldsymbol{u}}, S_{\boldsymbol{v}} : V \to \mathbb{F}^n$ as shown in the diagram below:

$$\begin{tikzcd} & \mathbb{F}^n \arrow[dd, "\Phi"] \\ V \arrow[ur, "S_{\boldsymbol{u}}"] \arrow[dr, "S_{\boldsymbol{v}}"'] & \\ & \mathbb{F}^n \end{tikzcd} \qquad \begin{tikzcd} & e_i \arrow[dd, mapsto, "\Phi"] \\ u_i \arrow[ur, mapsto, "S_{\boldsymbol{u}}"] \arrow[dr, mapsto, "S_{\boldsymbol{v}}"'] & \\ & S_{\boldsymbol{v}} u_i \end{tikzcd}$$

If we are given the coordinate vector x of $v \in V$ with respect to the basis $\boldsymbol{u}$, then $v = S_{\boldsymbol{u}}^{-1} x$ and the coordinate vector with respect to the basis $\boldsymbol{v}_2$ is given by $S_{\boldsymbol{v}} S_{\boldsymbol{u}}^{-1} x$. Hence the operator

$$\Phi := S_{\boldsymbol{v}} S_{\boldsymbol{u}}^{-1}$$

allows us to find the coordinate vector with respect to $\boldsymbol{v}$ given the one with respect to $\boldsymbol{u}$. We now want to find the matrix representation of Φ. Since Φ maps $\mathbb{F}^n$ to $\mathbb{F}^n$ the columns of that matrix are the images of the standard basis vectors $e_1, \dots, e_n$. From the definition of $S_{\boldsymbol{u}}$ we know that $S_{\boldsymbol{u}}^{-1} e_j = u_j$. Now every basis vector u_j can be represented in a unique way in the form

$$u_j = a_{1j} v_1 + a_{2j} v_2 + \cdots + a_{nj} v_n.$$

This is equivalent to saying that

$$\Phi e_j = S_{\boldsymbol{v}} S_{\boldsymbol{u}}^{-1} e_j = S_{\boldsymbol{v}} u_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj}. \end{bmatrix}$$

Hence we can identify Φ with the matrix that has the above coefficients in the columns, that is,

$$\Phi = [a_{ij}]_{1 \le i,j \le n}.$$

Definition 12.1 (transition matrix)**.** The matrix Φ defined above is called the *transition matrix* for the bases $(u_1, \dots, u_n)$ and $(v_1, \dots, v_n)$.

Remark 12.2. In practice we are often given the new basis $(v_1, \dots, v_n)$ by the coordinate vectors with respect to the old basis $(u_1, \dots, u_n)$. In that case it is easy to find Φ^{-1} because $S_{\boldsymbol{v}} v_i = e_i$ by definition of $S_{\boldsymbol{v}}$ and hence the columns of Φ^{-1} are just the coordinate vectors of $v_1, \dots, v_n$ with respect to the old basis $(u_1, \dots, u_n)$. To get Φ we have to invert the matrix Φ^{-1}.

Example 12.3. The vectors

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \qquad \text{and} \qquad v_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

form a basis of $\mathbb{R}^2$ (since they are linearly independent). We want to find the corresponding transition matrix. In this example $V = \mathbb{R}^2$ and the basis (u_1, u_2) is the standard basis (e_1, e_2). Hence the map $S_{\boldsymbol{u}}$ from above is the identity map. As seen above, the columns of the inverse of the transition matrix Φ are the basis vectors v_1 and v_2 with respect to the old basis (e_1, e_2). Hence

$$\Phi^{-1} = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$$

and therefore

$$\Phi = \frac{1}{7} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}.$$

A vector is given by

$$v = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

with respect to the standard basis. With respect to the new basis (v_1, v_2) its coordinate vector is

$$\Phi(v) = \frac{1}{7} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

We also see that

$$\Phi(v_1) = \frac{1}{7} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

and

$$\Phi(v_2) = \frac{1}{7} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

We should not be surprised to get the standard basis vectors since these are the coordinate vectors of the basis (v_1, v_2) with respect to the new basis (v_1, v_2) by definition of coordinate vectors. What we do by a change of basis is defining a new coordinate system, changing direction and scale of the axes.

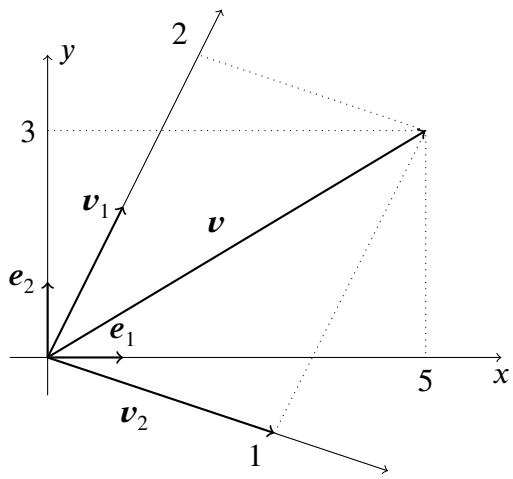


Figure 12.1: Change of basis and coordinate vectors.

Note that the axes in a coordinate system do not need to be perpendicular. We illustrate the change of basis in Figure 12.1. It can be seen there that the scale of the coordinates is such that the unit in the direction of v_1 is the length of v_1, and similarly for the direction of v_2. The vector v stays the same regardless of the basis chosen, but the components which are the projections parallel to the other basis vectors change.

Similarly to the coordinate vectors, the matrix representing a linear operator depends on the choice of the basis. Hence we want to find out how to compute the change of the matrix representation.

Let $\boldsymbol{v}_i = (v_{i1}, \dots, v_{in})$ $(i = 1, 2)$ be bases of V and $\boldsymbol{w}_i = (w_{i1}, \dots, w_{im})$ $(i = 1, 2)$ be bases of W. Let $T \in L(V, W)$ be a linear operator. The corresponding matrix representation with respect to the bases $\boldsymbol{v}_1$ in V and $\boldsymbol{w}_1$ in W we denote by B_1. We define B_2 similarly by using the bases $\boldsymbol{v}_2$ and $\boldsymbol{w}_2$ in V and W, respectively. As seen in the previous section we have isomorphisms $S_{\boldsymbol{v}_i} : V \to \mathbb{F}^n$ and $S_{\boldsymbol{w}_i} : W \to \mathbb{F}^m$ $(i = 1, 2)$. We can therefore define isomorphisms $\Phi : \mathbb{F}^n \to \mathbb{F}^n$ and $\Psi : \mathbb{F}^m \to \mathbb{F}^m$ by

$$\Phi := S_{\boldsymbol{v}_2} S_{\boldsymbol{v}_1}^{-1} \qquad \text{and} \qquad \Psi := S_{\boldsymbol{w}_2} S_{\boldsymbol{w}_1}^{-1}, \tag{12.1}$$

respectively. They are the transition matrices as defined above for the change of bases

in V and W. We can summarise this in the following commutative diagram.

$$\begin{tikzcd}
\mathbb{F}^n \arrow[rrr, "B_1"] \arrow[dd, "\Phi"'] & & & \mathbb{F}^m \arrow[dd, "\Psi"] \\
& V \arrow[ul, "S_{v_1}"'] \arrow[r, "T"] \arrow[dl, "S_{v_2}"] & W \arrow[ur, "S_{w_1}"] \arrow[dr, "S_{w_2}"'] & \\
\mathbb{F}^n \arrow[rrr, "B_2"'] & & & \mathbb{F}^m
\end{tikzcd} \tag{12.2}$$

Hence we can recover B_1 from B_2 and vice versa by the formulae

$$B_2 = \Psi B_1 \Phi^{-1} \qquad \text{and} \qquad B_1 = \Psi^{-1} B_2 \Phi. \tag{12.3}$$

The computation of Φ and Ψ is as explained before. Of course, dealing with linear operators we want their matrix representation to be as simple as possible. This amounts in choosing a basis (coordinate system) that is adapted to the problem under consideration. An example of such a choice of bases is the following theorem.

Theorem 12.4. *Suppose that V, W are vector spaces with* $\dim V = n$ *and* $\dim W = m$. *Let $T \in L(V, W)$ with* $\dim(\operatorname{im} T) = \ell$. *Then there exist bases $(v_1, \dots, v_n)$ of V and $(w_1, \dots, w_m)$ of W such that the matrix representation of A is of the form*

$$\begin{bmatrix} I_\ell & 0 \\ 0 & 0 \end{bmatrix},$$

where I_ℓ is the $\ell \times \ell$ identity matrix and the other blocks are zero matrices of appropriate shape.

Proof. Due to the rank-nullity theorem (Theorem 9.5) we can choose a basis $(v_1, \dots, v_n)$ of V such that $(Tv_1, \dots, Tv_\ell)$ is a basis of $\operatorname{im} T$ and $Tv_i = 0$ for $i = \ell + 1, \dots, n$. We set $w_i := Tv_i$ for $i = 1, \dots, \ell$ and complement this linearly independent set in W to a basis $(w_1, \dots, w_m)$ of W. This can be done because of Theorem 7.11. According to Remark 11.2(b) the columns of the matrix representation of T are the coordinate vectors of Tv_i with respect to the basis $(w_1, \dots, w_n)$. By construction of the basis we have $Tv_i = w_i$ for $i = 1, \dots, \ell$, so that the corresponding coordinate vector is e_i. Furthermore, again by construction of the basis $Tv_i = 0$ for $i = \ell + 1, \dots, n$. Hence the matrix has the required form. $\square$

Corollary 12.5. *Suppose that $A \in \mathrm{Mat}(m \times n; \mathbb{F})$. Then there exist invertible matrices $B \in \mathrm{Mat}(n \times n; \mathbb{F})$ and $C \in \mathrm{Mat}(m \times m; \mathbb{F})$ such that*

$$C^{-1}AB = \begin{bmatrix} I_\ell & 0 \\ 0 & 0 \end{bmatrix}$$

for some $0 \leq \ell \leq \min\{m, n\}$.

Proof. We can apply the theorem above to the linear operator $x \mapsto Ax$. The matrices B and C are the transition matrices for changing to a basis as the one used in the proof of the above theorem. □

The image of the linear operator defined by A is the span of the column vectors

$$a_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix} \in \mathbb{F}^n = \mathrm{Mat}(m \times 1; \mathbb{F})$$

of A, where $j = 1, \dots, n$. This is often referred to as the *column space* of A. Hence $\ell = \dim(\operatorname{im} A)$ is the dimension of the column space of A and called the *column rank* of A. If C, B are invertible matrices as in the above corollary, then $C^{-1}AB$ represents the same linear operator with respect to a different basis, so the dimension of the image and therefore the dimension of its column space is the same, namely ℓ. This means that the column ranks of A and $C^{-1}AB$ are equal.

We can also look at the *row space* of a matrix, that is, the span of the row vectors

$$r_i = \begin{bmatrix} a_{i1} & \dots & a_{in} \end{bmatrix} \in \mathrm{Mat}(1 \times n; \mathbb{F})$$

of the matrix A. The dimension of that vector space is called the *row rank* of A. If we consider the transposed matrix $A^T = [aji]_{1 \leq j \leq n, 1 \leq i \leq m}$ then the row space of A is isomorphic to the column space of A^T (by transposition of row vectors). Using properties of transposed matrices from first year we have

$$(C^{-1}AB)^T = B^T A^T (C^T)^{-1}.$$

Hence the row rank of A is the same as the row rank of $C^{-1}AB$ by the same argument as before. By the above corollary, the row and column ranks of $C^{-1}AB$ are the same, namely ℓ. This proves the following theorem.

Theorem 12.6. *For every $m \times n$ matrix A the column rank equals the row rank. They equal the dimension of the image of the linear operator $x \mapsto Ax$.*

This motivates the following definition.

Definition 12.7 (rank)**.** The *rank* of a matrix is the common dimension of the row and column spaces of that matrix.

If we look at a linear operator $T : V \to V$, that is, the domain and codomain are the same, we generally want to work with the same basis $\boldsymbol{v} := (v_1, \dots, v_n)$ in both. Then we have the following diagram.

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "S_{\boldsymbol{v}}"] & V \arrow[d, "S_{\boldsymbol{v}}"] \\ \mathbb{F}^n \arrow[r, "B"] & \mathbb{F}^n \end{tikzcd} \tag{12.4}$$

Hence we have $BS_{\boldsymbol{v}} = S_{\boldsymbol{v}}T$. We can also see what happens if we have two different bases $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ on V. Then we have the commutative diagram as shown below.

$$\begin{tikzcd} \mathbb{F}^n \arrow[rrr, "B_1"] \arrow[dd, "\Phi"] & & & \mathbb{F}^n \arrow[dd, "\Phi"] \\ & V \arrow[ul, "S_{\boldsymbol{v}_1}"] \arrow[r, "T"] \arrow[dl, "S_{\boldsymbol{v}_2}"] & V \arrow[ur, "S_{\boldsymbol{v}_1}"'] \arrow[dr, "S_{\boldsymbol{v}_2}"'] & \\ \mathbb{F}^n \arrow[rrr, "B_2"'] & & & \mathbb{F}^n \end{tikzcd} \tag{12.5}$$

In that case we have

$$B_2 = \Phi B_1 \Phi^{-1},$$

where Φ is the transition matrix from the basis $\boldsymbol{v}_1$ to $\boldsymbol{v}_2$. This motivates the following definition.

Definition 12.8 (similar matrices)**.** Two matrices $B_1, B_2 \in \mathrm{Mat}(n \times n; \mathbb{F})$ are called *similar* if there exists an invertible matrix $\Phi \in \mathrm{Mat}(n \times n; \mathbb{F})$ such that

$$B_2 = \Phi B_1 \Phi^{-1}$$

We write $B_2 \sim B_1$.

Similar matrices exactly correspond to matrix representations of the same linear operator with respect to different bases. In the proposition below we collect some elementary properties of similar matrices.

Proposition 12.9. *Similarity of matrices is an equivalence relation, that is,*

(i) $B \sim B$ *for all* $B \in \mathrm{Mat}(n \times n; \mathbb{F})$ *(reflexivity);*

(ii) $B_2 \sim B_1$ *implies* $B_1 \sim B_2$ *(symmetry);*

(iii) *If* $B_1 \sim B_2$ *and* $B_2 \sim B_3$*, then* $B_1 \sim B_3$ *(transitivity).*

Proof. (i) If we choose the identity $\Phi := I_n$, then $B = I_n B I_n^{-1}$, so $B \sim B$.

(ii) If $B_2 = \Phi B_1 \Phi^{-1}$, then by multiplying by Φ^{-1} from the left and by Φ from the right $B_1 = \Phi^{-1} B_2 \Phi = \Phi^{-1} B_2 (\Phi^{-1})^{-1}$, so $B_1 \sim B_2$.

(iii) Suppose there are invertible matrices Φ, Ψ such that $B_1 = \Phi B_2 \Phi^{-1}$ and $B_2 = \Psi B_3 \Psi^{-1}$. Hence

$$B_1 = \Phi B_2 \Phi^{-1} = B_1 = \Phi \Psi B_3 \Psi^{-1} \Phi^{-1} = B_1 = (\Phi\Psi) B_3 (\Phi\Psi)^{-1},$$

so $B_1 \sim B_3$. □

Sometimes, we do not want to change the basis in the domain of the linear transformation. In particular if we solve a system of linear equations we do want to apply a transformation to the solution, but we are happy to apply a transformation to the image to simplify the equation. This is what is achieved in steps when row-reducing a matrix. We explain this in the next section.

13 Reduced row-echelon form and elementary matrices

Consider $A \in \mathrm{Mat}(m \times n; \mathbb{F})$. Then A defines a linear operator $A : \mathbb{F}^n \to \mathbb{F}^m$. We want to choose a basis in $\mathbb{F}^m$ so that the new matrix B has a simpler form. The commutative diagram we have is then

$$\begin{tikzcd} \mathbb{F}^n \arrow[r, "A"] \arrow[d, "I_n"] & \mathbb{F}^m \arrow[d, "S"] \\ \mathbb{F}^n \arrow[r, "B = SA"] & \mathbb{F}^m \end{tikzcd}$$

If we consider the linear equation

$$Ax = b$$

then the solution x is from $\mathbb{F}^n$. By not changing the basis of $\mathbb{F}^n$ we do not change the vector x. Changing the basis in the codomain $\mathbb{F}^m$ means applying an invertible matrix $S \in \mathrm{Mat}(m \times m; \mathbb{F})$ and solving the equivalent equation

$$SAx = Sb,$$

without changing x. Since the standard basis $(e_1, \dots, e_n)$ spans $\mathbb{F}^n$ we have

$$\operatorname{im} A = \operatorname{span}(Ae_1, \dots, Ae_n) = \operatorname{span}(a_1, \dots, a_n) \tag{13.1}$$

where

$$a_i = Ae_i = \begin{bmatrix} a_{1i} \\ \vdots \\ a_{mi} \end{bmatrix},$$

$i = 1, \dots, n$, are the column vectors in the matrix A. We let $\ell := \dim(\operatorname{im} A)$. To avoid trivial cases we assume that $\ell \geq 1$, that is, $A \neq 0$. Hence we can select ℓ of the vectors

$a_1, \dots, a_n$ to form a basis of $\operatorname{im} A$. In general there will be many possibilities, and we want to choose one that gives a nice transformed matrix A. Clearly

$$\begin{aligned}\{0\} \subseteq \operatorname{span}(a_1) \subseteq \operatorname{span}(a_1, a_2) \subseteq \operatorname{span}(a_1, a_2, a_3) \\ \subseteq \operatorname{span}(a_1, a_1, a_3, a_4) \subseteq \cdots \subseteq \operatorname{span}(a_1, \dots, a_n).\end{aligned} \tag{13.2}$$

Since $\dim(\operatorname{im} A) = \ell$ there are precisely ℓ proper inclusions, namely when the dimension jumps by one. Hence for every proper inclusion

$$\operatorname{span}(a_1, \dots, a_{i-1}) \subsetneq \operatorname{span}(a_1, \dots, a_{i-1}, a_i)$$

we choose the vector a_i. This provides a basis $w_k := a_{i_k}$, $k = 1, \dots, \ell$ of $\operatorname{im} A$. According to Theorem 7.11 we can complement this to a basis $(w_1, \dots, w_m)$ of $\mathbb{F}^m$. Now, by choice of basis, the vectors a_i are linear combinations of $w_1, \dots, w_k$ if $i \leq i_k$, and therefore their coordinate vector have the form

$$b_i = \begin{bmatrix} b_{1i} \\ \vdots \\ b_{ki} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

with respect to the basis $(w_1, \dots w_k)$. Similarly, the coordinate vectors of a_i for $i \geq i_\ell$ are of the form

$$b_i = \begin{bmatrix} b_{1i} \\ \vdots \\ b_{\ell i} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

since a_i is a linear combination of $(w_1, \dots, w_\ell)$ in that case. Finally, also by choice of the basis, the component vector of a_{i_k} is $e_k \in \mathbb{F}^m$ for $k = 1, \dots, \ell$. Hence the matrix representation of A with respect to the standard basis in $\mathbb{F}^n$ and the basis $(w_1, \dots, w_m)$ in $\mathbb{F}^m$ is

$$\begin{array}{c}
\begin{array}{ccccccccccccc} e_1 & & & e_2 & & & & e_k & & & & e_\ell & \end{array} \\
\left[\begin{array}{ccccccccccccc}
0 & 1 & * & * & 0 & * & \dots & * & 0 & * & \dots & * & 0 & * & \dots \\
\vdots & & & 0 & 1 & * & \cdots & * & \cdot & * & \cdots & * & \cdot & * & \dots \\
\vdots & & & & & & & * & 0 & * & \cdots & * & \cdot & * & \cdots \\
\vdots & & & & & & & 0 & 1 & * & \cdots & * & \cdot & * & \cdots \\
\vdots & & & & & & & & & & & * & 0 & * & \cdots \\
\vdots & & & & & & & & & & & 0 & 1 & * & \cdots \\
\vdots & & & & & & & & & & & & & & \\
0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0
\end{array}\right]
\end{array}
\begin{array}{l} \\ \\ \\ \\ k\text{-th row} \\ \\ \ell\text{-th row} \\ \\ \\ \end{array}$$

where $*$ is any possibly non-zero entry in the matrix. All entries below the line are zero, and we have the standard basis vectors $e_k \in \mathbb{F}^m$ in column i_k for $k = 1, \dots, \ell$. This is the *reduced row-echelon form* of the matrix A as known from first year linear algebra. The matrix S for changing the basis in $\mathbb{F}^m$ is given by

$$S = \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix}^{-1}$$

as shown in Remark 12.2. (The inverse of the matrix with columns $w_1, \dots, w_m$.)

For instance, suppose that A is an 5×4 matrix with coefficients in $\mathbb{R}$, so that $n = 5$. If $\dim(\operatorname{im} A) = 3$ one possibility is that

$$\begin{aligned} \{0\} = \operatorname{span}(a_1) \subsetneq \operatorname{span}(a_1, a_2) = \operatorname{span}(a_1, a_2, a_3) \\ \subsetneq \operatorname{span}(a_1, a_2, a_3, a_4) \subsetneq \operatorname{span}(a_1, a_2, a_3, a_4, a_5) \end{aligned}$$

In this case we would choose (w_2, w_2, w_3), where $w_1 := a_2$, $w_2 := a_4$ and $w_3 := a_5$, and choose a vector w_4 to get a basis of $\mathbb{R}^4$. The reduced row-echelon form is then

$$\begin{bmatrix} 0 & 1 & b_{13} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where b_{13} may or may not be non-zero. If A is an invertible $n \times n$ matrix, then the columns are linearly independent since the n vectors span $\mathbb{F}^n$ (see Corollary 7.8). Hence the row-echelon form is the identity matrix I_n and $S = A^{-1}$.

Remark 13.1. Rather than a matrix we could have looked at a linear operator $T : V \to W$ between vector spaces of dimension n and m. The above argument shows that given a basis $(v_1, \dots, v_n)$ in V, there exists a basis of W such that the matrix representation of T has reduced row-echelon form. To see this replace the column vectors a_i by Tv_i in the above arguments.

Note that forming SA, that is, multiplying A by S from the left replaces every row of A by a linear combination of the rows of A. Indeed, if $S = [s_{ki}]_{1 \le k, i \le m}$, then the k-th row of SA is

$$\sum_{i=1}^{m} s_{ki} \begin{bmatrix} a_{i1} & a_{i2} & \dots & a_{in} \end{bmatrix}$$

which is clearly a linear combination of the m rows of A. It is not practicable to find these linear combinations in one step to get the reduced row-echelon form. However, we can find an algorithm which is doing that in stages, namely performing a sequence of *elementary row operations*. This is *Gaussian elimination* involving row operattions as known from first year linear algebra.

We now discuss these elementary row operations and what their meaning is in terms of matrix multiplication and change of basis. These row operations are also useful in the study of determinants and other properties of matrices. There are three commonly used row operations.

(I) Add λ times row j to row i ($i \neq j$). This corresponds to multiplication of A from the left with

$$
E_\lambda^{(ij)} = \begin{array}{c} \\ \\ \\ \\ \\ \\ \\ \\ \\ \end{array}
\begin{array}{c}
\begin{array}{cccccccccc} & & & & & j\text{-th column} & & & & \\ & & & & & \downarrow & & & & \end{array} \\
\begin{bmatrix}
1 & 0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 \\
0 & \ddots & & & & & & & & \vdots \\
\vdots & & \ddots & & & & & & & \vdots \\
\vdots & & & 1 & 0 & \cdots & 0 & \lambda & & \vdots \\
\vdots & & & & \ddots & & & 0 & & \vdots \\
\vdots & & & & & \ddots & & \vdots & & \vdots \\
\vdots & & & & & & \ddots & 0 & & \vdots \\
\vdots & & & & & & & 1 & \ddots & \vdots \\
\vdots & & & & & & & & \ddots & 0 \\
0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 & 1
\end{bmatrix}
\end{array}
\begin{array}{l} \\ \\ \\ \leftarrow i\text{-th row} \\ \\ \\ \\ \\ \\ \\ \end{array}
$$

The above is a matrix where $j > i$. If $j < i$, then the entry with λ is below the diagonal. In any case this corresponds to changing the basis in the codomain $\mathbb{F}^m$ from $(e_1, \dots, e_m)$ to the basis $(e_1, \dots, e_{i-1}, e_i - \lambda e_j, e_{i+1}, \dots, e_m)$. The matrix is clearly invertible since the inverse is a similar matrix with the entry ij replaced by $-\lambda$.

(II) Multiply row i by $\lambda \neq 0$. This corresponds to multiplication of A from the left with the diagonal matrix

$$
E_\lambda^{(i)} = \begin{bmatrix}
1 & 0 & \cdots & \cdots & \cdots & \cdots & 0 \\
0 & \ddots & & & & & \vdots \\
\vdots & & \ddots & & & & \vdots \\
\vdots & & & 1 & & & \vdots \\
\vdots & & & & \lambda & & \vdots \\
\vdots & & & & & 1 & \vdots \\
\vdots & & & & & & \ddots \\
\vdots & & & & & & 0 \\
0 & \cdots & \cdots & \cdots & \cdots & 0 & 1
\end{bmatrix}
\begin{array}{l} \\ \\ \\ \\ \leftarrow i\text{-th row} \\ \\ \\ \\ \\ \end{array}
$$

This corresponds to changing the basis in the codomain $\mathbb{F}^n$ from $(e_1, \dots, e_n)$ to the basis $(e_1, \dots, e_{i-1}, \lambda^{-1} e_i, e_{i+1}, \dots, e_n)$. The matrix $E_\lambda^{(i)}$ is invertible since the inverse is a similar matrix with λ replaced by λ^{-1}.

(III) Interchange row i and row j. This is given by the elemetary matrix

$$E_{ij} = \begin{array}{c} \left[\begin{array}{ccccccccccc} 1 & 0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 \\ 0 & \ddots & & & & & & & & & \vdots \\ \vdots & & 1 & & & & & & & & \vdots \\ \vdots & & & 0 & \cdots & \cdots & \cdots & 1 & & & \vdots \\ \vdots & & & \vdots & 1 & & & \vdots & & & \vdots \\ \vdots & & & \vdots & & \ddots & & \vdots & & & \vdots \\ \vdots & & & \vdots & & & 1 & \vdots & & & \vdots \\ \vdots & & & 1 & \cdots & \cdots & \cdots & 0 & & & \vdots \\ \vdots & & & & & & & & 1 & & \vdots \\ \vdots & & & & & & & & & \ddots & 0 \\ 0 & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & 0 & 1 \end{array}\right] \end{array} \begin{array}{l} \\ \\ \\ \\ \leftarrow i\text{-th row} \\ \\ \\ \\ \\ \leftarrow j\text{-th row} \\ \\ \\ \\ \end{array}$$

This corresponds to changing the basis in codomain $\mathbb{F}^n$ by swapping the i-th and the j-th basis vectors that is, from $(e_1, \dots, e_i, \dots, e_j, \dots, e_n)$ to the basis

$$(e_1, \dots, e_{i-1}, e_j, e_{i+1}, \dots, e_{j-1}, e_i, e_{j+1}, \dots, e_n).$$

The matrix is its own inverse.

Gaussian elimination then works as follows:

1. Identify the first non-zero column i_1;
2. In column i_1 identify a non-zero entry a_{ji_1} and multiply that row by $a_{ji_k}^{-1}$ to produce a 1.
3. Interchange rows 1 and j to have a one in the first row.
4. Use the first row to "eliminate" all non-zero coefficients in column i_1, that is, making them zero. This is achieved by adding a suitable multiple of the first row to all other rows.
5. Identify the next column i_2 with a non-zero coefficient in row j below row one and multiply that row by a suitable scalar to get a one.
6. Interchange rows 2 and j to get a one in row 2. Then use the second row to eliminate all other entries in that column in a similar way as done in column i_1.
7. Repeat that procedure with the next column i_3 having a non-zero entry below the second row and proceed as before.
8. Continue in a similar fashion until there are all zeros below the last row the elimination was performed.

The above is an outline of the algorithm suitable for theoretical purposes. In practice the choice of the row to eliminate elements above and below has many variations. In particular, one can sometimes take advantage of a row that has a one already. When numerical precision is required, then dividing rows by small numbers is not good, so one chooses the one so as to avoid that. We will not go into detail with this method called *partial pivoting*.

Each of the row operations performed corresponds to a multiplication of the A by an elementary matrix from the left. The result is clearly a matrix in reduced row-echelon form. Hence there exists a sequence of elementary matrices $(E_k)_{k=1,\dots,N}$ such that $E_N E_{N-1} \cdots E_2 E_1 A$. Hence $S := E_N E_{N-1} \cdots E_2 E_1 A$ is an invertible matrix such that SA is of reduced row-echelon form. We hence have the following theorem.

Theorem 13.2. *Let $A \in \mathrm{Mat}(m \times n, \mathbb{F})$ be a matrix. Then there exists a sequence of elementary matrices E_k, $k = 1, \dots, N$ of type* (I) *and* (II) *such that*

$$E_N E_{N-1} \cdots E_2 E_1 A$$

has reduced row-echelon form. Moreover, the reduced row-echelon form is unique, that is, if $S_1, S_2 \in \mathrm{Mat}(m \times m; \mathbb{F})$ are invertible and $S_1 A$ and $S_2 A$ are of reduced row echelon form, then $S_1 A = S_2 A$.

Proof. We only have to prove the uniqueness of the reduced row-echelon form. If $B = SA$ is of reduced row-echelon form with S an invertible matrix and $b_1, \dots, b_n$ are the columns of B, then we can again look at a chain similar to (13.2), namely

$$\begin{aligned}\{0\} \subseteq \mathrm{span}(b_1) \subseteq \mathrm{span}(b_1, b_2) \subseteq \mathrm{span}(b_1, b_2, b_3) \\ \subseteq \mathrm{span}(b_1, b_1, b_3, b_4) \subseteq \cdots \subseteq \mathrm{span}(b_1, \dots, b_n).\end{aligned} \tag{13.3}$$

Since $\dim(\mathrm{im}\, A) = \ell$ there are precisely ℓ proper inclusions, namely when the dimension jumps by one. By definition of B we have $b_i = Sa_i$ and since S defines an isomorphism from $\mathbb{F}^m$ to $\mathbb{F}^m$ it preserves linear dependence and independence (see Theorem 9.3). Hence the proper inclusions

$$\mathrm{span}(b_1, \dots, b_{i_k - 1}) \subsetneq \mathrm{span}(b_1, \dots, b_{i_k - 1}, b_{i_k})$$

occur at the same columns i_k, $k = 1, \dots, \ell$ as in (13.2). Moreover, for every k, each column vector a_i, $i \le i_k$, can be written as a linear combination of $(a_{i_1}, \dots, a_{i_k})$ in a unique fashion. A similar statement is true also for the column vectors of B. Since $b_i = Sa_i$ the linear combinations in both instances must be the same. The argument above also applies to all columns with $i > \ell$ which can be written as unique linear combinations of columns $i_1, \dots, i_\ell$ for both matrices. This completes the proof of uniqueness. □

Remarks 13.3. (a) The product $S := E_N E_{N-1} \cdots E_2 E_1$ is one possible choice for the matrix S. While the reduced row-echelon form is unique, this is in general not true for the matrix S since we could choose the basis elements $w_{\ell+1}, \dots, w_m$ freely.

(b) Using the reduced row-echelon form of a matrix it is easy to get a basis for the kernel (null space) of the matrix A. Recall that $\ker A$ is the set of all vectors $v \in \mathbb{F}^n$ such that $Av = 0$. Hence also $SAv = S0 = 0$. We know from Theorem 9.5 that $\dim(\mathrm{im}\, A) + \dim(\ker A) = n$ and so if $\dim(\mathrm{im}\, A) = \ell$ it is sufficient to find $n - \ell$ linearly independent vectors in $\ker A$. If B is the reduced row-echelon form of A we

use the columns b_j for $j \neq i_k$, $k = 1, \dots, \ell$. We form

$$v_j := e_j - \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_{1j} \\ 0 \\ \vdots \\ 0 \\ b_{2j} \\ 0 \\ \vdots \\ 0 \\ b_{\ell j} \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{array}{l} \\ \\ \\ \leftarrow i_1\text{-th component} \\ \\ \\ \\ \leftarrow i_2\text{-th component} \\ \\ \\ \\ \leftarrow i_\ell\text{-th component} \\ \\ \\ \\ \end{array}$$

in $\mathbb{F}^n$ with $j \neq i_k$ for $k = 1, \dots, \ell$. ($e_j \in \mathbb{F}^n$ is the j-th standard basis vector.) There are $n - \ell$ such vectors. They are clearly linearly independent since there is precisely one having a non-zero j-th component for each $j \neq i_k$. For instance, consider the 4×7 matrix

$$B = \begin{bmatrix} 0 & 1 & 0 & 3 & 2 & 0 & 5 \\ 0 & 0 & 1 & 4 & 6 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

in reduced row-echelon form. Here we have $\ell = 3$ and the columns with leading ones are columns $i_1 = 2$, $i_2 = 3$ and $i_3 = 6$. Hence we use columns 1, 4, 5 and 7 to construct a basis of the kernel which is a subspace of $\mathbb{R}^7$:

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 0 \\ -3 \\ -4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad v_5 = \begin{bmatrix} 0 \\ -2 \\ -6 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad v_7 = \begin{bmatrix} 0 \\ -5 \\ -2 \\ 0 \\ 0 \\ -3 \\ 1 \end{bmatrix}.$$

These vectors are linearly independent because precisely one has a non-zero entry in component 1, 4, 5 and 7.

Often we are not interested in the matrix S which brings A into row-echelon form. This is in particular the case if we solve a linear system of equations or find the kernel (null space) of a matrix. If we want to get S we can get it by augmenting A by the identity matrix, and then doing the row operations on the augmented matrix $[A|I_m]$.

The end product of the row operations is

$$S[A|I_m] = [SA|S]$$

so where the identity was we get S. This is useful if $m = n$ and is A an invertible matrix. Then A reduces to I_n, so if we do the row operations on the augmented matrix $[A|I_n]$ we get $[SA|S] = [I_n|S]$, that is, $SA = I_n$. This means that $S = A^{-1}$ is the inverse of A. We do not discuss this here further as this was part of first year linear algebra.

A consequence of the above considerations is the following theorem.

Theorem 13.4. *Suppose that $A \in \mathrm{Mat}(n \times n; \mathbb{F})$. Then the following assertions are equivalent.*

(i) *A is invertible;*

(ii) *The columns of A are linearly independent;*

(iii) *The rows of A are linearly independent;*

(iv) *The reduced row echelon form of A is I_n.*

Another consequence of our discussions on the reduced row echelon form is that row and column rank of a matrix are equal as already proved in Theorem 12.6.

Chapter IV

Determinants

Determinants appear naturally when solving linear systems of equations where the number of unknowns and the number of equations are equal. Determinants also turn out to be the (oriented) volume of the parallelepiped spanned by the colums of the matrix as we will see in Section 24.2.

14 First encounter with determinants

Determinants were introduced in first year linear algebra. For a 2×2 matrix it is given by

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

The determinant has the following properties:

1. The determinant is linear as a function of the rows (and also the columns). For instance look at the first row.

$$\begin{aligned} \det \begin{bmatrix} \lambda a_1 + \mu a_2 & \lambda b_1 + \mu b_2 \\ c & d \end{bmatrix} &= (\lambda a_1 + \mu a_2)d - (\lambda b_1 + \mu b_2)c \\ &= \lambda(a_1 d - b_1 c) + \mu(a_2 d - b_2 c) = \lambda \det \begin{bmatrix} a_1 & b_1 \\ c & d \end{bmatrix} + \mu \det \begin{bmatrix} a_2 & b_2 \\ c & d \end{bmatrix} \end{aligned}$$

   A similar calculation applies to the second row.

2. The determinant is zero if it has two equal rows. Indeed

$$\det \begin{bmatrix} a & b \\ a & b \end{bmatrix} = ab - ab = 0.$$

3. The determinant is normalised, that is, $\det I_2 = 1$:

$$\det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \cdot 1 - 0 \cdot 0 = 1.$$

From the first and second property it follows that A is invertible if and only if $\det A \neq 0$. Moreover, any function on the set of 2×2 matrices with the above properties must coincide with the above.

There is a geometric interpretation of determinants of matrices with real entries which implies very naturally the three properties above. In two dimensions it is the oriented area of the parallelogram spanned by the two column vectors $\boldsymbol{a}$, $\boldsymbol{b}$. A pair of vectors $\boldsymbol{a}$, $\boldsymbol{b}$ is positively oriented if the rotation from the first to the second through the smaller angle is anti-clockwise as shown in Figure 24.4. It is negatively oriented otherwise.

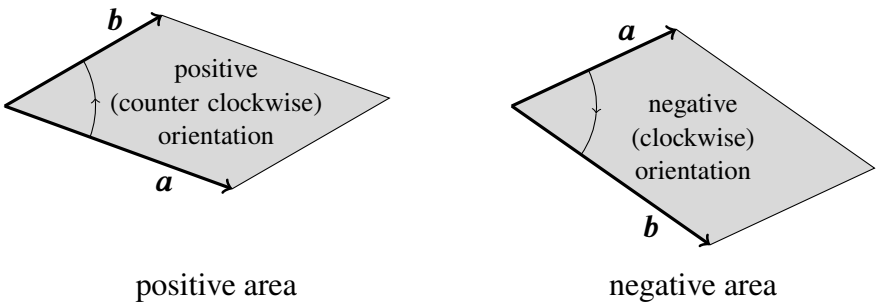


Figure 14.1: Oriented parallelograms spanned by ordered pairs of vectors $(\boldsymbol{a}, \boldsymbol{b})$

Property 1. about the determinant being zero for two equal vectors reflects the fact that a line segment has zero area. The normalisation 3. means that the square with side length one has area one. The linearity in 2. reflects the fact that volumes scale linearly when scaled in one direction. The additivity comes from the fact that parallelograms of the same height and base have the same area. The idea is illustrated in Figure 14.2.

The geometric interpretation of determinants extends to square matrices of any dimension. We will provide more detail in Section 24.2. The determinant of a 3×3

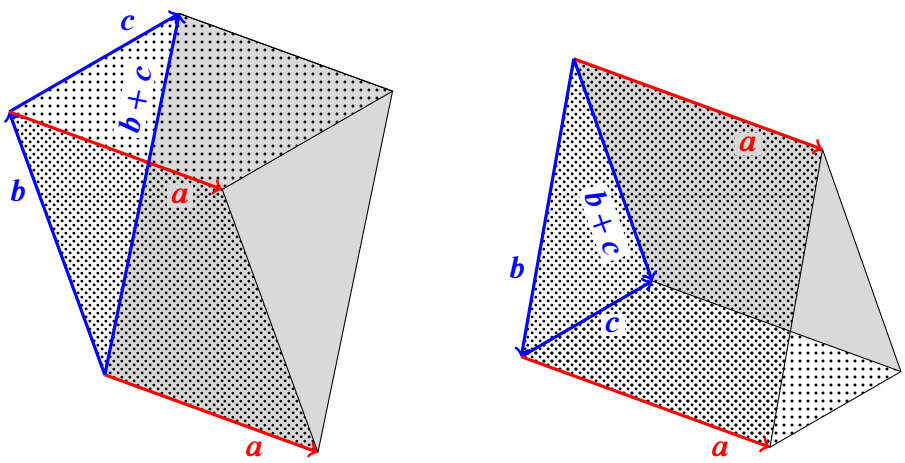


Figure 14.2: Additivity of area if $(\boldsymbol{a}, \boldsymbol{b})$ and $(\boldsymbol{a}, \boldsymbol{c})$ have the same or opposite orientations.

matrix can be defined in terms of 2×2 matrices as follows.

$$\det \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} = a_1 \det \begin{bmatrix} b_2 & c_2 \\ b_3 & c_3 \end{bmatrix} - a_2 \det \begin{bmatrix} b_1 & c_1 \\ b_3 & c_3 \end{bmatrix} + a_3 \det \begin{bmatrix} b_1 & c_1 \\ b_2 & c_2 \end{bmatrix}. \tag{14.1}$$

Geometrically, it is the sum of volumes, where the base parallelepiped is projected onto the coordinate planes in each coordinate direction and multiplied with the height. The minus sign comes from the different orientation of the coordinate system. We do not discuss this here in details. However, from the point of view of volumes it is no surprise that the determinant of a 3×3 matrix has the same three properties as a 2×2 matrix. In fact, if we assume that in the first row we have the linear combination

$$\begin{bmatrix} \lambda a_1 + \mu f_1 & \lambda b_1 + \mu g_1 & \lambda c_1 + \mu h_1 \end{bmatrix},$$

then in the second and third term in the expansion (14.1) we can use the linearity of the 2×2 matrix and for the first term we have

$$(\lambda a_1 + \mu f_1) \det \begin{bmatrix} b_2 & c_2 \\ b_3 & c_3 \end{bmatrix} = \lambda a_1 \det \begin{bmatrix} b_2 & c_2 \\ b_3 & c_3 \end{bmatrix} + \mu f_1 \det \begin{bmatrix} b_2 & c_2 \\ b_3 & c_3 \end{bmatrix}.$$

This proves the linearity in the first row. A similar argument applies to the other rows. If two rows are equal, one of the determinants in (14.1) has two equal rows and is therefore zero. The other two determinants in (14.1) have opposite sign. for instance if the first and second rows are equal, then

$$\det \begin{bmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{bmatrix} = a_1 \det \begin{bmatrix} b_1 & c_1 \\ b_3 & c_3 \end{bmatrix} - a_1 \det \begin{bmatrix} b_1 & c_1 \\ b_3 & c_3 \end{bmatrix} + a_3 \det \begin{bmatrix} b_1 & c_1 \\ b_1 & c_1 \end{bmatrix} = 0.$$

Finally, from (14.1) it is clear that $\det I_3 = 1$, so we have the same three basic properties as before.

Again, these properties uniquely determine the determinant. The above provides the main ideas for the axiomatic treatment of determinants in the next section: The determinant is a map on the space of square matrices with the three properties discussed above.

15 Definition and properties of determinants

As with vector spaces and fields we take an axiomatic approach to determinants. Based on only three requirements we derive all common properties of determinants. In particular this will lead to a formula to compute them as well. This formula, a generalisation of (14.1) will serve as a definition.

We only need to verify that it satisfies the basic axioms. The formula also shows that there is only one map on the set of square matrices that satisfies these axioms. We make the following definition. As discussed in the previous section, the three defining properties can be motivated by properties of n-dimensional volumes of (oriented) parallellepipeds.

Definition 15.1 (determinant)**.** A *determinant* is a map
$\det : \operatorname{Mat}(n \times n; \mathbb{F}) \to \mathbb{F}$ with the following properties:

(i) det is linear in every row of the matrix A. More precisely, if $r_i = [r_{i1} \dots r_{in}] \in \operatorname{Mat}(1 \times n; \mathbb{F})$ are the rows of A, and $s_k = [s_{k1} \dots s_{kn}]$, then

$$\det \begin{bmatrix} r_1 \\ \vdots \\ r_{k-1} \\ \lambda r_k + \mu s_k \\ r_{k+1} \\ \vdots \\ r_n \end{bmatrix} = \lambda \det \begin{bmatrix} r_1 \\ \vdots \\ r_{k-1} \\ r_k \\ r_{k+1} \\ \vdots \\ r_n \end{bmatrix} + \mu \det \begin{bmatrix} r_1 \\ \vdots \\ r_{k-1} \\ s_k \\ r_{k+1} \\ \vdots \\ r_n \end{bmatrix}$$

for all $\lambda, \mu \in \mathbb{F}$ and $k = 1, \dots, n$.

(ii) det is alternating, that is, if A has two equal rows, then $\det A = 0$;

(iii) det is normalised, that is, $\det I_n = 1$.

We also often write $|A| := \det A$.

Before we prove the existence of such a determinant we want to derive some consequences, assuming it exists. The properties we derive will naturally lead to a formula for the determinant, proving its existence.

Next we prove some properties following quite immediately from the definition.

Proposition 15.2. (a) *If B is a square matrix is obtained from the matrix A by interchanging two rows, then* $\det B = -\det A$.
(b) *If the rows of A are linearly dependent, then* $\det A = 0$.

Proof. (a) Assume that B is obtained from A by interchanging rows i and j, that is,

$$A = \begin{bmatrix} \vdots \\ r_i \\ \vdots \\ r_j \\ \vdots \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ r_i \\ \vdots \end{bmatrix}.$$

Due to the linearity and the alternating property of determinants

$$
\begin{aligned}
0 = \det \begin{bmatrix} \vdots \\ r_i + r_j \\ \vdots \\ r_j + r_i \\ \vdots \end{bmatrix} &= \det \begin{bmatrix} \vdots \\ r_i \\ \vdots \\ r_j + r_i \\ \vdots \end{bmatrix} + \det \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ r_j + r_i \\ \vdots \end{bmatrix} \\
&= \det \begin{bmatrix} \vdots \\ r_i \\ \vdots \\ r_j \\ \vdots \end{bmatrix} + \det \begin{bmatrix} \vdots \\ r_i \\ \vdots \\ r_i \\ \vdots \end{bmatrix} + \det \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ r_j \\ \vdots \end{bmatrix} + \det \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ r_i \\ \vdots \end{bmatrix} \\
&= \det A + 0 + 0 + \det B = \det A + \det B.
\end{aligned}
$$

Hence $\det B = -\det A$ as claimed.

(b) Suppose that the rows r_i, $i = 1, \dots, n$ of A are linearly dependent. Then one of the vectors, say r_k is a linear combination of the others. By interchanging two rows and taking into account (a) we can assume that this is the last row r_n. Hence there exist scalars $\lambda_i \in \mathbb{F}$ such that $r_n = \sum_{i=1}^{n-1} \lambda_i r_i$. By properties (i) and (ii) of determinants

$$
\det A = \det \begin{bmatrix} r_1 \\ \vdots \\ r_{n-1} \\ \sum_{i=1}^{n-1} \lambda_i r_i \end{bmatrix} = \sum_{i=1}^{n-1} \lambda_i \det \begin{bmatrix} r_1 \\ \vdots \\ r_{n-1} \\ r_i \end{bmatrix} = 0
$$

since each of the determinants in the last sum has two equal rows. □

To be able to derive more general properties of determinants we want to compute the determinants of elementary matrices as discussed in Section 13 (see page 57).

Lemma 15.3. *For the elementary matrices $E_\lambda^{(ij)}$ and $E_\lambda^{(i)}$ we have*

$$\det E_\lambda^{(ij)} = \det(E_\lambda^{(ij)})^T = 1 \tag{15.1}$$

$$\det E_\lambda^{(i)} = \det(E_\lambda^{(ij)})^T = \lambda \tag{15.2}$$

$$\det E_{ij} = \det E_{ij}^T = -1 \tag{15.3}$$

Moreover, if E is an elementary matrix and $A \in \mathrm{Mat}(n \times n; \mathbb{F})$, then

$$\det(EA) = (\det E)(\det A). \tag{15.4}$$

Proof. We consider the matrix A with rows $r_1, \dots, r_n$.

We start by $E_\lambda^{(ij)}$ which corresponds adding λ times row j to row i. Using (i) and (ii) in Definition 15.1 we get

$$
\det(E_\lambda^{(ij)} A) = \det \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ r_i + \lambda r_j \\ \vdots \end{bmatrix} = \det \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ r_i \\ \vdots \end{bmatrix} + \lambda \det \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ r_j \\ \vdots \end{bmatrix} = \det A + 0 = \det A.
$$

Applying the above in the special case $A = I_n$ and using the normalisation (iii) in Definition 15.1 we get (15.1) and hence (15.4) in case of $E = E_\lambda^{(ij)}$. The above also applies to the transposed since $(E_\lambda^{(ij)})^T = E_\lambda^{(ji)}$.

Next look at $E_\lambda^{(i)}$ which corresponds to multiplying row i by $\lambda \neq 0$. Using the linearity of the determinant again we get

$$\det(E_\lambda^{(i)} A) = \det \begin{bmatrix} \vdots \\ \lambda r_i \\ \vdots \end{bmatrix} = \lambda \det \begin{bmatrix} \vdots \\ r_i \\ \vdots \end{bmatrix} = \lambda \det A.$$

Applying the above in the special case $A = I_n$ we get (15.2) and hence (15.4) in case of $E = E_\lambda^{(i)}$. The matrix $E_\lambda^{(i)}$ is symmetric, so the transposed has the same determinant.

Finally, consider E_{ij} interchanging rows i and j of A. We have seen in Proposition 15.2 that $\det(E_{ij}A) = -\det A$. Applying this to $A = I_n$ we see that $\det E_{ij} = -\det I = -1$ from the normalisation property (iii) of determinants. In particluar (15.4) follows. The matrix E_{ij} is symmetric, so the above applies to the transposed. □

As a consequence we can prove the following corollary.

Proposition 15.4. *Let $A \in \mathrm{Mat}(n \times n; \mathbb{F})$ and B the reduced row-echelon form of A. Then there exist elementary matrices $E_1, \dots, E_N$ such that*

$$\begin{aligned} \det B &= \det(E_N)\det(E_{N-1}) \cdots \det(E_1)\det(A) \\ &= \det(E_N E_{N-1} \cdots E_1)(\det A). \end{aligned}$$

Moreover, $\det B = 1$ *if A is invertible and* $\det B = 0$ *otherwise.*

Proof. We know from Theorem 13.2 that there exist elementary matrices E_k, $k = 1, \dots, N$ such that $B = E_N \cdots E_1 A$ is in reduced row-echelon form. Repeatedly applying Lemma 15.3 we obtain

$$\begin{aligned} \det B &= \det(E_N E_{N-1} \cdots E_1 A) \\ &= \det(E_N)\det(E_{N-1} \cdots E_1 A) \\ &= \det(E_N)\det(E_{N-1})\det(E_{N-2} \cdots E_1 A) \\ &= \cdots = \det(E_N)\det(E_{N-1}) \cdots \det(E_1)\det(A). \end{aligned}$$

Using $A = I_n$ we get

$$\begin{aligned} \det(E_N E_{N-1} \cdots E_1) &= \det(E_N E_{N-1} \cdots E_1 I_n) \\ &= \det(E_N)\det(E_{N-1}) \cdots \det(E_1)\det I_n \end{aligned}$$

since $\det I_n = 1$ by definition. Finally note that if A is invertible, then the reduced row-echelon form is given by $B = I_n$, so $\det B = 1$. If A is not invertible, then B has a zero row and hence $\det B = 0$ by Proposition 15.2. □

From the above we can deduce the most important facts on matrices.

Theorem 15.5. *Let $A, B \in \mathrm{Mat}(n \times n; \mathbb{F})$. Then the following assertions are true.*

(i) $\det A \neq 0$ *if and only if A is invertible;*

(ii) $\det(AB) = (\det A)(\det B)$*;*

(iii) *If A is invertible, then* $\det(A^{-1}) = \dfrac{1}{\det A}$*;*

(iv) $\det(\lambda A) = \lambda^n \det A$ *for all $\lambda \in \mathbb{F}$;*

(v) $\det A = \det A^T$*, where A^T is the transposed matrix of A;*

(vi) *Properties* (i) *and* (ii) *in Definition* 15.1 *are valid if we replace rows by columns.*

(vii) *If B is obtained from A by interchanging two rows or two columns , then*

$$\det B = -\det A.$$

Proof. (i) We know that the reduced row-echelon form of A is invertible if and only if A is invertible. Hence by Proposition 15.4

$$\det(E_N)\det(E_{N-1})\cdots\det(E_1)(\det A) = \begin{cases} 1 & \text{if } A \text{ is invertible,} \\ 0 & \text{if } A \text{ is not invertible.} \end{cases}$$

Since $\det(E) \neq 0$ for elementary matrices by Lemma 15.3 we conclude that $\det A \neq 0$ if A is invertible and $\det A = 0$ otherwise.

(ii) If A is not invertible, then also AB is not invertible, so by (i) we get $\det(AB) = 0 = (\det A)(\det B)$. Hence assume that A is invertible. Then there exists a sequence of elementary matrices so that $E_N E_{N-1} \cdots E_1 A = I_n$ and therefore $A = E_1^{-1} E_2^{-1} \cdots E_N^{-1}$. Since E_k^{-1} are elementary matrices Proposition 15.4 implies that

$$\begin{aligned} \det(AB) &= \det(E_1^{-1} E_2^{-1} \cdots E_N^{-1} B) \\ &= \det(E_1^{-1} E_2^{-1} \cdots E_N^{-1})(\det B) \\ &= (\det A)(\det B). \end{aligned}$$

(iii) From (ii) and the normalisation of the determinant we have $1 = \det I_n = \det(AA^{-1}) = (\det A)(\det A^{-1})$. Hence (iii) follows.

(iv) This follows by the repeatedly applying the linearity of the determinant to each of the n rows of the matrix A.

(v) If A is not invertible, then A^T is not invertible either, so by (i) both have determinant zero. If A is invertible, then as shown in (ii) we can write A as a product of

elementary matrices $A = E_1 E_2 \cdots E_N$. Hence $A^T = E_N^T E_{N-1}^T \cdots E_1^T$ and therefore by Proposition 15.4 and Lemma 15.3

$$\begin{aligned}\det(A^T) &= \det(E_N^T)\det(E_{N-1}^T)\cdots\det(E_1^T)\\ &= \det(E_N)\det(E_{N-1})\cdots\det(E_1) = \det A.\end{aligned}$$

(vi) From (v) we can apply all properties concerning rows to columns by transposing the matrix.

(vii) The statement on columns has already been proved in Proposition 15.2. The corresponding claim for interchanging two columns follows from (vi). □

Remark 15.6. In the above theorem there is no statement about $\det(A+B)$. The reason is that there is no rule to compute the determinant of the sum of two matrices. In particular it is *not* true that $\det(A+B) = \det A + \det B$. As an example look at

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \qquad \text{and} \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Clearly $\det A = \det B = 0$ but

$$\det(A+B) = \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \neq 0 = \det A + \det B.$$

The following lemma will be useful for constructing determinants inductively in Theorem 15.11.

Lemma 15.7. *Suppose that $A \in \mathrm{Mat}(n,n;\mathbb{F})$, that $a \in \mathbb{F}$ and that b is a row vector with n components. Then the determinant of the block matrix*

$$\tilde{A} := \begin{bmatrix} a & b \\ 0 & A \end{bmatrix} \in \mathrm{Mat}(n+1, n+1; \mathbb{F})$$

is given by $\det \tilde{A} = a \det A$.

Proof. First we note that by the linearity of determinants in the first row

$$\det \tilde{A} = a \det \begin{bmatrix} 1 & 0 \\ 0 & A \end{bmatrix} + \det \begin{bmatrix} 0 & b \\ 0 & A \end{bmatrix}$$

Since all row vectors in the last matrix have a zero in their first component, the row space is an n dimensional. Since the matrix has $n+1$ row vectors must be linearly dependent and hence the determinant zero by Proposition 15.2. To prove the proposition we can therefore assume that $a = 1$ and $b = 0$.

According to Theorem 13.2 there exist a product $S = E_N \cdots E_1$ of elementary matrices E_k such that $B := SA$ is the reduced row-echelon form of A. We define the augmented matrices

$$\tilde{E}_k := \begin{bmatrix} 1 & 0 \\ 0 & E_k \end{bmatrix}$$

for $k = 1, \dots, N$. Then $\tilde{E}_k$ is an elementary matrix of the same kind as E_k and hence $\det \tilde{E}_k = \det E_k$ by Lemma 15.3. Defining $\tilde{S} := \tilde{E}_N \cdots \tilde{E}_1$ we conclude that $\det \tilde{S} = \det S \neq 0$. We also note that

$$\tilde{S}\tilde{A} = \begin{bmatrix} 1 & 0 \\ 0 & S \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & A \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & SA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & B \end{bmatrix}$$

The last matrix is the reduced row echelon form of $\tilde{A}$. If A is not invertible, then B and hence the last block matrix in the above identity has a row of zeros at the bottom. Thus $\det \tilde{A} = \det A = 0$. If A is invertible then $B = I_n$ and $\det A = \dfrac{1}{\det S}$ by Proposition 15.4. Also $\tilde{S}\tilde{A} = I_{n+1}$. Hence by the normalising property of determinants

$$1 = \det I_{n+1} = \det(\tilde{S}\tilde{A}) = (\det S)(\det \tilde{A})$$

Therefore, $\det \tilde{A} = \dfrac{1}{\det S} = \det A$ as claimed. □

We can easily compute the determinants of an upper or lower diagonal matrix.

Corollary 15.8. *Let A be upper or lower diagonal, that is, A is of the form*

$$\begin{bmatrix} \lambda_1 & * & \dots & * \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & * \\ 0 & \dots & 0 & \lambda_n \end{bmatrix} \quad \textit{or} \quad \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ * & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ * & \dots & * & \lambda_n \end{bmatrix}.$$

Then the determinant is the product of the diagonal elements, that is,

$$\det A = \lambda_1 \lambda_2 \cdots \lambda_n. \tag{15.5}$$

Proof. We only need to consider an upper triangular matrix, the result for a lower triangular matrix follows since transposed matrices have the same determinants (see Theorem 15.5(v)). Applying Lemma 15.7

$$\det \begin{bmatrix} \lambda_1 & * & \dots & * \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & * \\ 0 & \dots & 0 & \lambda_n \end{bmatrix} = \lambda_1 \det \begin{bmatrix} \lambda_2 & * & * \\ \vdots & \ddots & * \\ \dots & 0 & \lambda_n \end{bmatrix}.$$

Repeating a similar argument inductively we arrive at (15.5). □

The theory we derived is also very useful in computing determinants of larger matrices.

☞ *Remark* 15.9. In the proof of Proposition 15.4 we used row operations to compute the determinant of a matrix based on Lemma 15.3 by keeping track on the change of

determinant at each row operation. We can do that for any square matrix. By Proposition 15.4

$$\det A = \frac{\det B}{\det E_1 \det E_2 \cdots \det E_N},$$

where $E_1, \dots, E_n$ are the elementary matrices of the row operations used to reduce A. We can stop if B is of row-echelon form, not necessarily reduced row echelon form. To get to the reduced row-echelon form the only row operations needed are adding a multiple of one row to another row. We know that this does not change the determinant. Hence we can get $\det A$ by keeping track of the changes of determinant at each row operation:

- adding a multiple of a row to another row: no change of determinant.
- multiply a row by λ: $\det(EA) = \lambda$
- interchange two rows: $\det(EA) = -1$

As an example we compute $\det A = \det \begin{bmatrix} 5 & -1 & 1 \\ 3 & 1 & 1 \\ 6 & 2 & 1 \end{bmatrix}$:

Step	Row operation	Change in determinant
1.	$A \xrightarrow{R_3 \to R_3 - R_1} \begin{bmatrix} 5 & -1 & 1 \\ 3 & 1 & 1 \\ 1 & 3 & 0 \end{bmatrix}$	no change
2.	$\xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 1 \\ 5 & -1 & 1 \end{bmatrix}$	multiply by -1
3.	$\xrightarrow[R_3 \to R_3 - 5R_1]{R_2 \to R_2 - 3R_1} \begin{bmatrix} 1 & 3 & 0 \\ 0 & -8 & 1 \\ 0 & -16 & 1 \end{bmatrix}$	no change
4.	$\xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 1 & 3 & 0 \\ 0 & -8 & 1 \\ 0 & 0 & -1 \end{bmatrix}$	no change
5.	$\xrightarrow{R_2 \to -R_2/8} \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & -1/8 \\ 0 & 0 & -1 \end{bmatrix}$	multiply by $-\frac{1}{8}$
6.	$\xrightarrow{R_3 \to -R_3} \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} =: B$	multiply by -1

We therefore have

$$1 = \det B = (-1)\left(-\frac{1}{8}\right)(-1) \det A = -\frac{1}{8} \det A$$

and therefore $\det A = -8$. We could also stop doing row operations once we have an upper triangular matrix, but then we have take into account the corresponding determinant which can be computed by means of Corollary 15.8. In our example we would

stop at Step 4. If we do that

$$8 = 1(-1)(-8) = \det \begin{bmatrix} 1 & 3 & 0 \\ 0 & -8 & 1 \\ 0 & 0 & -1 \end{bmatrix} = (-1) \det A,$$

so $\det A = -8$ as before.

We now want to derive a formula for the determinant of an $n \times n$ matrix in terms of the determinants of suitable $(n-1) \times (n-1)$ matrices similar to the way a 3×3 matrix can be computed using 2×2 matrices. Hence let $A = [a_{ij}]_{1 \le i,j \le n}$ be an $n \times n$ matrix with columns a_k. Fix $j \in \{1, \dots, n\}$. By the linearity in the columns (see Theorem 15.5(vi)) we get

$$\det \begin{bmatrix} a_1 & \dots & a_{j-1} & \sum_{i=1}^n a_{ij} e_i & a_{j+1} & \dots & a_n \end{bmatrix} = \sum_{i=1}^n a_{ij} \det \begin{bmatrix} a_1 & \dots & a_{j-1} & e_i & a_{j+1} & \dots & a_n \end{bmatrix}.$$

If we write the matrix out in full we have

$$\det \begin{bmatrix} a_1 & \dots & a_{j-1} & e_i & a_{j+1} & \dots & a_n \end{bmatrix} = \det \begin{bmatrix} a_{11} & \dots & a_{1(j-1)} & 0 & a_{1(j+1)} & \dots & a_{1n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ \vdots & & \vdots & 0 & \vdots & & \vdots \\ a_{i1} & \dots & a_{i(j-1)} & 1 & a_{i(j+1)} & \dots & a_{in} \\ \vdots & & \vdots & 0 & \vdots & & \vdots \\ \vdots & & \vdots & \vdots & & & \vdots \\ a_{n1} & \dots & a_{n(j-1)} & 0 & a_{n(j+1)} & \dots & a_{nn} \end{bmatrix}. \quad (15.6)$$

We now bring the one at position ij to the top left corner of the matrix. This requires $i-1$ swaps of adjacent rows and $j-1$ swaps of adjacent columns. According to Theorem 15.5, each time we swap we have to multiply the determinant by -1. Thus the determinant changes by the factor $(-1)^{i+j-2} = (-1)^{i+j}$ and so

$$\det A = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det \begin{bmatrix} 1 & * \\ 0 & A_{ij} \end{bmatrix},$$

where A_{ij} is the $(n-1) \times (n-1)$ matrix we obtain from A by deleting the i-th row and the j-th column.

Definition 15.10 (minor)**.** The matrix

$$A_{ij} := \begin{bmatrix} a_{11} & \dots & a_{1j} & \dots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \dots & a_{ij} & \dots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \dots & a_{nj} & \dots & a_{nn} \end{bmatrix} \in \mathrm{Mat}\big((n-1)\times(n-1);\mathbb{F}\big). \tag{15.7}$$

is called a *minor* of the matrix A.

Now by Lemma 15.7 we have

$$\det \begin{bmatrix} 1 & * \\ 0 & A_{ij} \end{bmatrix} = \det A_{ij}$$

since the 1×1 matrix $[1]$ has determinant one. Hence for every $j \in \{1, \dots, n\}$

$$\det A = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det A_{ij}. \tag{15.8}$$

This is Laplace's formula for the determinant. A special case is expansion (14.1) for a 3×3 matrix. We have not yet proved the existence of a determinant in general, but with (15.8) we have a a candidate to give a definition by induction by n. In fact this is the *only* possibility!

Theorem 15.11 (Weierstrass)**.** *There exists a unique function*

$$\det : \ \mathrm{Mat}(n \times n; \mathbb{F}) \to \mathbb{F}$$

satisfying the three basic properties from Definition 15.1*. Moreover the determinant of* $A = [a_{ij}]_{1 \le i,j \le n}$ *is given by Laplace's formula*

$$\det A = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det A_{ij} = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det A_{ij}. \tag{15.9}$$

Proof. The uniqueness of the determinant follows from (15.8) since every function satisfying (i)–(iii) in Definition 15.1 must be of that form. The second equality in (15.9) follows by transposing A since $\det A = \det A^T$. For the existence we do a proof by induction on n. If $n = 1$ there is only one "row" of the 1×1 matrix $[a]$. The determinant is a linear map and therefore given by $\det[a] := ma$ for some $m \in \mathbb{F}$. Requirement (iii) means that $\det[1] = m1 = 1$, so $m = 1$. Hence $\det[a] = a$ is the only possibility. For the induction step we assume that $\det : \ \mathrm{Mat}((n-1) \times (n-1), \mathbb{F})$ is given. We then define det by (15.8) with $j = 1$ for an $n \times n$ matrix. We check conditions (i)–(iii) from Definition 15.1.

(i) Linearity as a function of the rows: Let r_i, $i = 1, \dots n$ be the rows of A. Suppose we look at row k is given by $\lambda r_k + \mu s_k$ where $\lambda, \mu \in \mathbb{F}$ and

$$r_k = \begin{bmatrix} r_{1k} & \tilde{r}_k \end{bmatrix} \quad \text{and} \quad s_k = \begin{bmatrix} s_{1k} & \tilde{s}_k \end{bmatrix}$$

with $\tilde{r}_k, \tilde{s}_k$ are row vectors with $n-1$ components. Now from the induction assumption, the i-th term in (15.8) is assumption

$$\begin{aligned}(-1)^{i+1} a_{i1} \det A_{i1} &= \det \begin{bmatrix} \vdots \\ \lambda \tilde{r}_k + \mu \tilde{s}_k \\ \vdots \end{bmatrix} \\ &= \lambda (-1)^{i+1} a_{i1} \det \begin{bmatrix} \vdots \\ \tilde{r}_k \\ \vdots \end{bmatrix} + \mu (-1)^{i+1} a_{i1} \det \begin{bmatrix} \vdots \\ \tilde{s}_k \\ \vdots \end{bmatrix}\end{aligned}$$

if $i \neq k$ and

$$(-1)^{k+1} (\lambda r_{1k} + \mu s_{1k}) \det \begin{bmatrix} \vdots \\ \cancel{\lambda \tilde{r}_k + \mu \tilde{s}_k} \\ \vdots \end{bmatrix}$$

if $i = k$. Hence each term in (15.8) is either linear in the determinant or in the coefficient. This completes the proof of property (i).

(ii) Assume that rows j and k are equal, where $j < k$. Then the matrices A_{i1} have two equal rows $r_j = r_k$ for $i \neq j, k$ and hence have zero determinant by the induction assumption. Hence (15.8) reduces to

$$\det A = (-1)^{j+1} \det A_{1j} + (-1)^{k+1} \det A_{1k}.$$

The rows of A_{1j} and A_{1k} are the same except that the $(k-1)$-st row of A_{1j} is equal to the j-th row of A_{1k}:

$$A_{1j} = \begin{bmatrix} \vdots \\ \cancel{r_j} \\ \vdots \\ r_k \\ \vdots \end{bmatrix} \leftarrow \text{row } k-1$$

and

$$A_{1k} = \begin{bmatrix} \vdots \\ r_j \\ \vdots \\ \cancel{r_k} \\ \vdots \end{bmatrix} \leftarrow \text{row } j$$

We swap adjacent rows to bring these rows to the top to form a matrix B. For A_{1j} the number of swaps required is $k-2$, and for A_{1k} the number of swaps needed is $j-1$. Hence,

$$\det A_{1j} = (-1)^{k-2} \det B \quad \text{and} \quad \det A_{1k} = (-1)^{j-1} \det B$$

and therefore

$$\det A = (-1)^{j+1} \det A_{1j} + (-1)^{k+1} \det A_{1k}$$
$$= (-1)^{j+k-1} \det B + (-1)^{k+j} \det B = 0$$

as claimed.

(iii) In the special case of I_n the sum (15.8) has only one term, namely

$$\det I_n = 1 \det I_{n-1} = 1$$

by the induction assumption. This completes the proof of the theorem. □

Remark 15.12. Laplace's formula is an expansion of determinants along a row or column of a matrix. The pattern for the sign of the terms ij is the following:

$$\begin{bmatrix} + & - & + & - & \dots \\ - & + & - & + & \dots \\ + & - & + & - & \dots \\ - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

In practice we choose a row or column with as many zeros as possible to compute a determinant.

16 The determinant of an endomorphism

Our original motivation to look at matrices was that they are representations of linear operators between finite dimensional vector spaces. We now want to make a connection from determinants of matrices to linear operators. Hence let V be a vector space over $\mathbb{F}$ with $\dim V = n$ and $T : V \to V$ a linear operator. We can then choose a basis $(v_1, \dots, v_n)$ of V. Let B be the matrix representing V with respect to that basis as in diagram (12.4). We can then try and define

$$\det T := \det B.$$

However, we need to be careful and check whether or not the definition depends on the choice of the basis. We know from (12.5) that the matrix representing T with respect to the new basis is similar, that is, it is given by $\Phi B \Phi^{-1}$. By Theorem 15.5 we have that

$$\det(\Phi B \Phi^{-1}) = \det \Phi \det B \det \Phi^{-1} = \det \Phi \det B \frac{1}{\det \Phi} = \det B$$

This proves the following result.

Proposition 16.1. *Similar matrices have the same determinant.*

Moreover it allows us to define the determinant of an endomorphism.

Definition 16.2. Let $\dim V = n < \infty$ and $T : V \to V$ be linear. Let B be the matrix representation of T with respect to some basis in V. We then define

$$\det T := \det B.$$

17 Cramer's rule and the adjugate matrix

Determinants naturally appear in the solution of linear equations. We assume $A \in \mathrm{Mat}(n \times n; \mathbb{F})$ and $x = (x_1, \dots, x_n) \in \mathbb{F}^n$ is a solution of

$$Ax = b,$$

where $b \in \mathbb{F}^n$. Suppose the columns of A are $a_1, \dots, a_n$. For $j = 1, \dots, n$ we then look at the matrices

$$B_j = \begin{bmatrix} a_1 & \dots & a_{j-1} & b & a_{j+1} & \dots & a_n \end{bmatrix},$$

where we replace column j in A by b. If x is a solution of $Ax = b$, then

$$b = \sum_{i=1}^{n} x_i a_i.$$

By the linearity of the determinant in the columns (Theorem 15.5) we have

$$\begin{aligned} \det B_j &= \det \begin{bmatrix} a_1 & \dots & a_{j-1} & \sum_{i=1}^{n} x_i a_i & a_{j+1} & \dots & a_n \end{bmatrix} \\ &= \sum_{i=1}^{n} x_i \det \begin{bmatrix} a_1 & \dots & a_{j-1} & a_i & a_{j+1} & \dots & a_n \end{bmatrix} \end{aligned}$$

for all $j = 1, \dots, n$. From Theorem 15.5 we also know that the determinant of a matrix with two equal columns is zero, so

$$\det \begin{bmatrix} a_1 & \dots & a_{j-1} & a_i & a_{j+1} & \dots & a_n \end{bmatrix} = 0$$

for all $i \neq j$ and

$$\det \begin{bmatrix} a_1 & \dots & a_{j-1} & a_j & a_{j+1} & \dots & a_n \end{bmatrix} = \det A$$

for $i = j$. Hence the above sum reduces to

$$\det B_j = x_j \det A.$$

This allows us to compute the solution of $Ax = b$ in terms of determinants, a method often referred to as *Cramer's rule*.

Proposition 17.1 (Cramer's rule)**.** *Let A be an invertible $n \times n$ matrix with coefficients in $\mathbb{F}$ and $b \in \mathbb{F}^n$, then the solution $x = (x_1, \dots, x_n)$ of $Ax = b$ is given by*

$$x_j = \frac{\det B_j}{\det A} = \frac{1}{\det A} \begin{vmatrix} a_{11} & \dots & a_{1(j-1)} & b_1 & a_{1(j-1)} & \dots & a_{1n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \dots & a_{n(j-1)} & b_n & a_{n(j-1)} & \dots & a_{nn} \end{vmatrix}$$

for $j = 1, \dots n$.

The above can be used to give an expression for the inverse matrix A^{-1}. Recall that the i-th column of the matrix is given by $c_i = A^{-1}e_i$. Observe that c_i is the solution to the equation

$$Ac_i = e_i.$$

That solution can be written in terms of Cramer's rule

$$c_{ji} = \frac{\det \begin{bmatrix} a_1 & \dots & a_{j-1} & e_i & a_{j+1} & \dots & a_n \end{bmatrix}}{\det A}.$$

The determinant at the top is the same as (15.6) and hence

$$c_{ji} = \frac{(-1)^{i+j} \det A_{ij}}{\det A},$$

where A_{ji} are the minors of A is as defined in (15.7). From the above we get

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} |A_{11}| & -|A_{21}| & |A_{31}| & \dots & (-1)^{n+1}|A_{n1}| \\ -|A_{12}| & |A_{22}| & -|A_{32}| & \dots & (-1)^{n+2}|A_{n2}| \\ |A_{13}| & -|A_{23}| & |A_{33}| & \dots & (-1)^{n+3}|A_{n3}| \\ \vdots & & & & \vdots \\ (-1)^{1+n}|A_{1n}| & \dots\dots\dots\dots & & & |A_{nn}| \end{bmatrix}$$

We obtain the above matrix by replacing a_{ij} in A by $(-1)^{i+j}|A_{ij}|$ and then transposing it. The matrix on the right hand side has a name.

Definition 17.2. The matrix

$$\operatorname{adj}(A) := \left[(-1)^{i+j}|A_{ij}|\right]_{1 \le i,j \le n}^T = \begin{bmatrix} |A_{11}| & -|A_{12}| & \dots & (-1)^{1+n}|A_{1n}| \\ -|A_{21}| & |A_{22}| & \dots & (-1)^{2+n}|A_{2n}| \\ |A_{31}| & -|A_{32}| & \dots & (-1)^{3+n}|A_{3n}| \\ \vdots & & & \vdots \\ (-1)^{n+1}|A_{n1}| & \dots\dots\dots & & |A_{nn}| \end{bmatrix}^T$$

is called the *adjugate matrix* of A, where $|A_{ij}|$ is the determinant of the minor A_{ij}.

From our discussions we have the following theorem.

Theorem 17.3. *For an invertible matrix A we have*

$$A^{-1} = \frac{1}{|A|}\,\mathrm{adj}(A),$$

where $\mathrm{adj}(A)$ *is the adjugate matrix of A.*

The above formula for the inverse of a matrix is useful for theoretical purposes, but except for 2×2 and perhaps 3×3 matrices *not* a practical method to compute the inverse of a matrix. This is better done by row operations. The reason is that for large matrices, the determinant is "expensive" to compute, and for $\mathrm{adj}(A)$ we have to compute many of them. In the case of a 2×2 matrix the formula provides a very easy way to get the inverse of a matrix, namely

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

We put negative signs on the off diagonal elements, interchange the diagonal elements and divide by the determinant.

Chapter V

Eigenvalues and eigenvectors

In many areas of mathematics the theory of eigenvalues plays a key role. In particular the ideas are taken further in algebra, they are used to analyse stability/instability or bifurcations in non-linear differential equations. Here we introduce the basic theory in finite dimensional spaces.

18 Eigenvalues and eigenvectors

Let V be a vector space over $\mathbb{F}$ and $T : V \to V$ a linear operator. We are interested in the set of $\lambda \in \mathbb{F}$ so that $\lambda I - T$ is not invertible. If V is finite dimensional we know from 9.6 that $\lambda I - T$ is invertible if and only if it is injective. Hence if $\lambda I - T$ is not invertible there exists $v \in V$ such that $v \neq 0$ and $\lambda v - Tv = 0$. This leads to the following definition

Definition 18.1 (eigenvalue)**.** Let $T : V \to V$ be a linear operator. We call $\lambda \in \mathbb{F}$ an *eigenvalue* of T if there exists $v \in V$ such that $v \neq 0$ and

$$Tv = \lambda v.$$

If that is the case v is called an *eigenvector to the eigenvalue* λ. The set

$$V_\lambda := \{v \in V : Tv = \lambda v\}$$

is called the *eigenspace corresponding to* λ.

Remark 18.2. The eigenspace V_λ corresponding to the eigenvalue λ of T is a subspace of V. Indeed, if $v_1, v_2 \in V_\lambda$ and $\mu_1, \mu_2 \in \mathbb{F}$, then

$$T(\mu_1 v_1 + \mu_2 v_2) = \mu_1 T v_1 + \mu_2 T v_2 = \lambda \mu_1 v_1 + \lambda \mu_2 v_2 = \lambda(\mu_1 v_1 + \mu_2 v_2),$$

so $\mu_1 v_1 + \mu_2 v_2 \in V_\lambda$ as well.

Remark 18.3. As seen above, if V is finite dimensional, then λ is an eigenvalue of t if and only if $\lambda I - T$ is not invertible. By Theorem 15.5 and the discussion in Section 16 this is the case if and only if $\det(\lambda I - T) = 0$. The latter is a polynomial in λ.

Definition 18.4 (characteristic polynomial)**.** Let $T : V \to V$ be an endomorphism and $\dim V < \infty$. We call

$$p_T(\lambda) := \det(\lambda I - T)$$

the *characteristic polynomial* of T.

Before we discuss the existence of eigenvalues we prove a few properties. We start by proving linear independence of eigenvectors corresponding to different eigenvalues.

Theorem 18.5. *Let $T : V \to V$ be a linear operator with distinct eigenvalues $\lambda_1, \dots, \lambda_p$. Let $v_i \in V_{\lambda_i}$ be an eigenvector of T corresponding to λ_i. Then the family $(v_1, \dots, v_p)$ is linearly independent.*

Proof. We give a proof by induction in p. If $p = 1$, then (v_1) is linearly independent since by definition eigenvectors are always non-zero. Suppose that $\mu_1, \dots, \mu_p \in \mathbb{F}$ are such that

$$\mu_1 v_1 + \mu_2 v_2 + \cdots + \mu_p v_p = 0. \tag{18.1}$$

and assume that $(v_1, \dots, v_{p-1})$ is linearly independent. Before we proceed note that

$$(T - \lambda_k I)v_i = Tv_i - \lambda_k v_i = \lambda_i v_i - \lambda_k v_i = (\lambda_i - \lambda_k)v_i$$

for $i, k = 1, \dots, p$. Hence if we apply $(\lambda_1 I - T)$ to (18.1) we get

$$\begin{aligned} 0 = (T - \lambda_1 I)(\mu_1 v_1 + \mu_2 v_2 + \cdots + \mu_p v_p) \\ = 0 + \mu_2(\lambda_2 - \lambda_1)v_2 + \cdots + \mu_p(\lambda_p - \lambda_1)v_p. \end{aligned}$$

Then we apply $T - \lambda_2 I$ to get

$$0 = \mu_3(\lambda_3 - \lambda_1)(\lambda_3 - \lambda_2)v_2 + \cdots + \mu_p(\lambda_p - \lambda_1)(\lambda_p - \lambda_2)v_p.$$

Then apply $T - \lambda_3 I, T - \lambda_4 I$ and finally $T - \lambda_{p-1} I$ to get

$$0 = \mu_p(\lambda_p - \lambda_1)(\lambda_p - \lambda_2) \cdots (\lambda_p - \lambda_{p-1})v_p.$$

Because all eigenvalues are distinct and v_p is an eigenvector and therefore non-zero, the only possibility for the above product to be zero is that $\mu_p = 0$. Hence, (18.1) reduces to

$$\mu_1 v_1 + \mu_2 v_2 + \cdots + \mu_{p-1} v_{p-1} = 0. \tag{18.2}$$

By the induction assumption $(v_1, \dots, v_{p-1})$ is linearly independent and therefore $\mu_1 = \cdots = \mu_{p-1} = 0$ as well. This completes the proof of the theorem. □

If $\lambda_1, \dots, \lambda_p$ are distinct eigenvalues of T we can look at the sum

$$W := V_{\lambda_1} + V_{\lambda_2} + \cdots + V_{\lambda_p} = \{v_1 + v_2 + \cdots + v_p : v_i \in V_{\lambda_i}, i = 1, \dots, p\}.$$

If $v \in W$ there is a unique representation of the form

$$w = v_1 + v_2 + \cdots + v_p$$

with $v_i \in V_{\lambda_i}, i = 1, \dots, p$. This follows as in case of the sum of two and generalises the notion of direct sums as discussed in Section 10.

Corollary 18.6. *Suppose that $\lambda_1, \dots, \lambda_p$ are distinct eigenvalues of T and V_{λ_i} the corresponding eigenspaces. Then*

$$W = V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_p}$$

is a direct sum.

In the above theorem we did not assume that V is finite dimensional, but if it is, we can get a limit on the number of eigenvalues.

Corollary 18.7. *Suppose that V is a vector space with $\dim V = n < \infty$ and that $T : V \to V$ is a linear operator. Then T has at most n distinct eigenvalues.*

Proof. Let $\lambda_1, \dots, \lambda_p$ be the distinct eigenvalues of T. For each eigenvalue λ_i choose an eigenvector v_i. By the previous theorem $(v_1, \dots, v_p)$ is linearly independent. Since $\dim V = n$ the largest linearly independent family has n elements. Hence $p \leq n$ as claimed. □

The above corollary gives us an upper estimate for the number of eigenvalues. However, do eigenvalues exist at all? In general the answer is no. Consider for instance a rotation by an angle $\theta \neq k\pi$ for any $k \in \mathbb{Z}$. Indeed, if v is an eigenvector to the eigenvalue λ we get $Tv = \lambda v$, which means Tv is parallel to v. However, for a rotation this is for sure not the case. We know that $\lambda I - T$ is not invertible if and only if $\det(\lambda I - T) = 0$. To compute the determinant we choose a basis in V and look at the matrix representation of T. For instance if T is a rotation, then the matrix representation is (see Example 11.3(b))

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

The characteristic polynomial is then

$$\begin{aligned} p_T(\lambda) = \det \begin{bmatrix} \lambda - \cos\theta & \sin\theta \\ -\sin\theta & \lambda - \cos\theta \end{bmatrix} &= (\lambda - \cos\theta)^2 + \sin^2\theta \\ &= \lambda^2 - 2\lambda\cos\theta + \cos^2\theta + \sin^2\theta = \lambda^2 - 2\lambda\cos\theta + 1. \end{aligned}$$

The zero's of that polynomial are

$$\lambda = \cos\theta \pm \sqrt{\cos^2\theta - 1} = \cos\theta \pm \sqrt{-\sin^2\theta} = \cos\theta \pm i|\sin\theta|.$$

The above is real if and only if $\theta = k\pi$, $k \in \mathbb{Z}$. What we learn from the above is that it depends on the underlying field whether or not T has an eigenvalue. The property $\mathbb{C}$ has and not $\mathbb{R}$ is that every polynomial with coefficients in $\mathbb{C}$ has a zero. This is the *fundamental theorem of algebra*, a proof of which we include here for completeness. We give a very recent elementary proof (see [1]) which only relies on de Moivre's theorem and the fact that a continuous real valued function has a minimum on any bounded closed set. We first prove a lemma valid for polynomials with coefficients in $\mathbb{R}$ or $\mathbb{C}$.

Lemma 18.8. *If $p(z)$ is a non-constant polynomial with coefficients in $\mathbb{C}$, then $|p(z)| \to \infty$ as $|z| \to \infty$ and $|p(z)|$ has an absolute minimum on $\mathbb{C}$.*

Proof. Let $p(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$ a non-constant polynomial with coefficients $a_k \in \mathbb{C}$, $k = 0, \ldots, n$ and $a_n \neq 0$. To prove that $|p(z)| \to \infty$ as $|z| \to \infty$ note that

$$\begin{aligned} |a_n z^n| &= |p(z) - a_{n-1}z^{n-1} - \cdots - a_1 z - a_0| \\ &\leq |p(z)| + |a_{n-1}||z|^{n-1} + \cdots + |a_1||z| + |a_0| \end{aligned}$$

and therefore

$$\frac{|p(z)|}{|z|^n} \geq |a_n| - \left(|a_{n-1}||z|^{-1} + \cdots + |a_1||z|^{-(n-1)} + |a_0||z|^{-n}\right).$$

for all $z \in \mathbb{C}$. The right hand side of the above inequality converges to zero as $|z| \to \infty$, so there exists $R > 0$ so that

$$\frac{|p(z)|}{|z|^n} \geq \frac{|a_n|}{2}$$

for all $z \in \mathbb{C}$ with $|z| \geq R$. Hence

$$|p(z)| \geq \frac{|a_n|}{2}|z|^n$$

whenever $|z| \geq R$ and so $|p(z)| \to \infty$ as $|z| \to \infty$. In particular any absolute minimum of $|p(z)|$ must be attained at a point with $|z_0| \leq R$, that is, in the closed bounded set $\{z \in \mathbb{C} : |z| \leq R\}$. As $|p(z)|$ is continuous $|p(z)|$ attains an absolute minimum at some $z_0 \in \mathbb{C}$ with $|z_0| \leq R$. □

Theorem 18.9 (Fundamental Theorem of Algebra)**.** *Every non-constant polynomial with coefficients in $\mathbb{C}$ has a root in $\mathbb{C}$.*

Proof. If $p(z)$ is a non-constant polynomial, then according to Lemma 18.8 $|p(z)|$ has an absolute minimum at some point $z_0 \in \mathbb{C}$. To simplify the proof we introduce the polynomial $P(z) := p(z + z_0)$ (just use the binomial theorem to multiply all terms $(z - z_0)^k$ out). Then $P(0) = p(z_0)$, which means that we can assume without loss of generality that $|p(0)|$ is the minimum of $|p(z)|$ by replacing p by P. Writing p for P again we want to show that $p(0) = 0$. We represent $p(z)$ in the form

$$p(z) = q(z)z^k + p(0)$$

for some $k \in \{1, \dots, n\}$, where $q(z)$ is a polynomial of degree $n - k$ with $q(0) \neq 0$. By construction $|p(z)| \geq |p(0)|$ for all $z \in \mathbb{C}$ and so

$$\begin{aligned} 0 \leq |p(z)|^2 - |p(0)|^2 &= \Big|q(z)z^k + p(0)\Big|^2 - |p(0)|^2 \\ &= |q(z)|^2|z|^{2k} + 2\operatorname{Re}\big[q(z)\overline{p(0)}z^k\big] + |p(0)|^2 - |p(0)|^2 \\ &= |q(z)|^2|z|^{2k} + 2\operatorname{Re}\big[q(z)\overline{p(0)}z^k\big]. \end{aligned}$$

Writing $z = re^{it}$ in modulus-argument form we get

$$0 \leq |q(re^{it})|^2 r^{2k} + 2r^k \operatorname{Re}\big[q(re^{it})\overline{p(0)}e^{ikt}\big].$$

If we divide by $2r^k > 0$ and let $r \to 0$ we obtain

$$0 \leq |q(re^{it})|^2 r^k + 2\operatorname{Re}\big[q(re^{it})\overline{p(0)}e^{ikt}\big] \to 2\operatorname{Re}\big[q(0)\overline{p(0)}e^{ikt}\big].$$

Hence we see that

$$0 \leq \operatorname{Re}\big[q(0)\overline{p(0)}e^{ikt}\big] \tag{18.3}$$

for all $t \in \mathbb{R}$. We claim that this implies that $q(0)p(0) = 0$. As $q(0) \neq 0$ we then conclude that $p(0) = 0$, so the polynomial p has a zero as claimed. To complete the proof we write $q(0)\overline{p(0)}$ in modulus argument form

$$q(0)\overline{p(0)} = |q(0)p(0)|e^{is}$$

so that from (18.3)

$$0 \leq \operatorname{Re}\big[|q(0)p(0)|e^{is}e^{ikt}\big] = |q(0)p(0)| \operatorname{Re}\big[e^{i(kt+s)}\big] = |q(0)p(0)| \cos(kt + s)$$

for all $t \in \mathbb{R}$. As $k \geq 1$ this is only possible if $|q(0)p(0)| = 0$. As $q(0) \neq 0$ we must have $p(0) = 0$, completing the proof of the theorem. □

There are other fields with that property, so we introduce the following definition.

Definition 18.10 (algebraically closed)**.** A field $\mathbb{F}$ is called *algebraically closed* if every non-constant polynomial in $\mathbb{F}[X]$ has a zero in $\mathbb{F}$.

In such fields we have the following theorem about the existence of eigenvalues.

Theorem 18.11. *Suppose that V is a finite dimensional vector space over $\mathbb{F}$ and that $\mathbb{F}$ is algebraically closed. Then every linear operator $T : V \to V$ has an eigenvalue.*

Proof. Since $\mathbb{F}$ is algebraically closed, the characteristic polynomial $p_T(\lambda) = \det(\lambda I - T)$ has a zero, and therefore T has an eigenvalue. $\square$

The if $\dim V = n$, then the characteristic polynomial p_T has degree n, so the fundamental theorem of algebra implies that p_T has n zeros. However, these zeros may be repeated, so they are not all distinct. Consider for instance a linear operator given by the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Then the characteristic polynomial is

$$p(\lambda) = \det(\lambda I - A) = \det \begin{bmatrix} \lambda - 2 & -1 \\ 0 & \lambda - 2 \end{bmatrix} = (\lambda - 2)^2.$$

Hence $\lambda = 2$ is a repeated root. To find an eigenvector we look at the matrix

$$2I - A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Hence the eigenspace of the eigenvalue λ is

$$V_2 = \left\{ t \begin{bmatrix} 1 \\ 0 \end{bmatrix} : t \in \mathbb{R} \right\}.$$

Even though $\lambda = 2$ has multiplicity two as a polynomial, the dimension of its eigenspace is only one. As a consequence we cannot expect the eigenspaces to span the whole space in general. For this reason we introduce the following definitions.

Definition 18.12 (geometric/algebraic multiplicity)**.** Let $T : V \to V$ be a linear operator with eigenvalue λ and corresponding eigenspace V_λ.

(i) We call $\dim V_\lambda$ the *geometric multiplicity* of λ.

(ii) If V is finite dimensional, we call the multiplicity of λ as a zero of the characteristic polynomial the *algebraic multiplicity* of λ.

If $\dim V < \infty$, $\mathbb{F}$ is algebraically closed and $\lambda_1, \dots, \lambda_p$ are the distinct eigenvalues of T, then

$$V_{\lambda_1} \oplus V_{\lambda_2} \oplus \cdots \oplus V_{\lambda_p} \subseteq V$$

is possibly a proper inclusion. The reason for that is that the algebraic multiplicity of one or several eigenvalues may be larger than the geometric multiplicity. We will show that the geometric multiplicity is always less or equal to the algebraic multiplicity (see Corollary 21.4 on page 98).

As the above example shows, the geometric multiplicity can be strictly less than the algebraic multiplicity for one or more eigenvalues. In the next section we discuss the case the eigenvalues span the whole space.

19 Diagonalisable matrices

Let $T : V \to V$ be linear, and that the eigenvectors of T span V. Then we can choose a basis $(v_1, \dots, v_n)$ of eigenvectors of T. Assuming v_i is an eigenvector to the eigenvalue λ_i we have $Tv_i = \lambda_i v_i$, so the coordinate vector of Tv_i with respect to the basis $(v_1, \dots, v_n)$ is $\lambda_i e_i$. Hence the matrix representation with respect to the new basis is the diagonal matrix

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & \lambda_n \end{bmatrix}.$$

If T is given by the matrix A with respect to some other basis and v_i are given by the coordinate vectors with respect to that basis The corresponding transition matrix from the standard basis to the basis of eigenvectors is then given by (see Remark 12.2)

$$\Phi := \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix}^{-1}.$$

Hence

$$D = \Phi A \Phi^{-1} \qquad \text{and} \qquad A = \Phi^{-1} D \Phi$$

as seen from the diagram below.

$$\begin{tikzcd} \mathbb{F}^n \arrow[r, "A"] \arrow[d, "\Phi"] & \mathbb{F}^n \arrow[d, "\Phi"] \\ \mathbb{F}^n \arrow[r, "D"] & \mathbb{F}^n \end{tikzcd}$$

We call such an operator *diagonalisable*. The above proves the following theorem.

Theorem 19.1. *Suppose V is a finite dimensional vector space over $\mathbb{F}$ and $T : V \to V$ a linear operator. Then T is diagonalisable if and only if V has a basis of eigenvectors of V.*

☞ *Remark* 19.2. The considerations above show that a basis with respect to which A becomes diagonal consists of a basis of eigenvectors of A, and the diagonal elements are the eigenvalues of A.

There are several cases, where it is guaranteed that a matrix is diagonalisable. One is if all eigenvalues are simple.

Proposition 19.3. *Suppose that V is a vector space over $\mathbb{F}$ with $\dim V = n$ and that and $T : V \to V$ a linear operator. If T has n distinct eigenvalues, then T is diagonalisable.*

Proof. Let $v_1, \dots, v_n$ be the eigenvectors to the distinct eigenvalues $\lambda_1, \dots, \lambda_n$ of T. By Theorem 18.5 the family of eigenvectors $(v_1, \dots, v_n)$ is linearly independent. As $\dim V = n$ they form a basis of V (see Corollary 7.8), so T is diagonalisable by the previous theorem. □

Example 19.4. We want to find a basis of $\mathbb{R}^3$ such that the matrix

$$A := \begin{bmatrix} -1 & 0 & 1 \\ 6 & -1 & -6 \\ -2 & 4 & 8 \end{bmatrix}$$

becomes diagonal. We also want to compute the corresponding transition matrices for the change of basis. To get the eigenvalues we compute the characteristic polynomial by expanding along the top row (because there is a zero). Some calculation yields

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} -1-\lambda & 0 & 1 \\ 6 & -1-\lambda & -6 \\ -2 & 4 & 8-\lambda \end{bmatrix} \\ &= -(1+\lambda) \det \begin{bmatrix} -(1+\lambda) & -6 \\ 4 & 8-\lambda \end{bmatrix} + 1 \det \begin{bmatrix} 6 & -(1+\lambda) \\ -2 & 4 \end{bmatrix} \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6. \end{aligned}$$

Hence we need to find the solutions of $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$. One solution obviously is $\lambda = 1$. Dividing the polynomial by $(\lambda - 1)$ we get

$$\lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3) = 0$$

so that the other eigenvalues are $\lambda = 2$ and $\lambda = 3$. Hence we have three distinct eigenvalues, and therefore the matrix A is diagonalisable. To find a basis for which A becomes diagonal we need a basis of eigenvectors of A. we determine an eigenvector to the eigenvalue $\lambda = 1$ by row reducing

$$A - I = \begin{bmatrix} -1-1 & 0 & 1 \\ 6 & -1-1 & -6 \\ -2 & 4 & 8-1 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 1 \\ 6 & -2 & -6 \\ -2 & 4 & 7 \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 + 3R_1} \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -3 \\ 0 & 4 & 6 \end{bmatrix} \xrightarrow{R_3 \to R_3 + 2R_2} \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[R_2 \to -\frac{R_2}{2}]{R_1 \to -\frac{R_1}{2}} \begin{bmatrix} 1 & 0 & -\frac{1}{2} \\ 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

Hence

$$v_1 := \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$$

is an eigenvector corresponding to the eigenvalue $\lambda_1 = 1$. For an eigenvector corresponding to $\lambda_2 = 2$ we do the reduction

$$A - 2I = \begin{bmatrix} -3 & 0 & 1 \\ 6 & -3 & -6 \\ -2 & 4 & 6 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow -\frac{R_3}{2}} \begin{bmatrix} 1 & -2 & -3 \\ 6 & -3 & -6 \\ -3 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 - 6R_1} \begin{bmatrix} 1 & -2 & -3 \\ 0 & 9 & 12 \\ 0 & -6 & -8 \end{bmatrix} \xrightarrow[R_3 \to -\frac{R_3}{6}]{R_2 \to \frac{R_2}{9}} \begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & \frac{4}{3} \\ 0 & 1 & \frac{4}{3} \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_2]{R_1 \to R_1 + 2R_2} \begin{bmatrix} 1 & 0 & -\frac{1}{3} \\ 0 & 1 & \frac{4}{3} \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore

$$v_2 := \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$$

is an eigenvector to the eigenvalue $\lambda_2 - 2$. For $\lambda_3 = 3$ we do the reduction

$$A - 3I = \begin{bmatrix} -4 & 0 & 1 \\ 6 & -4 & -6 \\ -2 & 4 & 5 \end{bmatrix} \xrightarrow{-R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & -4 & -5 \\ 6 & -4 & -6 \\ -4 & 0 & 1 \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 - 3R_1} \begin{bmatrix} 2 & -4 & -5 \\ 0 & 8 & 9 \\ 0 & -8 & -9 \end{bmatrix} \xrightarrow[R_3 \to -\frac{R_3}{8}]{\substack{R_1 \to \frac{R_1}{2} \\ R_2 \to \frac{R_2}{8}}} \begin{bmatrix} 1 & -2 & -\frac{5}{2} \\ 0 & 1 & \frac{9}{8} \\ 0 & 1 & \frac{9}{8} \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_2]{R_1 \to R_1 + 2R_2} \begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & \frac{9}{8} \\ 0 & 0 & 0 \end{bmatrix}.$$

and therefore

$$v_3 := \begin{bmatrix} 2 \\ -9 \\ 8 \end{bmatrix}$$

is an eigenvector to the eigenvalue $\lambda_3 = 3$. According to the above, A becomes diagonal with respect to the basis

$$(v_1, v_2, v_3) = \left(\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -9 \\ 8 \end{bmatrix} \right)$$

and the matrix is

$$D := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Note that the eigenvalues in the diagonal have to be in the same order as the corresponding eigenvectors in the basis! The columns of the transition matrix Φ from the new to the old basis are the basis elements given above, and the one from the old to the new basis its inverse, that is,

$$\Phi^{-1} = \begin{bmatrix} 1 & 1 & 2 \\ -3 & -4 & -9 \\ 2 & 3 & 8 \end{bmatrix} \qquad \text{and} \qquad \Phi = \begin{bmatrix} 5 & 2 & 1 \\ -6 & -4 & -3 \\ 1 & 1 & 1 \end{bmatrix}.$$

The inverse is obtained by row reducing the augmented matrix $[\Phi \mid I]$ as usual. Hence we have

$$A = \Phi^{-1} D \Phi \qquad \text{or} \qquad D = \Phi A \Phi^{-1}.$$

We next give an example in a vector space over a finite field.

Example 19.5. Consider the matrix $A \in \mathrm{Mat}(2 \times 2; \mathbb{Z}_3)$ given by

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

As this is an upper diagonal matrix its eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$. As there are two distinct eigenvalues the matrix can be diagonalised. We need the eigenvectors. First note that

$$A - I = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

so

$$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

is an eigenvector (remember we do $\mathbb{Z}_3$-arithmetic). Similarly,

$$A - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \xrightarrow{R_2 \to R_2 + R_1} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

so

$$v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

is an eigenvector to the eigenvalue 2. Hence the diagonalised matrix is

$$D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

if we choose the basis v_1, v_2). The transition matrices from new to old and old to new basis are

$$\Phi = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix} \qquad \text{and} \qquad \Phi^{-1} = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix},$$

respectively.

The next example illustrates the fact that not all matrices are diagonalisable.

Example 19.6. Consider the matrix

$$A = \begin{bmatrix} 5 & -5 & -6 \\ 0 & -1 & 0 \\ 3 & -2 & -4 \end{bmatrix}.$$

The characteristic polynomial is given by

$$\begin{aligned} \det(A - \lambda I) &= \det \begin{bmatrix} 5-\lambda & -5 & -6 \\ 0 & -1-\lambda & 0 \\ 3 & -2 & -4-\lambda \end{bmatrix} \\ &= (5-\lambda) \det \begin{bmatrix} -(1+\lambda) & 0 \\ -2 & -(4+\lambda) \end{bmatrix} + 3 \det \begin{bmatrix} -5 & -6 \\ -(1+\lambda) & 0 \end{bmatrix} \\ &= -(1+\lambda)(\lambda^2 - \lambda - 2) = -(\lambda+1)^2(\lambda-4). \end{aligned}$$

Hence, $\lambda_1 = -1$ is a double eigenvalue and $\lambda_2 = 2$ a simple eigenvalue. We determine the eigenvectors corresponding to -1 by doing the row reduction

$$A + I = \begin{bmatrix} 6 & -5 & -6 \\ 0 & 0 & 0 \\ 3 & -2 & -3 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 3 & -2 & -3 \\ 0 & 0 & 0 \\ 6 & -5 & -6 \end{bmatrix} \xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 3 & -2 & -3 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1 - 2R_3} \begin{bmatrix} 3 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \xrightarrow[R_2 \leftrightarrow -R_3]{R_1 \to \frac{R_2}{3}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence the eigenspace to $\lambda_1 = -1$ is spanned by the eigenvector

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

The eigenvectors corresponding to 2 are obtained from the row reduction

$$A - 2I = \begin{bmatrix} 3 & -5 & -6 \\ 0 & -3 & 0 \\ 3 & -2 & -6 \end{bmatrix} \xrightarrow{R_3 \to R_3 + R_2} \begin{bmatrix} 3 & -5 & -6 \\ 0 & -3 & 0 \\ 3 & -5 & -6 \end{bmatrix} \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to -\frac{R_2}{3}} \begin{bmatrix} 3 & -5 & -6 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1 + 5R_2} \begin{bmatrix} 3 & 0 & -6 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \to \frac{R_1}{3}} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and so the vector

$$v_2 := \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

spans the eigenspace. Hence there are only two linearly independent eigenvectors, which means that A is not diagonalisable.

In the next sections we investigate how the above can be modified if T is not diagonalisable. For that we need another look at kernels and images.

20 Generalised kernels and images

For a linear operator $T : V \to V$ on a finite dimensional space V we know from Theorem 9.5 that

$$\dim V = \dim(\operatorname{im} T) + \dim(\ker T).$$

Unfortunately, that does not imply that $V = \operatorname{im} T \oplus \ker T$ necessarily. As an example look at $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by the matrix

$$T = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The image is the span of the column vectors, and so is is the line

$$t\begin{bmatrix}1\\0\end{bmatrix}, \quad t \in \mathbb{R}.$$

Clearly $\ker T$ is given by the same subspace and therefore

$$\operatorname{im} T + \ker T \neq \mathbb{R}^2.$$

Hence in general $\ker T \cap \operatorname{im} T \neq \{0\}$. We nevertheless want to find a decomposition of V into a direct sum associated with the operator T. For that we look at "powers" of T. We define

$$T^k := \underbrace{T \circ T \circ \ldots \circ T}_{k \text{ times}},$$

that is, apply k times the operator T. For convenience we let $T^0 := I$ (the identity) and $T^1 = T$. Applying Theorem 9.5 to T^k we still get

$$\dim V = \dim(\operatorname{im} T^k) + \dim(\ker T^k)$$

for all $k \in \mathbb{N}$. Our aim is to show that $\operatorname{im} T^k \cap \ker T^k = \{0\}$ if k is large enough, and for sure if $k = \dim V < \infty$. Because the dimensions of image and kernel add up to the full dimension we therefore have a direct sum decomposition $V = \operatorname{im} T^k \oplus \ker T^k$. We first prove some facts on $\operatorname{im} T^k$ and $\ker T^k$.

Proposition 20.1. *Let V be a vector space over $\mathbb{F}$ and $T \in L(V)$ an endomorphism. Then the following assertions hold.*

(i) $\{0\} = \ker T^0 \subseteq \ker T^1 \subseteq \ker T^2 \subseteq \ker T^3 \subseteq \ldots$;

(ii) $V = \operatorname{im} T^0 \supseteq \operatorname{im} T^1 \supseteq \operatorname{im} T^2 \supseteq \operatorname{im} T^3 \supseteq \ldots$;

(iii) *If* $\ker T^k = \ker T^{k+1}$, *then* $\ker T^k = \ker T^j$ *for all* $j > k$;

(iv) *If* $\operatorname{im} T^k = \operatorname{im} T^{k+1}$, *then* $\operatorname{im} T^k = \operatorname{im} T^j$ *for all* $j > k$;

Proof. (i) Let $k \in \mathbb{N}$. If $v \in \ker T^k$, then $T^{k+1}v = T(T^k u) = T0 = 0$, so $v \in \ker T^{k+1}$ as claimed.

(ii) Let $k \in \mathbb{N}$. If $v \in \operatorname{im} T^{k+1}$, then there exists $u \in V$ such that $v = T^{k+1}u$. Hence $v = T^k(Tu)$, so $v \in \operatorname{im} T^k$ as claimed.

(iii) We prove by induction that $\ker T^j = \ker T^{j+1}$ for all $j \geq k$. For $j = k$ we have $\ker T^k = \ker T^{k+1}$ by assumption. Hence assume that $j \geq k$ and $\ker T^j = \ker T^{j+1}$. To prove that $\ker T^{j+1} = \ker T^{j+2}$ let $v \in \ker T^{j+2}$. Then $0 = T^{j+2}v = T^{j+1}(Tv)$, so $Tv \in \ker T^{j+1}$. By the induction assumption $Tv \in \ker T^j$ and so $T^j(Tv) = T^{j+1}v = 0$. Hence $v \in \ker T^{j+1}$ and so $\ker T^{j+2} \subseteq \ker T^{j+1}$. By (i) equality follows.

(iv) We prove by induction that $\operatorname{im} T^j = \operatorname{im} T^{j+1}$ for all $j \geq k$. For $j = k$ we have $\operatorname{im} T^k = \operatorname{im} T^{k+1}$ by assumption. To prove that $\operatorname{im} T^{j+1} = \operatorname{im} T^{j+2}$ let $v \in \operatorname{im} T^{j+1}$.

Then there exists $u \in V$ with $v = T^{j+1}u$. By induction assumption there exists $w \in V$ with $T^{j+1}w = T^j u$, and so $v = T(T^j u) = T(T^{j+1}w) = T^{j+2}w$. Hence $u \in \operatorname{im} T^{j+2}$, proving that $\operatorname{im} T^{j+1} \subseteq \operatorname{im} T^{j+2}$. By (ii) equality follows. $\square$

If V is finite dimensional we can say more about direct sum decompositions.

Theorem 20.2. *Let V be a vector space with $\dim V = n < \infty$ and $T \in L(V)$ an endomorphism. Then the following assertions hold.*

(i) $\ker T^k = \ker T^{k+1}$ *if and only if* $\operatorname{im} T^k = \operatorname{im} T^{k+1}$

(ii) *There exists a smallest number $\nu \in \{0, \dots, n\}$ such that* $\ker T^\nu = \ker T^j$ *for all $j \geq \nu$;*

(iii) $V = \operatorname{im} T^\nu \oplus \ker T^\nu$

(iv) $T : \operatorname{im} T^\nu \to \operatorname{im} T^\nu$ *is an isomorphism and* $T : \ker T^\nu \to \ker T^\nu$ *and* $T^\nu = 0$ *on* $\ker T^\nu$.

Proof. (i) From Proposition 20.1 we know $\ker T^k \subseteq \ker T^{k+1}$ and $\operatorname{im} T^k \supseteq \operatorname{im} T^{k+1}$. Hence there is equality if and only if the dimension stays constant. Applying Theorem 9.5 to T^k we obtain $n = \dim(\ker T^k) + \dim(\operatorname{im} T^k)$ for all $k \in \mathbb{N}$, so the dimension stays constant if and only if equality holds for kernel and image.

(ii) By Proposition 20.1(i) we have

$$0 = \dim(\ker T^0) \leq \dim(\ker T^1) \leq \dim(\ker T^2) \leq \cdots \leq \dim V = n \tag{20.1}$$

with strict inequality if and only if there is a proper inclusion of the corresponding kernels. As $\dim(V) = n$ there can only be at most n strict inequalities, so there must be equality in (20.1) for $k \geq n$. Hence there is a minimal $\nu \in \{0, 1, \dots, n\}$ such that $\ker T^\nu = \ker T^{\nu+1}$. By Proposition 20.1(iii) we have

$$\{0\} = \ker T^0 \subsetneq \ker T^1 \subsetneq \ker T^2 \subsetneq \cdots \subsetneq \ker T^\nu = \ker T^{\nu+1} = \ker T^{\nu+2} = \dots .$$

In particular $\ker T^\nu = \ker T^j$ for all $j \geq \nu$.

(iii) We know that $n = \dim(\ker T^\nu) + \dim(\operatorname{im} T^\nu)$, so we only need to show that $\ker T^\nu \cap \operatorname{im} T^\nu = \{0\}$. Let $v \in \ker T^\nu \cap \operatorname{im} T^\nu$. As $v \in \operatorname{im} T^\nu$ there exists $u \in V$ with $v = T^\nu u$. Now $T^\nu v = T^{2\nu} u = 0$ since $v \in \ker T^\nu$. Thus $u \in \ker T^{2\nu}$. By (ii) we have $\ker T^{2\nu} = \ker T^\nu$ and so $v = T^\nu u = 0$ as claimed.

(iv) If $u \in \operatorname{im} T^\nu$, then $Tu \in \operatorname{im} T^{\nu+1}$. By (ii) we have $\operatorname{im} T^\nu = \operatorname{im} T^{\nu+1}$, and so $Tu \in \operatorname{im} T^\nu$ as claimed. To prove that $T : \operatorname{im} T^\nu \to \operatorname{im} T^\nu$ is surjective we use that by (ii) $\operatorname{im} T^\nu = \operatorname{im} T^{\nu+1}$. Hence if $v \in \operatorname{im} T^\nu$, there exists $u \in V$ with $v = T^{\nu+1} u = T(T^\nu u)$, and so v is the image of some $w = T^\nu u \in \operatorname{im} T^\nu$. Hence $T : \operatorname{im} T^\nu \to \operatorname{im} T^\nu$ is surjective and therefore bijective by Corollary 9.6 since $\operatorname{im} T^\nu$ is finite dimensional. Finally let $w \in \ker T^\nu$. Then by (ii) $w \in \ker T^\nu$ and therefore $T^\nu w = 0$ as claimed. $\square$

A linear operator T such that $T^\nu = 0$ has a special name.

Definition 20.3 (nilpotent operator)**.** A linear operator $T : V \to V$ is called *nilpotent* if there exists $\nu \in \mathbb{N}$ such that $T^\nu = 0$. Similar terminology applies to matrices.

An example of a nilpotent matrix is

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

since

$$A^3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

As a consequence of the above theorem we can find a simple matrix representation of T with respect to a suitable basis in V.

Corollary 20.4. *Let V be a vector space with $\dim V = n < \infty$ and $T \in L(V)$ an endomorphism. Suppose that ν is the number from Theorem 20.2(ii) and let $\ell := \dim(\operatorname{im} T^\nu)$. Then there exist a basis of V such that the matrix representing T has the form*

$$A = \begin{bmatrix} A_1 & 0 \\ 0 & A_2 \end{bmatrix},$$

where $A_1 \in \operatorname{Mat}(\ell \times \ell; \mathbb{F})$ is invertible and $A_2 \in \operatorname{Mat}((n-\ell) \times (n-\ell); \mathbb{F})$ is nilpotent.

Proof. According to Theorem 20.2 we have $V = \operatorname{im} T^n \oplus \ker T^n$. We choose a basis $(u_1, \dots, u_\ell)$ if $\operatorname{im} T^n$ and a basis $(w_1, \dots, w_{n-\ell})$ for $\ker T^n$. Then the family

$$(v_1, \dots, v_\ell, w_1, \dots, w_{n-\ell})$$

is a basis of V. By Theorem 20.2 we have $Tv_i \in \operatorname{im} T^n$ for $i = 1, \dots, \ell$. Define A_1 to be the matrix representation of T restricted to $\operatorname{im} T^n$. Since this restriction of T is invertible A_1 is invertible. Similarly define A_2 to be the matrix representation of T restricted to $\ker T^n$. Then A_2 is clearly nilpotent. The matrix representation of T of V then clearly has the required form. □

21 Generalised eigenvectors and spectral decomposition

We have seen that eigenvectors to different eigenvalues are linearly independent, but do not necessarily span the whole space. Eigenvectors are non-zero elements of $\ker(\lambda I - T)$. We now look at elements of the kernel of powers of $\lambda I - T$.

Definition 21.1 (generalised eigenvector)**.** Let $T : V \to V$ be an endomorphism, $\lambda \in \mathbb{F}$. A non-zero vector v is called a *generalised eigenvector* to the eigenvalue λ if $(\lambda I - T)^k v = 0$, that is, $v \in \ker(\lambda I - T)^k$ for some integer $k \geq 1$.

We have looked at kernels and images of T^k in Theorem 20.2. Applying these results to $\lambda I - T$ for any eigenvalue of T we get the following proposition.

Proposition 21.2. *Let V be a vector space with $\dim V = n < \infty$ and let $T \in L(V)$ be an endomorphism. Let λ be an eigenvalue of T and set*

$$U_\lambda := \ker(\lambda I - T)^n \quad \textit{and} \qquad W_\lambda := \operatorname{im}(\lambda I - T)^n.$$

Then the following assertions hold.

(i) *There exists a smallest number $\nu \in \{1, \dots, n\}$ such that $U_\lambda = \ker(\lambda I - T)^j$ for all $j \geq \nu$;*

(ii) $V = U_\lambda \oplus W_\lambda$

(iii) $T : W_\lambda \to W_\lambda$ *and λ is not an eigenvalue of T restricted to U_λ;*

(iv) $T : U_\lambda \to U_\lambda$ *and $(\lambda I - T)^\nu = 0$ on U_λ.*

Proof. The assertions follow directly by applying Theorem 20.2 with T replaced by $\lambda I - T$. The only thing to check is that U_λ and W_λ are invariant under T and not just $\lambda I - T$. If $u \in U_\lambda$, then $\lambda u \in U_\lambda$ since U_λ is a subspace and hence $Tu = \lambda u - (\lambda I - T)u \in U_\lambda$. A similar argument applies for W_λ. That λ is not an eigenvalue of T restricted to W_λ just means that $\lambda I - T$ is invertible on W_λ. □

If $\mathbb{F}$ is algebraically closed we can decompose V into a direct sum of generalised eigenspaces.

Theorem 21.3 (spectral decomposition)**.** *Let V be a vector space with $\dim V = n < \infty$ over the algebraically closed field $\mathbb{F}$. Let $T \in L(V)$ be an endomorphism and $\lambda_1, \dots, \lambda_p$ be its distinct eigenvalues. For $i = 1, \dots, p$ denote the generalised eigenspaces of λ_i by*

$$U_i := \ker(\lambda_i I - T)^n$$

assume that the algebraic multiplicity of λ_i is m_i. Then

$$V = U_1 \oplus U_2 \oplus U_3 \oplus \cdots \oplus U_p.$$

Moreover, for every $i = 1, \dots, p$

(i) $T : U_i \to U_i$;

(ii) *λ_i is the only eigenvalue of T restricted to U_i;*

(iii) $\dim(U_i) = m_i$;

(iv) *$\lambda_i I - T$ restricted to U_i is nilpotent;*

(v) *λ_i is not an eigenvalue of T restricted to U_j if $j \neq i$.*

Finally, there exists a basis of V such that the matrix representation of T with respect to that basis is a block matrix of the form

$$A = \begin{bmatrix} A_1 & 0 & \dots & 0 \\ 0 & A_2 & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \dots & 0 & A_p, \end{bmatrix} \tag{21.1}$$

where $A_i \in \mathrm{Mat}(m_i \times m_i; \mathbb{F})$, $i = 1, \dots, p$ are square matrices.

Proof. Since $\mathbb{F}$ is algebraically closed T has an eigenvalue by Theorem 18.11. We give a proof by induction by p. If $p = 1$, that is, T only has one eigenvalue λ_1, then the assertions follow directly from Proposition 21.2. Since λ_1 is the only eigenvalue its algebraic multiplicity is clearly n, which is the same as the dimension of the space.

Suppose now that assertions (i)–(v) of the theorem hold for an operator with p distinct eigenvalues that T has $p + 1$ eigenvalues $\lambda_1, \dots, \lambda_p, \lambda_{p+1}$. By Proposition 21.2 we have

$$V = U_{p+1} \oplus W,$$

where $U_{p+1} = \ker(\lambda_{p+1} I - T)^n$ and $W := \mathrm{im}(\lambda_{p+1} I - T)^n$, and $\lambda_1, \dots, \lambda_p$ are the distinct eigenvalues of T restricted to W. Hence we can apply the induction assumption to conclude that $W = U_1 \oplus \cdots \oplus U_p$ with all properties of T restricted to U_i as stated in the theorem for $i = 1, \dots, p$. Also by Proposition 21.2 λ_{p+1} is the only eigenvalue

of T restricted to U_{p+1}. By the induction assumption $\dim U_i = m_i$ for $i = 1, \dots, p$. In particular, $\dim(V) - \dim(W) = m_{p+1}$, which coincides with the dimension of U_{p+1}.

To prove the last statement on the matrix representation of T choose a basis $(u_{i1}, \dots, u_{im_i})$ in each of the generalised eigenspaces U_i. Then

$$(u_{11}, \dots, u_{1m_1}, u_{21}, \dots, u_{2m_2}, \dots, u_{p1}, \dots, u_{pm_p}) \tag{21.2}$$

is a basis of V. Since $Tu_{ik} \in U_i$ for every $u_{ik} \in U_i$ for every $k = 1, \dots, m_i$ and $i = 1, \dots, p$ This completes the proof of the theorem. □

As a direct consequence of (iii) and since $\ker(\lambda_i I - T) \subseteq \ker(\lambda_i I - T)^n$ we get the following inequality between the geometric and algebraic multiplicities of an eigenvalue.

Corollary 21.4. *If λ is an eigenvalue of the linear operator T, then its geometric multiplicity is less or equal to its algebraic multiplicity.*

Remark 21.5. If $\ker(\lambda_i I - T)^n = \ker(\lambda_i I - T)$, for every $i = 1, \dots, p$, then the generalised eigenspaces coincide with the eigenspaces, and every block A_i is a diagonal matrix. This is exactly the situation considered in Theorem 19.1. In general A_i are not diagonal matrices. However, we can split them into a diagonalisable and a nilpotent operator. This is the *Jordan decomposition theorem*.

Theorem 21.6 (Jordan decomposition)**.** *Let V be a vector space with $\dim V = n < \infty$ over the algebraically closed field $\mathbb{F}$. Let $T \in L(V)$ be an endomorphism. Then there exist $D, N \in L(V)$ with the following properties.*

(i) *$T = D + N$;*

(ii) *D is diagonalisable and N is nilpotent;*

(iii) *D and N commute, that is, $DN = ND$.*

Proof. Let $\lambda_1, \dots, \lambda_p$ be the distinct eigenvalues of T. As before, denote the generalised eigenspaces of λ_i by

$$U_i := \ker(\lambda_i I - T)^n$$

and assume that $\dim U_i = m_i$ for $i = 1, \dots, p$. From the above theorem we have $T : U_i \to U_i$. Hence it is sufficient to define D and N on each of the generalised eigenspaces U_i separately. For every $i = 1, \dots, p$ we define

$$\begin{aligned} D|_{U_i} &:= \lambda_i I, \\ N|_{U_i} &:= T - \lambda_i I. \end{aligned}$$

As $V = U_1 \oplus \cdots \oplus U_p$ the operators D and N are well defined. Clearly $T = D + N$. For $u \in U_i$ and $i = 1, \dots, p$ we have $Du = \lambda_i u$, so every $u \in U_i$ is an eigenvalue of D.

Hence U is diagonalisable by choosing a basis like the basis (21.2), the operator D is diagonal. Also, by definition of U_i

$$(\lambda_i I - T)^{m_i} u = 0$$

Hence by the above theorem. Finally,

$$DNu = \lambda_i(T - \lambda_i I)u = (T - \lambda_i I)\lambda_i u = NDu,$$

so D and N commute. Since the operators is completely determined by how they act on U_i this completes the proof of the theorem. □

Remarks 21.7. (a) Another equivalent way to define D and N is the following. As $V = U_1 \oplus \cdots \oplus U_p$ every $v \in V$ has a unique representation $v = u_1 + u_2 + \cdots + u_p$ with $u_i \in U_i$ for $i = 1, \dots, p$. Hence

$$\begin{aligned} Dv &:= \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_p u_p \\ Nv &:= (T - \lambda_1 I)u_1 + (T - \lambda_2 I)u_2 + \cdots + (T - \lambda_p I)u_p \end{aligned}$$

(b) With a more careful choice of basis of U_i it is possible to arrange it so that A_i in (21.1) that $A_i = D_i + N_i$ with

$$D_i = \begin{bmatrix} \lambda_i & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_i \end{bmatrix}$$

and the nilpotent matrix N_i being a block matrix with diagonal blocks of the form

$$\begin{bmatrix} 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & 0 \\ \vdots & & \ddots & 1 \\ 0 & \dots\dots & & 0 \end{bmatrix}$$

and zero otherwise. This may include 1×1 blocks of zeros. For instance

$$N_i = \left[\begin{array}{ccc|cc|cc} 0 & 1 & 0 & & & & \\ 0 & 0 & 1 & & & & \\ 0 & 0 & 0 & & & & \\ \hline & & & 0 & 1 & & \\ & & & 0 & 0 & & \\ \hline & & & & & 0 & \\ & & & & & & 0 \end{array}\right],$$

where all entries not shown are zero. Doing the decomposition for every block A_i we get the decomposition $A = D + N$ for the above choice of basis.

(c) One can show that the properties of D and N in Theorem 21.6 uniquely determine D and N.

Corollary 21.8. *Let V be a vector space over $\mathbb{F}$, where $\mathbb{F}$ is algebraically closed. Suppose that $\lambda_1, \dots, \lambda_p$ are the distinct eigenvalues of the linear operator $T : V \to V$ and that*

$$\ker(\lambda_i I - T)^2 = \ker(\lambda_i I - T) \tag{21.3}$$

for all $i = 1, \dots, n$. Then T is diagonalisable.

Proof. By Proposition 21.2 the identity (21.3) implies that the eigenspace $V_i \ker(\lambda_i I - T)$ equals the generalised eigenspace for every $i = 1, \dots, p$. Hence by Theorem 21.3

$$V = V_1 \oplus V_2 \oplus \cdots \oplus V_p.$$

Hence V has a basis of eigenvectors of T and therefore is diagonalisable by Theorem 19.1. □

Chapter VI

Orthogonal transformations

22 Inner product spaces

In $\mathbb{R}^n$ we do not just have addition and multiplication by scalars, but also the dot product. The dot product is useful to express geometric facts such as angles between vectors and the magnitude of a vector. We want to generalise the concept to vector spaces over $\mathbb{C}$. We could also replace $\mathbb{R}$ to $\mathbb{C}$ but $\mathbb{C}$ has the advantage of being algebraically closed. When working with linear operators and their eigenvalues this is a distinct advantage.

Definition 22.1 (inner product and norm)**.** Let V be a vector space over $\mathbb{C}$. A map

$$\langle \cdot\,, \cdot \rangle : V \times V \to \mathbb{C}$$

is called an *inner product* if the following properties hold:

(i) $\langle v, v \rangle \geq 0$ with equality if and only if $v = 0$ (positivity);

(ii) $\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$ (anti-symmetry);

(iii) $\langle \lambda u + \mu v, w \rangle = \lambda \langle u, w \rangle + \mu \langle v, w \rangle$ (linearity in the first argument).

We call

$$\|u\| := \sqrt{\langle u, u \rangle}$$

the *norm* induced by the inner product.

In the above definition we just require linearity in the first argument. We almost have linearity in the second argument except for the complex conjugates. Indeed by (ii) and (iii) we have

$$\begin{aligned} \langle w, \lambda u + \mu v \rangle = \overline{\langle \lambda u + \mu v, w \rangle} &= \overline{\lambda \langle u, w \rangle + \mu \langle v, w \rangle} \\ &= \bar{\lambda} \overline{\langle u, w \rangle} + \bar{\mu} \overline{\langle v, w \rangle} = \bar{\lambda} \langle w, u \rangle + \bar{\mu} \langle w, v \rangle \end{aligned} \tag{22.1}$$

In particular

$$\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle, \tag{22.2}$$

so if we bring a scalar to the front from the second argument we need to take the complex conjugate. The main example we use is the standard inner product in $\mathbb{C}^n$.

Definition 22.2 (Standard inner product, Euclidean norm)**.** If $u = (z_1, \dots, z_n)$ and $v := (w_1, \dots, w_n)$ are vectors in $\mathbb{C}^n$ we set

$$\langle u, v\rangle = \sum_{k=1}^{n} z_k \bar{w}_k$$

(note the complex conjugate in the second factor!). We call it the *standard inner product*. We define the *Euclidean norm* on $\mathbb{C}^n$ by

$$\|u\| := \sqrt{\langle u, u\rangle} = \sqrt{\sum_{k=1}^{n} |z_k|^2}.$$

It is rather evident from the definition that the above defines an inner product on $\mathbb{C}^n$. If $u \in \mathbb{R}^n$ this is the usual dot product. For vectors in the plane the dot product is given by

$$u \cdot v = \|u\|\|v\| \cos\theta,$$

where θ is the angle between the vectors u and v. Since $|\cos\theta| \leq 1$ we get

$$|u \cdot v| \leq \|u\|\|v\||\cos\theta| \leq \|u\|\|v\|.$$

It turns out that the above inequality follows from the three basic properties of an inner product as listed in Definition 22.1. This is the *Cauchy-Schwarz inequality*.

Theorem 22.3 (Cauchy Schwarz inequality)**.** *Suppose V is an inner product space. Then*

$$|\langle u, v\rangle| \leq \|u\|\|v\|$$

with equality if and only if u and v are linearly dependent.

Proof. Consider the polynomial $p(t) = \|u - t\langle u, v\rangle v\|^2$ with real coefficients which has at most one zero. Let now $u, v \in V$ and $t \in \mathbb{R}$. Using the basic properties of the inner product and (22.1) we have

$$\begin{aligned}
0 \leq p(t) &= \|u - t\langle u, v\rangle v\|^2 = \big\langle u - t\langle u, v\rangle v, u - t\langle u, v\rangle v\big\rangle \\
&= \langle u, u\rangle - \big\langle u, t\langle u, v\rangle v\big\rangle - \big\langle t\langle u, v\rangle v, u\big\rangle + \big\langle t\langle u, v\rangle v, t\langle u, v\rangle v\big\rangle \\
&= \|u\|^2 - t\overline{\langle u, v\rangle}\langle u, v\rangle - t\langle u, v\rangle\langle v, u\rangle + t^2\overline{\langle u, v\rangle}\langle u, v\rangle\langle v, v\rangle \\
&= \|u\|^2 - 2t|\langle u, v\rangle|^2 + t^2|\langle u, v\rangle|^2\|v\|^2.
\end{aligned}$$

The above is a non-negative quadratic with real coefficients. This is only possible if its discriminant satisfies

$$|\langle u, w\rangle|^4 - |\langle u, w\rangle|^2\|u\|^2\|w\|^2 \leq 0.$$

If we rearrange the inequality the Cauchy-Schwarz inequality follows. There is equality if and only if the discriminant is zero, which means that there exists $t \in \mathbb{R}$ with $0 = p(t) = \|u - t\langle u, v\rangle v\|^2$. The latter only happens if u and v are linearly dependent. □

Compare the above to the more geometric proof for the dot product in $\mathbb{R}^N$ given in vector calculus. As a consequence of the Cauchy-Schwarz inequality we get the usual properties of the Euclidean norm.

Theorem 22.4 (norm properties)**.** *Suppose V is an inner product space. Then for all $u, v \in V$ and $\lambda \in \mathbb{C}$*

(i) $\|u\| \geq 0$ *with equality if and only if* $u = 0$*;*

(ii) $\|\lambda u\| = |\lambda|\|u\|$*;*

(iii) $\|u + v\| \leq \|u\| + \|v\|$ *(triangle inequality);*

Proof. (i) follows from the definition of an inner product and the norm.

(ii) Using the properties of an inner product and (22.1) we get

$$\|\lambda u\|^2 = \langle \lambda u, \lambda u\rangle = \lambda\langle u, \lambda u\rangle = \lambda\bar{\lambda}\langle u, u\rangle = |\lambda|^2\|u\|^2$$

as claimed.

(iii) By the properties of the inner product

$$\begin{aligned}\|u + v\|^2 = \langle u + v, u + v\rangle &= \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2 \\ &= \|u\|^2 + \langle u, v\rangle + \overline{\langle u, v\rangle} + \|v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u, v\rangle + \|v\|^2.\end{aligned}$$

Using the Cauchy-Schwarz inequality $\mathrm{Re}\langle u, v\rangle \leq |\langle u, v\rangle| \leq \|u\|\|v\|$, so

$$\|u + v\|^2 \leq \|u\|^2 + 2\|u\|\|v\| + \|v\|^2 = \big(\|u\| + \|v\|\big)^2.$$

Now the required inequality follows by taking square roots. □

23 Orthonormal bases

In the plane and in space we usually use *rectangular* coordinate systems. Such a coordinate system is determined by the unit vectors defining the coordinate axes. These unit vectors are pairwise orthogonal. In first year linear algebra you learnt that vectors u, v are orthogonal if and only if their dot product $u \cdot v = 0$. This motivates the following definition.

Definition 23.1 (orthogonal, orthonormal family)**.** Let V be an inner product space.

(i) The vectors $u, v \in V$ are called *orthogonal* if $\langle u, v\rangle = 0$.

(ii) Let $(v_i)_{i\in I}$ be a family in V. The family is called an *orthogonal family* if $\langle v_i, v_j\rangle = 0$ whenever $i \neq j$. It is called an *orthonormal family* if in addition $\|u_i\| = 1$ for all $i \in I$.

(iii) If an orthonormal family is a basis of V we say it is an *orthonormal basis*.

Orthogonal and orthonormal families have many nice properties. In particular they are always linearly independent.

Proposition 23.2. *Let V be an inner product space and $(v_i)_{i\in I}$ an orthogonal family with $v_i \neq 0$ for all $i \in I$. Then $(v_i)_{i\in I}$ is linearly independent. Moreover, every orthonormal family is linearly independent.*

Proof. We look at a finite orthogonal family $(v_1, \dots, v_n)$ with $v_k \neq 0$ for all $k = 1, \dots, n$. Suppose the scalars $\lambda_1, \dots, \lambda_n$ are such that

$$0 = \lambda v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n.$$

We have to prove that $\lambda_k = 0$ for all $k = 1, \dots, n$. By the linearity of the inner product in the first argument and since $\langle v_i, v_k\rangle = 0$ if $i \neq k$ we get

$$\begin{aligned} 0 &= \langle \lambda v_1 + \lambda_2 v_2 + \cdots + \lambda_n v_n, v_k\rangle \\ &= \lambda_1\langle v_1, v_k\rangle + \cdots + \lambda_k\langle v_k, v_k\rangle + \cdots + \lambda_n\langle v_n, v_k\rangle = \lambda_k\|u_k\|^2. \end{aligned}$$

As $u_k \neq 0$ we therefore have $\lambda_k = 0$. Doing the above with every $k = 1, \dots, n$ we get $\lambda_1 = \cdots = \lambda_n = 0$. Hence $(v_1, \dots, v_n)$ is linearly independent. By definition an arbitrary family is linearly independent if every finite sub-family is linearly independent (see Definition 6.1). Finally, if the family is orthonormal, zero is not in the family so we can apply what we just proved. □

Next we see how to represent the inner product with respect to an orthonormal basis.

Proposition 23.3. *Let $(v_1, \dots, v_n)$ be an orthonormal basis in the inner product space V. Let $u, v \in V$ and $z, w \in \mathbb{C}^n$ the corresponding coordinate vectors. Then*

$$\langle u, v\rangle = \langle z, w\rangle_{\mathbb{C}^n},$$

where $\langle z, w\rangle_{\mathbb{C}^n}$ is the standard inner product in $\mathbb{C}^n$ and $\langle u, v\rangle$ the inner product in V.

Proof. Suppose the coordinate vectors of u and v are $z = (\lambda_1, \dots, \lambda_n)$ and $w = (\mu_1, \dots, \mu_n)$, respectively. Then by the algebraic rules for the inner product

$$\langle u, v\rangle = \Big\langle \sum_{k=i}^{n} \lambda_i v_i, \sum_{k=j}^{n} \mu_j v_j \Big\rangle = \sum_{k=i}^{n} \lambda_i \Big\langle v_i, \sum_{k=j}^{n} \mu_j v_j \Big\rangle$$
$$= \sum_{k=i}^{n} \sum_{j=1}^{n} \lambda_i \bar{\mu}_j \langle v_i, v_j\rangle = \sum_{k=i}^{n} \lambda_i \bar{\mu}_i = \langle z, w\rangle_{\mathbb{C}^n}$$

as claimed. □

In particular, two vectors in V are orthogonal if and only if the corresponding coordinate vectors are orthogonal. Also, the norms are preserved.

We now want to derive properties of the transition matrix between two orthonormal bases $\boldsymbol{v} = (v_1, \dots, v_n)$ and $\boldsymbol{w} = (w_1, \dots, w_n)$. The columns of the transition matrix $\Phi = [t_{ij}] \in \mathrm{Mat}(n \times n; \mathbb{C})$ from $\boldsymbol{v}$ to $\boldsymbol{w}$ are the coordinate vectors

$$t_i = \begin{bmatrix} t_{i1} \\ \vdots \\ t_{in} \end{bmatrix} \in \mathbb{C}^n$$

of v_i with respect to $\boldsymbol{w}$. Since the vectors v_i are pairwise orthogonal the same is true for the coordinate vectors t_i as shown in the above proposition. Moreover, since $\|v_i\| = 1$ also $\|t_i\| = 1$. Hence the columns of the matrix Φ are an orthonormal basis of $\mathbb{C}^n$. By the orthogonality

$$\langle t_i, t_k\rangle = \sum_{j=1}^{n} t_{ij} \bar{t}_{kj} = \delta_{ik},$$

where δ_{ik} is the Kronecker delta. This means that

$$\Phi \bar{\Phi}^T = I_n,$$

where $\bar{\Phi}^T$ is the *conjugate transposed* of Φ defined by

$$\bar{\Phi}^T := \begin{bmatrix} \bar{t}_{11} & \bar{t}_{21} & \dots & \bar{t}_{n1} \\ \bar{t}_{12} & \bar{t}_{22} & & \bar{t}_{n2} \\ \vdots & & & \vdots \\ \bar{t}_{1n} & \bar{t}_{2n} & \dots & \bar{t}_{nn} \end{bmatrix}$$

In particular

$$\Phi^{-1} = \bar{\Phi}^T.$$

Since matrices with that property play an important role the have a special name.

Definition 23.4 (orthogonal/unitary matrix)**.** A matrix $U \in \mathrm{Mat}(n \times n; \mathbb{C})$ is called *unitary* if $U\bar{U}^T = I$, that is, $U^{-1} = \bar{U}^T$. If U has real coefficients and $U^{-1} = U^T$, then the matrix is called *orthogonal*

As shown above unitary matrices are precisely those whose columns form an orthonormal basis in $\mathbb{C}^n$ and the orthogonal matrices are those whose columns form an orthonormal basis in $\mathbb{R}^n$. Moreover, both are transition matrices between different orthonormal bases of $\mathbb{C}^n$ and $\mathbb{R}^n$, respectively.

24 Gram-Schmidt orthogonalisation and consequences

In the previous section we discussed orthonormal families and bases without actually showing that they exist. The purpose of this section is in particular to show that every finite dimensional inner product space V has an orthonormal basis. We also use this to derive the QR-factorisation of matrices and a geometric interpretation of determinants.

Given an inner product space V, the idea is to take any basis and "orthogonalise" it by a process called *Gram-Schmidt orthogonalisation*. In fact, we do not really need a basis, but consider a linearly independent family $(w_1, \dots, w_n)$ in V and the subspaces

$$V_k := \operatorname{span}(w_1, \dots, w_k)$$

for $k = 1, \dots, n$. We inductively construct an orthonormal family $(v_1, \dots, v_n)$ in V such that

$$V_k = \operatorname{span}(v_1, \dots, v_k)$$

for $k = 1, \dots, n$. The idea is to take the orthogonal projection u_k of w_k onto V_{k-1}, where for convenience we set $V_0 := \{0\}$. We then expect $w_k - u_k$ to be orthogonal to V_k as illustrated in Figure 24.1. We then just normalise the vectors $w_k - u_k$ so that they have norm one, that is,

$$v_k := \frac{w_k - u_k}{\|w_k - u_k\|}.$$

We note also that $u_1 = 0$ since we project w_1 onto $V_0 = \{0\}$.

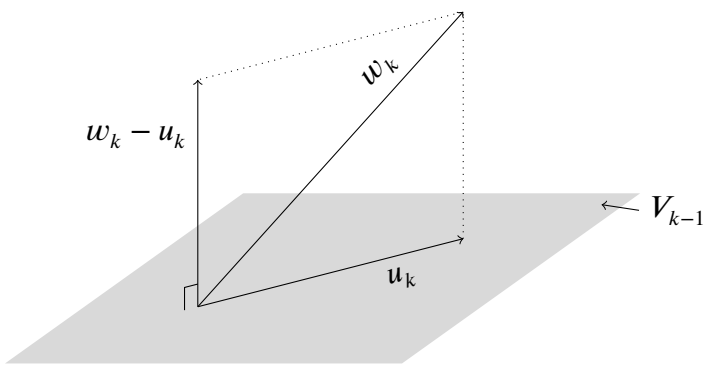


Figure 24.1: Orthogonal projection of w_k onto V_{k-1}

The next task is to find a way to compute the vectors u_k and v_k inductively. This also shows the existence of the orthogonal projections considered above. We first set

$$u_1 := 0 \qquad \text{and} \qquad v_1 := \frac{w_1 - u_1}{\|w_1 - u_1\|} = \frac{w_1}{\|w_1\|}.$$

Then clearly $\|v_1\| = 1$ and $V_1 = \operatorname{span}(w_1) = \operatorname{span}(v_1)$. Next we construct u_2 and v_2 by taking the projection of w_2 into the direction of v_1. We define

$$u_2 := \langle w_2, v_1 \rangle v_1$$

and

$$v_2 := \frac{w_2 - \langle w_2, v_1 \rangle v_1}{\|w_2 - \langle w_2, v_1 \rangle v_1\|} = \frac{w_2 - u_2}{\|w_2 - u_2\|}$$

as shown in Figure 24.2. Clearly $\|v_2\| = 1$. As $\langle v_1, v_1 \rangle = \|v_1\| = 1$, by the linearity of the inner product in the first argument we get

$$\langle w_2 - u_2, v_1 \rangle = \langle w_2, v_1 \rangle - \langle w_2, v_1 \rangle \langle v_1, v_1 \rangle = \langle w_2, v_1 \rangle - \langle w_2, v_1 \rangle = 0$$

so v_1 and v_2 are orthogonal. From the definition of v_2 it is clear that v_2 is expressed as a linear combination of (v_1, w_2) such that the coefficient of w_2 is non-zero. Hence the replacement lemma (Lemma 7.2) implies that (v_1, v_2) is a basis of V_2. Hence $\operatorname{span}(w_1, w_2) = \operatorname{span}(v_1, v_2)$.

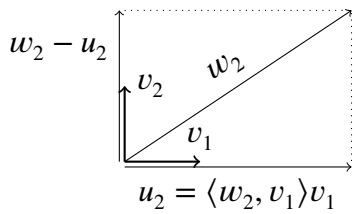


Figure 24.2: Projection of w_2 into the direction of v_1.

Next we get a vector v_3 orthogonal to v_1 and v_2, which means it is orthogonal to the subspace V_2. The projection of w_3 onto that plane is given by

$$u_3 := \langle w_3, v_1 \rangle v_1 + \langle w_3, v_2 \rangle v_2.$$

and we expect

$$v_3 = \frac{w_3 - (\langle w_3, v_1 \rangle v_1 + \langle w_3, v_2 \rangle v_2)}{\|w_3 - (\langle w_3, v_1 \rangle v_1 + \langle w_3, v_2 \rangle v_2)\|} = \frac{w_3 - u_3}{\|w_3 - u_3\|}.$$

to be orthogonal to V_2 as illustrated in Figure 24.3. By the linear independence of (v_1, v_2, w_3) we have that $u_3 \neq 0$. Since the family (v_1, v_2) is orthonormal, by construction

$$\begin{aligned}\langle w_3 - u_3, v_i \rangle &= \big\langle w_3 - \big(\langle w_3, v_1 \rangle v_1 + \langle w_3, v_2 \rangle v_2\big), v_i \big\rangle \\ &= \langle w_3, v_i \rangle - \langle w_3, v_1 \rangle \langle v_1, v_i \rangle - \langle w_3, v_2 \rangle \langle v_2, v_i \rangle = \langle w_3, v_i \rangle - \langle w_3, v_i \rangle = 0.\end{aligned}$$

for $i = 1, 2$ and hence (v_1, v_2, v_3) is orthonormal. Moreover, v_3 is expressed in terms of the basis (v_1, v_2, w_3) of V_3 with the coefficient of w_3 being non-zero. by the replacement lemma (v_1, v_2, v_3) is a basis of V_3.

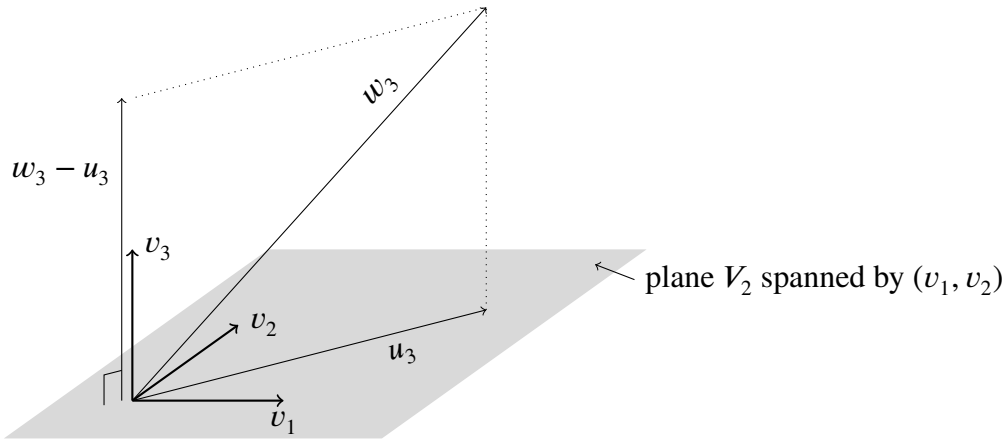


Figure 24.3: Orthogonal projection of w_3 onto V_2.

We continue in a similar manner. Assume we have constructed the orthonormal family $(v_1, \dots, v_k)$ so that $(v_1, \dots, v_k)$ is a basis of V_k. The projection of w_{k+1} onto V_k is then given by

$$u_{k+1} := \sum_{j=1}^{k} \langle w_{k+1}, v_j \rangle v_j \tag{24.1}$$

and we set

$$v_{k+1} = \frac{w_{k+1} - \sum_{j=1}^{k} \langle w_{k+1}, v_j \rangle v_j}{\left\| w_{k+1} - \sum_{j=1}^{k} \langle w_{k+1}, v_j \rangle v_j \right\|} = \frac{w_{k+1} - u_{k+1}}{\| w_{k+1} - u_{k+1} \|}. \tag{24.2}$$

The vector v_{k+1} is then orthogonal to $v_1, \dots v_k$ since for $i = 1, \dots, k$ we have

$$\begin{aligned} \langle w_{k+1} - u_{k+1}, v_i \rangle &= \left\langle w_{k+1} - \sum_{j=1}^{k} \langle w_{k+1}, v_j \rangle v_j, v_i \right\rangle \\ &= \langle w_{k+1}, v_i \rangle - \sum_{j=1}^{k} \langle w_{k+1}, v_j \rangle \langle v_j, v_i \rangle = \langle w_{k+1}, v_i \rangle - \langle w_{k+1}, v_i \rangle = 0. \end{aligned}$$

Here we use that $\langle v_j, v_i \rangle = \delta_{ij}$ (the Kronecker delta) by assumption on $(v_1, \dots, v_k)$. Hence $(v_1, \dots, v_k, v_{k+1})$ is an orthonormal family. By the replacement lemma $(v_1, \dots, v_{k+1})$ is a basis of V_{k+1} since the coefficient of w_{k+1} in the definition of v_{k+1} is obviously non-zero. We do the above construction inductively until $k + 1 = n$. We therefore proved the following theorem.

Theorem 24.1 (Gram-Schmidt orthogonalisation)**.** *Let V be an inner product space and $(w_1, \dots, w_n)$ a linearly independent family. Then there exists an orthonormal family $(v_1, \dots, v_n)$ such that*

$$\operatorname{span}(w_1, \dots, w_k) = \operatorname{span}(v_1, \dots, v_k)$$

for all $k = 1, \dots, n$. The vectors v_k are given inductively by

$$v_1 = \frac{w_1}{\|w_1\|} \qquad \text{and} \qquad v_{k+1} = \frac{w_{k+1} - \sum_{j=1}^{k} \langle w_{k+1}, v_j \rangle v_j}{\left\| w_{k+1} - \sum_{j=1}^{k} \langle w_{k+1}, v_j \rangle v_j \right\|}$$

for $k = 1, \dots, n-1$. In particular, every finite dimensional inner product space has an orthonormal basis.

Example 24.2. Consider the vectors

$$w_1 = \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix} \quad w_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 3 \end{bmatrix} \quad w_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ -1 \end{bmatrix} \tag{24.3}$$

in $\mathbb{R}^4$. We want to apply the Gram-Schmidt orthogonalisation process. First we have $\|w_1\| = \sqrt{4+1+1+1} = \sqrt{7}$, so that

$$v_1 = \frac{1}{\sqrt{7}} w_1 = \frac{1}{\sqrt{7}} \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix}.$$

Next we compute the projection of w_2 into the direction of $\boldsymbol{v}_2$. It is given by

$$u_2 = \langle w_2, v_1 \rangle v_1 = \frac{1}{7}(2 - 1 + 3) \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \frac{4}{7} \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix}$$

and so

$$w_2 - u_2 = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 3 \end{bmatrix} - \frac{4}{7} \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} -8 \\ 10 \\ 11 \\ 17 \end{bmatrix}.$$

To get v_2 we need to normalise that vector by

$$\|w_2 - u_2\| = \frac{1}{7}\sqrt{64 + 100 + 121 + 289} = \frac{1}{7}\sqrt{574}.$$

Hence

$$v_2 = \frac{1}{\sqrt{574}} \begin{bmatrix} -8 \\ 10 \\ 11 \\ 17 \end{bmatrix}.$$

Next we compute the projection u_3 of w_3 onto the plane spanned by v_1 and v_2. It is given by

$$\begin{aligned} u_3 &= \langle w_3, v_1 \rangle v_1 + \langle w_3, v_2 \rangle v_2 \\ &= \frac{1}{7}(2+2-1-1) \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix} + \frac{1}{574}(-8+20+11-17) \begin{bmatrix} -8 \\ 10 \\ 11 \\ 17 \end{bmatrix} \\ &= \frac{2}{7} \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix} + \frac{6}{574} \begin{bmatrix} -8 \\ 10 \\ 11 \\ 17 \end{bmatrix} = \frac{82}{287} \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix} + \frac{3}{287} \begin{bmatrix} -8 \\ 10 \\ 11 \\ 17 \end{bmatrix} = \frac{1}{41} \begin{bmatrix} 20 \\ 16 \\ -7 \\ 19 \end{bmatrix}. \end{aligned}$$

Hence

$$w_3 - u_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ -1 \end{bmatrix} - \frac{1}{41} \begin{bmatrix} 20 \\ 16 \\ -7 \\ 19 \end{bmatrix} = \frac{1}{41} \begin{bmatrix} 41 \\ 82 \\ 41 \\ -41 \end{bmatrix} - \frac{1}{41} \begin{bmatrix} 20 \\ 16 \\ -7 \\ 19 \end{bmatrix} = \frac{3}{41} \begin{bmatrix} 7 \\ 22 \\ 16 \\ -20 \end{bmatrix}$$

and therefore

$$v_3 = \frac{w_3 - u_3}{\|w_3 - u_3\|} = \frac{1}{\sqrt{1189}} \begin{bmatrix} 7 \\ 22 \\ 16 \\ -20 \end{bmatrix}.$$

In summary we have the orthogonalised system

$$v_1 = \frac{1}{\sqrt{7}} \begin{bmatrix} 2 \\ 1 \\ -1 \\ 1 \end{bmatrix} \quad v_2 = \frac{1}{\sqrt{574}} \begin{bmatrix} -8 \\ 10 \\ 11 \\ 17 \end{bmatrix} \quad v_3 = \frac{1}{\sqrt{1189}} \begin{bmatrix} 7 \\ 22 \\ 16 \\ -20 \end{bmatrix}. \tag{24.4}$$

Example 24.3. Consider the vector space $V := C([-1,1])$ of all continuous real valued functions on the interval $[-1,1]$. Then

$$\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)\,dx$$

defines an inner product on V. The corresponding norm is

$$\|f\|_2 = \sqrt{\int_{-1}^{1} |f(x)|^2\,dx}.$$

We know that $(1, x, x^2, x^3)$ is a linearly independent family. We apply the Gram-Schmidt orthogonalisation process to that family. Let $w_1(x) := 1$, $w_2(x) := x$, $w_3(x) := x^2$ and $w_4(x) := x^3$ for all $x \in [0, 1]$. First note that

$$\|w_1\|_2 = \sqrt{\int_{-1}^{1} 1^2 \, dx} = \sqrt{2},$$

so

$$v_1(x) = \frac{1}{\sqrt{2}}.$$

Next we have

$$u_2 = \langle u_2, v_1 \rangle v_1 = \int_{-1}^{1} \frac{1}{\sqrt{2}} x \, dx \cdot \frac{1}{\sqrt{2}} = 0$$

and so

$$w_2 - u_2 = w_2 = x.$$

Hence

$$\|w_2\|_2 = \sqrt{\int_{-1}^{1} x^2 \, dx} = \sqrt{\frac{2}{3}}$$

and therefore

$$v_2 = \frac{w_2}{\|w_2\|_2} = \sqrt{\frac{3}{2}} x$$

Next

$$u_3 = \langle w_3, v_1 \rangle v_1 + \langle w_3, v_2 \rangle v_2 = \int_{-1}^{1} x^2 \, dx \cdot \frac{1}{2} + \frac{3}{2} \int_{-1}^{1} x^2 \cdot x \, dx \cdot x = \frac{2}{3}\frac{1}{2} + 0x = \frac{1}{3},$$

and so

$$w_3 - u_3 = x^2 - \frac{1}{3}.$$

Now

$$\begin{aligned} \|w_3 - u_3\|_2 = \sqrt{\int_{-1}^{1} \left(x^2 - \frac{1}{3}\right)^2 dx} &= \sqrt{\int_{-1}^{1} x^4 - \frac{2}{3}x^2 + \frac{1}{9} \, dx} \\ &= \sqrt{\frac{2}{5} - \frac{4}{9} + \frac{2}{9}} = \frac{2}{3}\sqrt{\frac{2}{5}} \end{aligned}$$

and therefore

$$v_3 = \frac{w_3 - u_3}{\|w_3 - u_3\|_2} = \frac{1}{2}\sqrt{\frac{5}{2}}\left(3x^2 - 1\right)$$

To compute v_4 we need

$$\begin{aligned} u_4 &= \langle w_3, v_1 \rangle v_1 + \langle w_3, v_2 \rangle v_2 + \langle w_4, v_3 \rangle v_3 \\ &\quad \frac{1}{2} \int_{-1}^{1} x^3 \, dx + \frac{3}{2} \int_{-1}^{1} x^3 \cdot x \, dx \cdot x + \frac{5}{8} \int_{-1}^{1} x^3 (3x^2 - 1) \, dx (3x^2 - 1) \\ &\qquad = 0 + \frac{3}{5}x + \frac{5}{8} \cdot 0(3x^2 - 1) = \frac{3}{5}x. \end{aligned}$$

Then

$$w_4 - u_4 = x^3 - \frac{3}{5}x = \frac{1}{5}\big(5x^3 - 3x\big)$$

and

$$\|w_4 - u_4\|_2 = \frac{1}{5}\sqrt{\int_{-1}^{1}(5x^3 - 3x)^2\,dx} = \frac{2}{5}\sqrt{\frac{2}{7}},$$

so

$$v_4 = \frac{w_4 - u_4}{\|w_4 - u_4\|_2} = \frac{1}{2}\sqrt{\frac{7}{2}}\big(5x^3 - 3x\big).$$

Hence the orthogonalised system is

$$v_1 = \frac{1}{\sqrt{2}}, \quad v_2 = \sqrt{\frac{3}{2}}x, \quad v_3 = \frac{1}{2}\sqrt{\frac{5}{2}}\big(3x^2 - 1\big), \quad v_4 = \frac{1}{2}\sqrt{\frac{7}{2}}\big(5x^3 - 3x\big).$$

These polynomials are (up to a multiplicative constant) the *Legendre polynomials* and examples of *orthogonal polynomials*. These polynomials arise naturally when solving certain differential equations arising in mathematical physics and elsewhere. In case of the Legendre polynomials, they solve the differential equation

$$\frac{d}{dx}\Big((1 + x^2)\frac{dP}{dx}\Big) = -n(n + 1)P$$

The orthogonality relations play an important rôle in the theory of ordinary and partial differential equations.

Next we derive some consequences of the Gram-Schmidt process.

24.1 QR-Factorisation

The Gram-Schmidt process is an algorithm that transforms the basis $(w_1, \dots, w_k)$ of $V_k = \operatorname{span}(w_1, \dots, w_k)$ into an orthonormal basis $(v_1, \dots, v_k)$. The transition matrix can be obtained by rearranging (24.2). Solving for w_{k+1} we get

$$\begin{aligned}
w_1 &= \|w_1 - u_1\|v_1 = \|w_1\|v_1 \\
w_2 &= \langle w_2, v_1\rangle v_1 + \|w_2 - u_2\|v_2 \\
w_3 &= \langle w_3, v_1\rangle v_1 + \langle w_3, v_2\rangle v_2 + \|w_3 - u_3\|v_3 \\
&\vdots \\
w_n &= \langle w_n, v_1\rangle v_1 + \langle w_n, v_2\rangle v_2 + \cdots + \langle w_n, v_{n-1}\rangle v_{n-1} + \|w_n - u_n\|v_n.
\end{aligned} \tag{24.5}$$

Hence the coordinate vector of w_k with respect to the basis $(v_1, \dots, v_n)$ is

$$\begin{bmatrix} \langle w_k, v_1\rangle \\ \vdots \\ \langle w_k, v_{k-1}\rangle \\ \|w_k - u_k\| \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Hence the transition matrix from $(w_1, \dots, w_n)$ to the orthonormal basis $(v_1, \dots, v_n)$ is given by

$$R := \begin{bmatrix} \|w_1\| & \langle w_2, v_1\rangle & \langle w_3, v_1\rangle & \vdots & \langle w_{n-1}, v_1\rangle & \langle w_n, v_1\rangle \\ 0 & \|w_2 - u_2\| & \langle w_3, v_2\rangle & \vdots & \vdots & \vdots \\ \vdots & 0 & \|w_3 - u_3\| & \vdots & \vdots & \vdots \\ \vdots & & 0 & \vdots & \langle w_{n-1}, v_{n-2}\rangle & \vdots \\ \vdots & & & 0 & \|w_{n-1} - u_{n-1}\| & \langle w_n, v_{n-1}\rangle \\ 0 & \dots\dots\dots\dots\dots\dots\dots\dots & & & 0 & \|w_n - u_n\| \end{bmatrix} \tag{24.6}$$

Let $n \leq m$ and consider a matrix $A \in \mathrm{Mat}(m \times n; \mathbb{K})$ of rank n, that is, a matrix with n linearly independent columns $w_1, \dots, w_n \in \mathbb{K}^m$. We then apply the Gram-Schmidt process to these column vectors as above and obtain orthonormal vectors $(v_1, \dots, v_n)$. Then we form the transition matrix R as done above. That transition matrix can also be used to factorise A. We form the matrix

$$Q := \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}$$

with columns $v_1, \dots, v_n$. We then deduce from (24.5) and (24.6) that $A = QR$, which leads to the following theorem.

Theorem 24.4 (QR-factorisation)**.** *For every matrix $A \in \mathrm{Mat}(m \times n; \mathbb{C})$ there exist a matrix $Q \in \mathrm{Mat}(m \times n; \mathbb{C})$ with orthonormal columns and an upper triangular matrix $R \in \mathrm{Mat}(n \times n; \mathbb{C})$ such that $A = QR$.*

Example 24.5. We consider the 4×3 matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 2 & 2 \\ -1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

whose columns are the vectors w_1, w_2, w_3 from Example 24.2. The orthogonalised system turns out to be (24.4). According to the construction of the QR-factorisation these vectors form the columns of Q, so

$$Q = \begin{bmatrix} \frac{2}{\sqrt{7}} & -\frac{8}{\sqrt{574}} & \frac{7}{\sqrt{1189}} \\ \frac{1}{\sqrt{7}} & \frac{10}{\sqrt{574}} & \frac{22}{\sqrt{1189}} \\ -\frac{1}{\sqrt{7}} & \frac{11}{\sqrt{574}} & \frac{16}{\sqrt{1189}} \\ \frac{1}{\sqrt{7}} & \frac{17}{\sqrt{574}} & -\frac{20}{\sqrt{1189}} \end{bmatrix}.$$

The matrix R is given by (24.6), and so we can use the information computed in Example 24.2. Hence

$$R = \begin{bmatrix} \|w_1\| & \langle w_2, v_1\rangle & \langle w_3, v_1\rangle \\ 0 & \|w_2 - u_2\| & \langle w_3, v_2\rangle \\ 0 & 0 & \|w_3 - u_3\| \end{bmatrix} = \begin{bmatrix} \sqrt{7} & \frac{4}{\sqrt{7}} & \frac{2}{\sqrt{7}} \\ 0 & \frac{\sqrt{574}}{7} & \frac{6}{\sqrt{574}} \\ 0 & 0 & \frac{3\sqrt{1189}}{41} \end{bmatrix}.$$

Hence the QR-decomposition is

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 2 & 2 \\ -1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{7}} & -\frac{8}{\sqrt{574}} & \frac{7}{\sqrt{1189}} \\ \frac{1}{\sqrt{7}} & \frac{10}{\sqrt{574}} & \frac{22}{\sqrt{1189}} \\ -\frac{1}{\sqrt{7}} & \frac{11}{\sqrt{574}} & \frac{16}{\sqrt{1189}} \\ \frac{1}{\sqrt{7}} & \frac{17}{\sqrt{574}} & -\frac{20}{\sqrt{1189}} \end{bmatrix} \begin{bmatrix} \sqrt{7} & \frac{4}{\sqrt{7}} & \frac{2}{\sqrt{7}} \\ 0 & \frac{\sqrt{574}}{7} & \frac{6}{\sqrt{574}} \\ 0 & 0 & \frac{3\sqrt{1189}}{41} \end{bmatrix} = QR.$$

We next use the QR-factorisation to derive a geometric interpretation of determinants.

24.2 Geometric interpretation of determinants

In $\mathbb{R}^n$ consider a family of linearly independent vectors $w_1, \dots, w_k$. If $k = 1$ we are interested in the length of the vector. If $k = 2$, the two vectors span a parallelogram and we are interested in its area. If $k = 3$ the vectors span a parallelepiped, and we are interested in computing its volume. These objects are illustrated in Figure 24.4 We are

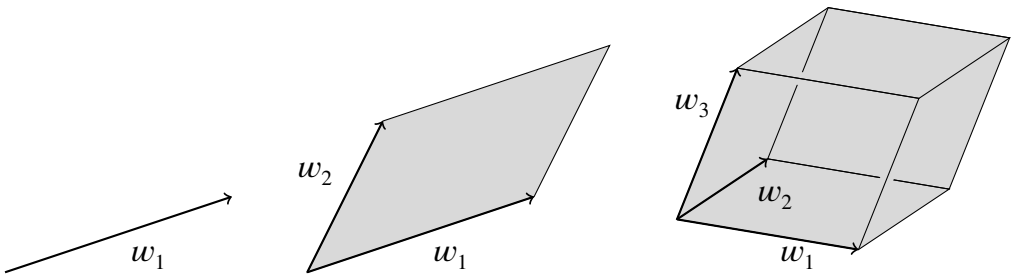


Figure 24.4: Parallelepipeds spanned by one, two and three vectors.

more generally interested in parallelepipeds spanned by k vectors $w_1, \dots, w_k$ in $\mathbb{R}^n$ and the corresponding k-dimensional volume. The parallelepiped is formally the set

$$\left\{\lambda_1 w_1 + \cdots + \lambda_k w_k : 0 \le \lambda_j \le 1,\ 1 \le j \le k\right\}$$

To find its volume we transform it into a rectangular box without changing the volume. The k-dimensional "volume" of a rectangular box is the product of the lengths of the edges as in one, two and three dimensions. To compute the volume of a parallelepiped we transform it into a rectangle of the same volume by using the fact that the volume is given by the formula "base times height." This involves projections into the direction vertical to a $(k-1)$-dimensional subspace as in the Gram-Schmidt process. If we have two vectors w_1, w_2, we look at the projection u_2 of w_2 in the direction of w_1. Then the length of $w_2 - u_2$ is the height of the parallelogram, so that

$$\text{area of parallelogram} = \|w_1\|\|w_2 - u_2\|.$$

The idea is illustrated in Figure 24.5.

We use a similar idea for three vectors w_1, w_2, w_3. To find the height of the parallelepiped we compute the projection u_3 of w_3 onto the plane spanned by w_1 and w_2.

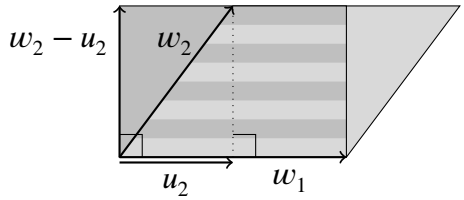


Figure 24.5: Rectangle of same area as parallelogram

Then $\|w_3 - u_3\|$ is the height of the parallelepiped. We know from above how to compute the area of the parallelogram at the base, so

$$\text{volume of parallelepiped} = \|w_1\|\|w_2 - u_2\|\|w_3 - u_3\|.$$

The idea is illustrated in Figure 24.6.

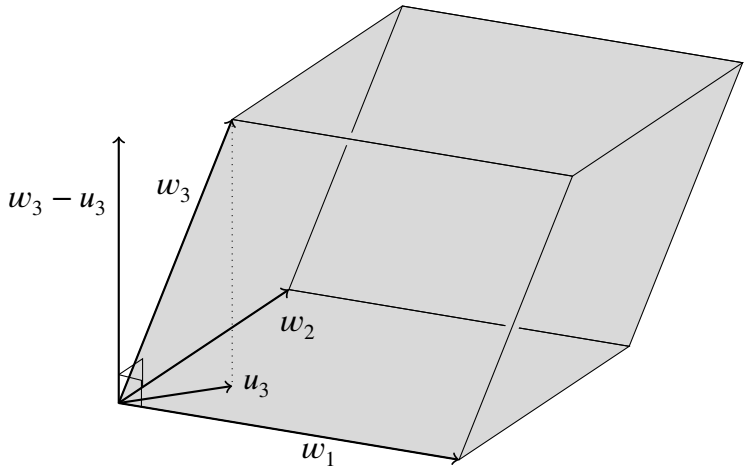


Figure 24.6: Volume of a parallelepiped spanned by w_1, w_2, w_3.

Generalising the idea we see that the n-dimensional volume of a parallelepiped spanned by n vectors $w_1, \dots, w_n$ is given by

$$\det(R) = \|w_1\|\|w_2 - u_2\|\|w_3 - u_3\| \cdots \|w_n - u_n\|$$

where is R the matrix given by (24.6). This is regardless of the dimension of the underlying space the vectors are embedded in. We want to use $w_1, \dots, w_n$ to compute that volume. for that we set

$$J := \begin{bmatrix} w_1 & \dots & w_n \end{bmatrix}.$$

From the previous section we know that $J = QR$ with Q having orthogonal columns. Hence we see that

$$J^T J = (QR)^T QR = R^T Q^T QR = R^T I_n R = R^T R$$

and therefore

$$\det(J^T J) = \det(R^T R) = \det(R^T)\det(R) = \big(\det(R)\big)^2$$

is the square of the volume of the parallelepiped spanned by $w_1, \dots, w_n$. This proves the following theorem.

Theorem 24.6. *Let $w_1, \dots, w_n$ be vectors in an inner product space V and set*

$$J := \begin{bmatrix} w_1 & \dots & w_n \end{bmatrix}.$$

Then

$$\sqrt{\det(J^T J)}$$

is the volume of the parallelepiped spanned by $w_1, \dots, w_n$. If $\dim V = n$, then the volume is given by

$$|\det J|.$$

The last statement follows since for a square matrix

$$\det(J^T J) = \det(J^T) \det J = (\det J)^2.$$

☞ *Remark* 24.7. (a) Above we assumed that the vectors $w_1, \dots w_n$ are linearly independent. If they are linearly independent the n-dimensional volume of the parallelepiped spanned by these vectors is zero because they lie in an $(n-1)$-dimensional subspace. If that is the case $\det(J^T J) = 0$ as well because the rank of J is at most $n-1$.

(b) The formula derived above also applies for one single vector w_1. We simply have $J = w_1$ and $J^T J = w_1^T w_1 = \|w_1\|^2$, which is the "one-dimensional volume", that is length.

Examples 24.8. (a) The area of the parallelogram spanned in $\mathbb{R}^2$ by $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$ is given by

$$\left| \det \begin{bmatrix} 3 & 2 \\ 1 & 5 \end{bmatrix} \right| = 13.$$

The absolute value is relevant if the determinant is negative. The significance of a positive or negative determinant is that it gives one of two possible orientations of the pair (or in general n-tuple) of vectors. We will not discuss that here.

(b) To compute the area of the parallelogram in $\mathbb{R}^3$ spanned by $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$ we form the matrix

$$J := \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 1 & 2 \end{bmatrix}$$

and then compute

$$J^T J = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 7 \\ 7 & 14 \end{bmatrix}.$$

Note that the matrix $J^T J$ is always symmetric, which simplifies its computation. The required area is then given by

$$\sqrt{\det(J^T J)} = \sqrt{35}.$$

Alternatively we could compute the area by taking the cross product of the two vectors and then take its norm. That is however quite a bit more work. The advantage of the above method is also that it works regardless of the dimension of the underlying space.

(c) We compute the (three dimensional) volume of the parallelepiped spanned by the vectors

$$\begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ -1 \\ 0 \\ 3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}$$

in $\mathbb{R}^4$. To do so we form the matrix

$$J := \begin{bmatrix} 1 & 2 & -2 \\ 0 & -1 & 1 \\ -1 & 0 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

and then compute

$$J^T J = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 2 & -1 & 0 & 3 \\ -2 & 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -2 \\ 0 & -1 & 1 \\ -1 & 0 & 3 \\ 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 8 & -3 \\ 8 & 14 & -2 \\ -3 & -2 & 15 \end{bmatrix}.$$

Again note that the matrix is symmetric, so only the above (or below) diagonal elements have to be computed. Expanding $\det(J^T J)$ along the first column we get

$$\begin{aligned} \det(J^T J) &= 6 \det \begin{bmatrix} 14 & -2 \\ -2 & 15 \end{bmatrix} - 8 \det \begin{bmatrix} 8 & -3 \\ -2 & 15 \end{bmatrix} - 3 \det \begin{bmatrix} 8 & -3 \\ 14 & -2 \end{bmatrix} \\ &= 6 \times 206 - 8 \times 114 - 3 \times 26 = 246. \end{aligned}$$

Hence the required volume is

$$\sqrt{\det(J^T J)} = \sqrt{246} \approx 15.68.$$

25 Orthogonal Diagonalisation

We know that not all linear operators are diagonalisable. Here we want to look at a special but important class of operators, namely Hermitian (self-adjoint) operators. They turn out to be diagonalisable.

Definition 25.1 (Hermitian operator)**.** Let V be an inner product space and $T : V \to V$. Then T is called a Hermitian (or self-adjoint) operator if

$$\langle Tu, v\rangle = \langle u, Tv\rangle$$

for all $u, v \in V$.

Examples of Hermitian operators are real symmetric matrices or Hermitian matrices $A \in \mathrm{Mat}(n \times n; \mathbb{C})$ such that

$$\bar{A}^T = A.$$

Indeed, if $z, w \in \mathbb{C}^n$ and $A = [a_{ij}]$, then by definition of the standard inner product

$$\langle Az, w\rangle = \sum_{i=1}^{n}\sum_{j=1} a_{ij} z_j \bar{w}_i = \sum_{j=1}^{n}\sum_{i=1} \overline{\bar{a}_{ij} w_i} z_j = \langle z, \bar{A}^T w\rangle = \langle z, Aw\rangle.$$

Example 25.2. The matrix

$$A = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}$$

is a Hermitian matrix. Since

$$\bar{A}^T = \begin{bmatrix} 2 & \overline{-i} \\ \bar{i} & 2 \end{bmatrix} = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix} = A$$

Note that the diagonal entries of a Hermitian matrix are always real since $a_{ii} = \bar{a}_{ii}$ by definition of a Hermitian matrix, and so $a_{ii} \in \mathbb{R}$.

Theorem 25.3. *Let T be a Hermitian operator on the inner product space V.*

(i) *All eigenvalues of T are real;*

(ii) *Eigenvectors of T to different eigenvalues are orthogonal;*

(iii) *If* $\dim V = n$*, then V has an orthonormal basis of eigenvectors of T and hence T is diagonalisable.*

Proof. (i) If $\lambda \in \mathbb{C}$ is an eigenvalue and u a corresponding eigenvector, then by definition of a Hermitian operator and the anti-symmetry of the inner product

$$\lambda\|u\|^2 = \langle Tu, u\rangle = \langle u, Tu\rangle = \overline{\langle Tu, u\rangle} = \bar{\lambda}\|u\|^2.$$

Any complex number that coincides with its complex conjugate must be real, so $\lambda \in \mathbb{R}$.

(ii) Let λ, μ be distinct eigenvalues of T and u, v corresponding eigenvectors. Then by (i) $\lambda, \mu \in \mathbb{R}$ and by definition of a Hermitian operator

$$\lambda\langle u, v\rangle = \langle Tu, v\rangle = \langle u, Tv\rangle = \mu\langle u, v\rangle.$$

Since by assumption $\lambda \neq \mu$ the above identity can only be true if $\langle u, v\rangle = 0$, that is, the eigenvectors are orthogonal.

(iii) By definition of a Hermitian operator, for any $\lambda \in \mathbb{R}$ and $u, v \in V$

$$\langle \lambda u - Tu, v\rangle = \langle \lambda u, v\rangle - \langle Tu, v\rangle = \langle u, \lambda v\rangle - \langle u, Tv\rangle = \langle u, \lambda v - Tv\rangle.$$

Hence $\lambda I - T$ is a Hermitian operator as well. By (ii) every eigenvalue λ is real and so we have

$$\|(\lambda I - T)u\|^2 = \langle (\lambda I - T)u, (\lambda I - T)u\rangle = \langle (\lambda I - T)^2 u, u\rangle$$

for all $u \in U$. Hence if $u \in \ker(\lambda I - T)^2$, then $u \in \ker(\lambda I - T)$. This implies that $\ker(\lambda I - T)^2 = \ker(\lambda I - T)$. By Corollary 21.8 T is diagonalisable since $\mathbb{C}$ is algebraically closed. By Theorem 24.1, in each eigenspace we can choose an orthonormal basis. Since eigenvectors to different eigenvalues are orthogonal there exists an orthonormal basis of eigenvectors. □

As a consequence, every Hermitian matrix is similar to a diagonal matrix with real entries, and the similarity is via a unitary matrix.

Theorem 25.4 (orthogonal diagonalisation)**.** *Let $A \in \mathrm{Mat}(n \times n, \mathbb{C})$ be a Hermitian matrix. Then there exists a unitary matrix $U \in \mathrm{Mat}(n \times n, \mathbb{C})$ such that*

$$A = UD\bar{U}^T.$$

where D is a diagonal matrix. More precisely, if $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A counted according to their algebraic multiplicity, then there exists an orthonormal basis $(v_1, \dots, v_n)$ of corresponding eigenvectors so that

$$D = \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix} \qquad \text{and} \qquad U = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}^{-1}.$$

Proof. First note that the map $z \mapsto Az$ defines a Hermitian operator on $\mathbb{C}^n$. By Theorem 25.3 all eigenvalues $\lambda_1, \dots, \lambda_n$ of A are real and there exists an orthonormal basis $(v_1, \dots, v_n)$ of $\mathbb{C}^n$ of eigenvectors of A. We assume that v_i is an eigenvector corresponding to λ_i. Then U is a unitary matrix and therefore $U^{-1} = \bar{U}^T$. □

Remark 25.5. In the case of a real symmetric matrix U is an orthogonal matrix, hence the name orthogonal diagonalisation.

Example 25.6. The matrix

$$A = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix}$$

is Hermitian. To compute the eigenvalues note that

$$\det(\lambda I - A) = (\lambda - 2)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3),$$

so that $\lambda_1 = 1$ and $\lambda_2 = 3$. To get the eigenvectors we do a row reduction

$$I - A = \begin{bmatrix} -1 & -i \\ i & -1 \end{bmatrix} \xrightarrow[R_1 \to -R_1]{R_2 \to R_2 + iR_1} \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$$

and so

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}$$

is an eigenvector corresponding to $\lambda_1 = 1$ with $\|v_1\| = 1$. Similarly we get and eigenvector to $\lambda_2 = 3$.

$$3I - A = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} \xrightarrow{R_2 \to R_2 - iR_1} \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$$

and so

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix}$$

is an eigenvector corresponding to $\lambda_2 = 3$ with $\|v_2\| = 1$. The theory tells us that the eigenvectors are orthogonal, but to convince ourselves we check this as well:

$$\left\langle \begin{bmatrix} 1 \\ i \end{bmatrix}, \begin{bmatrix} 1 \\ -i \end{bmatrix} \right\rangle = 1 \cdot \bar{1} + i \cdot (\overline{-i}) = 1 + i^2 = 0.$$

Hence,

$$\left(\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} \end{bmatrix} \right)$$

is an orthonormal basis of eigenvectors of A and

$$U := \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix}$$

is the transition matrix from the standard basis to the basis of eigenvectors. The inverse of the unitary matrix U is very easy to get. Since the columns form an orthonormal basis, the inverse is the conjugate transposed. We then have

$$\begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix}$$

and also

$$\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{bmatrix}.$$

Bibliography

[1] Oswaldo Rio Branco de Oliveira. "The Fundamental Theorem of Algebra: An Elementary and Direct Proof". In: *The Mathematical Intelligencer* 33.2 (2011), pp. 1–2. DOI: 10.1007/s00283-011-9199-2.

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